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Exercise Workbook

Introduction to Electric Power Conversion

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Introduction

Electric power conversion is a central theme in the technological development of electrical engineering disciplines and represents one of the cornerstones of the ongoing energy transition. The growing integration of renewable sources, the electrification of transport and the advancement of storage systems make this subject not only highly relevant but also strategic for the education of future engineers. Given this importance, a wide range of textbooks and resources are available that thoroughly address the theoretical and practical aspects of electric energy conversion.

In recent years, teaching methods have been further enriched by the availability of video lectures and blended learning approaches, which provide flexibility and immediacy in accessing content. Digital platforms undoubtedly offer significant advantages, such as the possibility of monitoring which parts of a resource have been effectively viewed or revisited by students, thus providing useful feedback to instructors. However, these tools also involve certain risks: the learning process may be perceived as reduced to simply watching a multimedia resource — at times in accelerated mode — without truly engaging the student. Numerous pedagogical approaches emphasize that learning is more effective when it actively engages multiple senses and requires direct effort from the learner. In this respect, the passive consumption of a multimedia resource may risk “passivizing” the student, limiting the depth of the educational experience.

This workbook has been conceived precisely with the aim of fostering active student engagement through a series of exercises designed to guide learners in the analysis and calculation of concrete cases. Drawing graphs, performing calculations by hand and developing diagrams with pen and paper complement computer-based simulations, creating a truly in-

tegrated approach. The underlying conviction is that the combination of traditional and digital tools provides undeniable educational benefits: on the one hand, it enhances the understanding of the concepts covered, and on the other, it supports long-term retention and assimilation of the most important concepts.

This workbook assumes prior knowledge of basic electrical engineering, in particular the ability to analyze electrical circuits both in steady-state conditions and in periodic (general) regimes. Short theoretical notes are provided throughout the text, but readers are referred to specialized textbooks for a more comprehensive treatment of the underlying theory.

The focus is limited to static conversion circuits, commonly referred to as “power electronics”. The topics covered are intended as a first introduction to the subject and therefore only fundamental configurations are presented and discussed. For numerical simulations, the software PLECS is suggested, and a short introduction to its use is provided in the appendix.

Further improvements to the text are under development and I will be therefore grateful to anyone who would like to report errors, suggestions or inaccuracies to me at luigi.alberti@unipd.it.

A printed version of this workbook is available. It also includes solutions to every exercise and is printed on demand on white paper with premium colour quality. For more information visit the page: <https://luigialberti.github.io/publications/>.

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Part I

Exercises

Circuits in periodic steady-state

In this chapter circuits operating in periodic steady-state (not necessarily sinusoidal) are considered. Such circuits are particularly important in power conversion equipment and they exhibit some specific properties that are here analyzed by mean of some exercises. At first the concepts of periodic waveforms and the main related quantities such as average value and root mean square (rms) value are reviewed.

Periodic waveforms, average and rms values

The voltages and currents in a circuit operating in periodic steady-state can be described mathematically by:

$$\begin{aligned}v(t + T) &= v(t) \\i(t + T) &= i(t)\end{aligned}$$

This means that all the quantities in the circuit repeated themselves identical after a the period T and that a potential transient has already finished. For periodic quantities it is interesting to consider the average value, defined as the average of the considered quantity over the period T . Considering the voltage, for example, the average value V is computed as:

$$V = \frac{1}{T} \int_{t_0}^{t_0+T} v(t) dt$$

Since the quantity is periodic, the initial time for integration t_0 can be chosen arbitrarily, provided that the circuit is operating in steady-state. It is also of interest to consider the average value of the product of periodic quantities such as the instantaneous electrical power $p(t) = v(t) \cdot i(t)$ which can be written as:

$$P = \frac{1}{T} \int_{t_0}^{t_0+T} p(t) dt = \frac{1}{T} \int_{t_0}^{t_0+T} v(t) \cdot i(t) dt$$

Given that $v(t)$ and $i(t)$ are periodic with the same period T , $p(t)$ is also periodic and P has the important and significant meaning of average work passing the electric port in the period T . In general, the period of $p(t)$ is not the same as for $v(t)$ and $i(t)$.

The root mean square (rms) of a quantity is defined as the square root of the average

value of a squared quantity. For example for a voltage:

$$V_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} v^2(t) dt}$$

It is useful to think about this definition by remembering the meaning of the “rms” acronym step by step. Considering a generic function f :

- the square of f is considered, $f^2(t)$
- the average value of $f^2(t)$ is computed over the period, $\frac{1}{T} \int_{t_0}^{t_0+T} f^2(t) dt$
- the square root of such a value is then computed to obtain $F_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} f^2(t) dt}$

Consider these steps when solving the following exercises. When a waveform can be decomposed into more elementary orthogonal components whose rms values are known, the total rms value can be computed as:

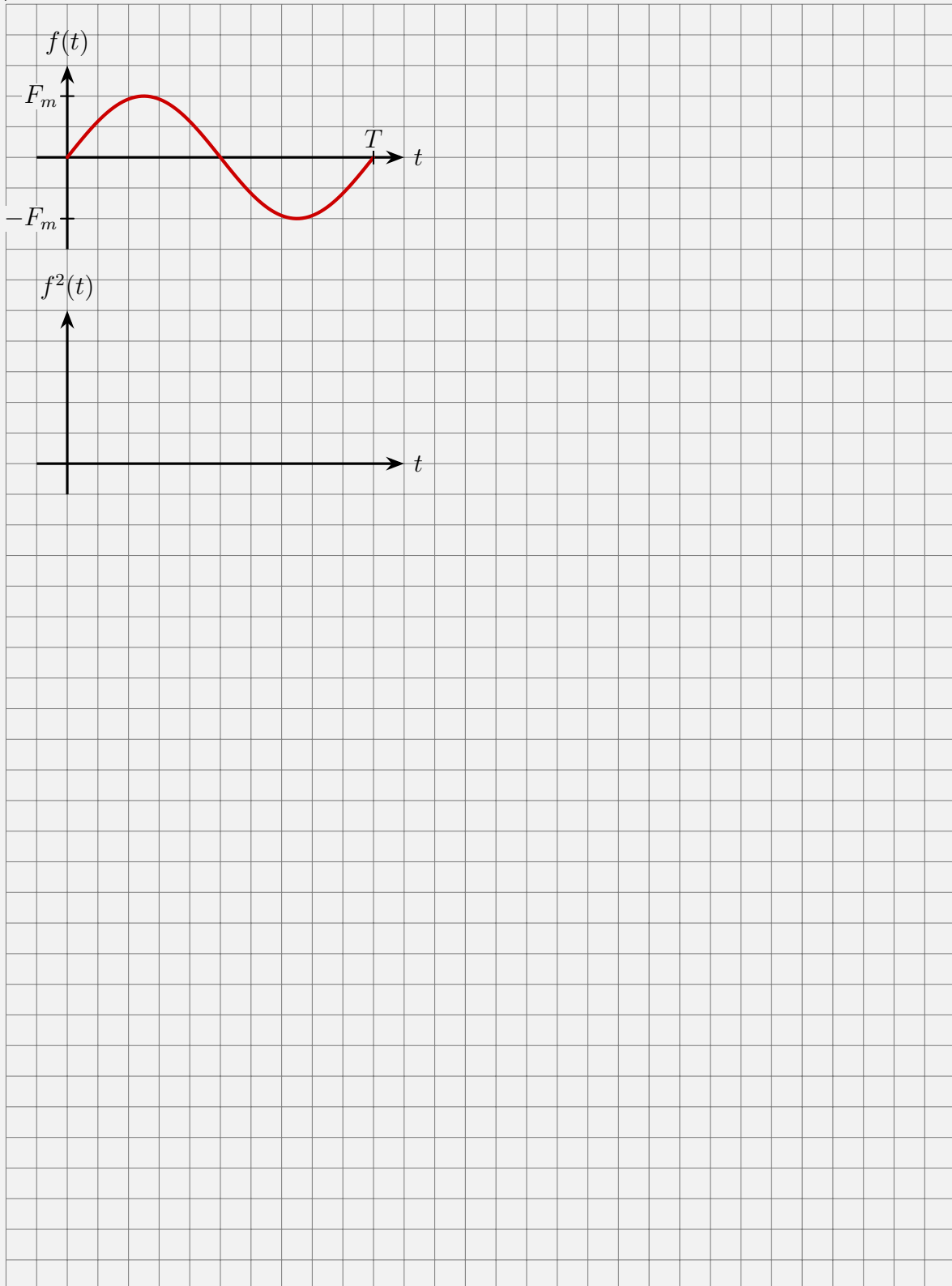
$$V_{rms} = \sqrt{V_{1,rms}^2 + V_{2,rms}^2 + \dots} = \sqrt{\sum_{n=1}^N V_{n,rms}^2} \quad (1.1)$$

For example, this is the case when a Fourier series expansion is adopted to describe periodic functions.

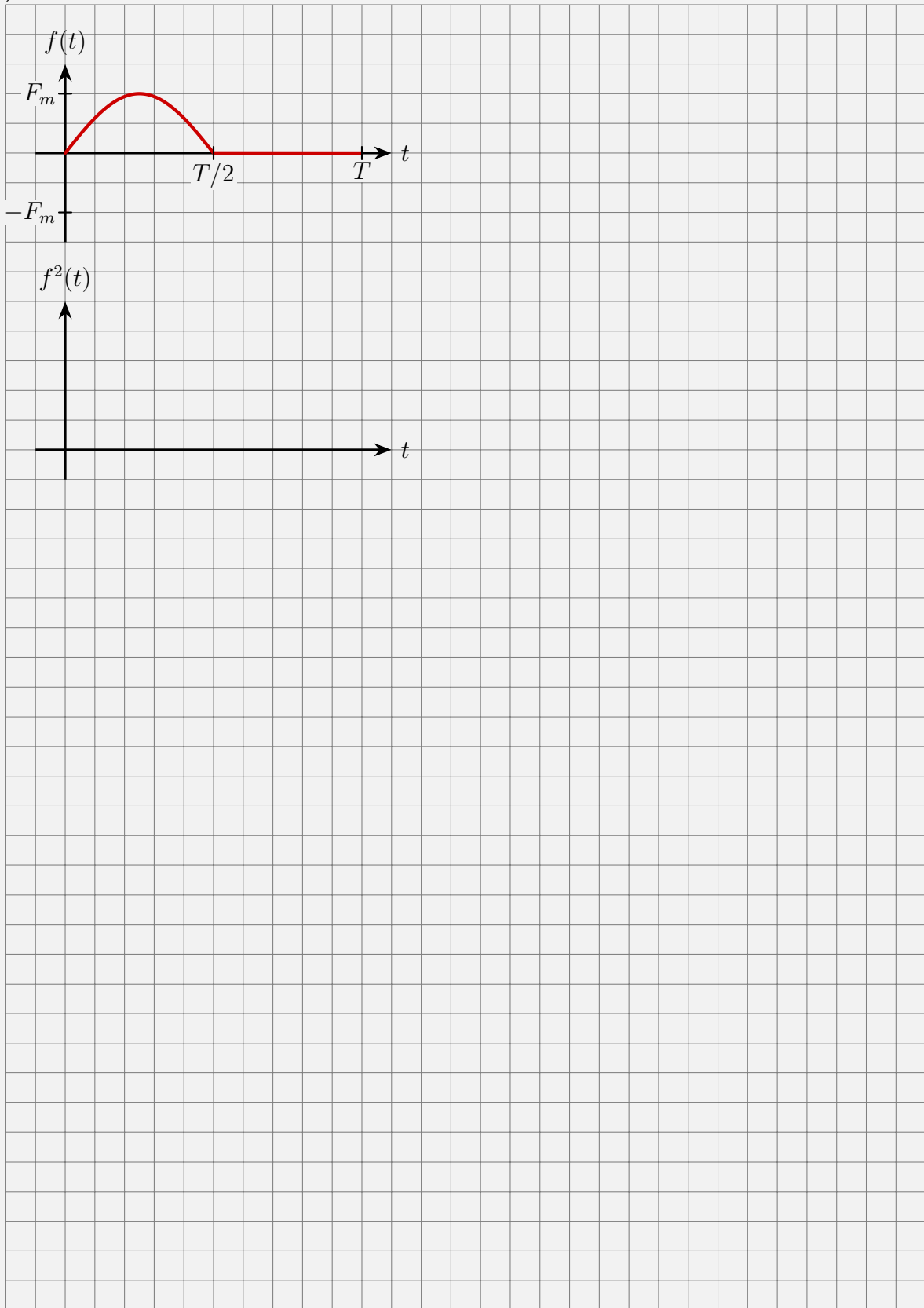
Exercise 1.1: compute the average and rms value of the following waveforms.

1.1

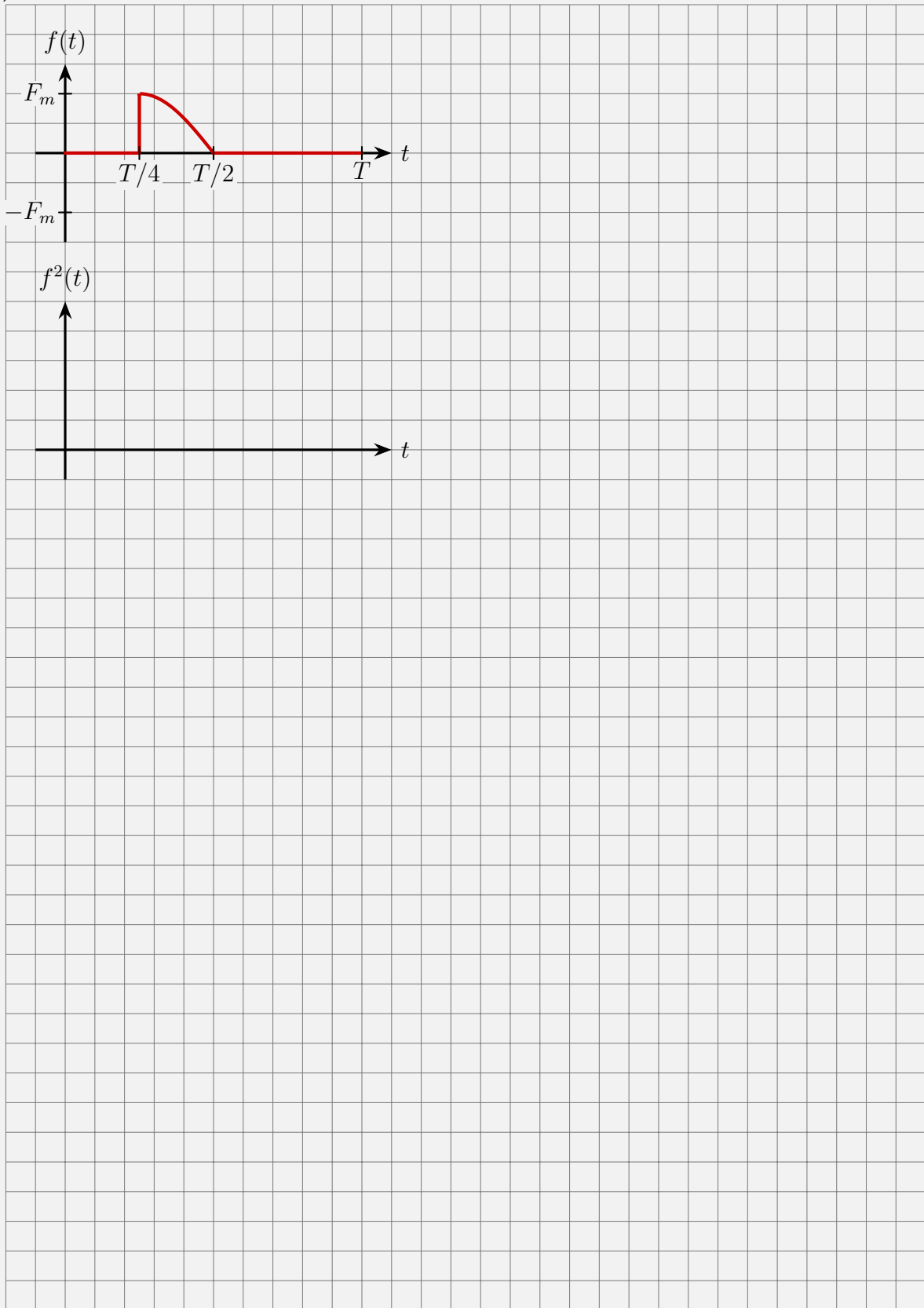
a)



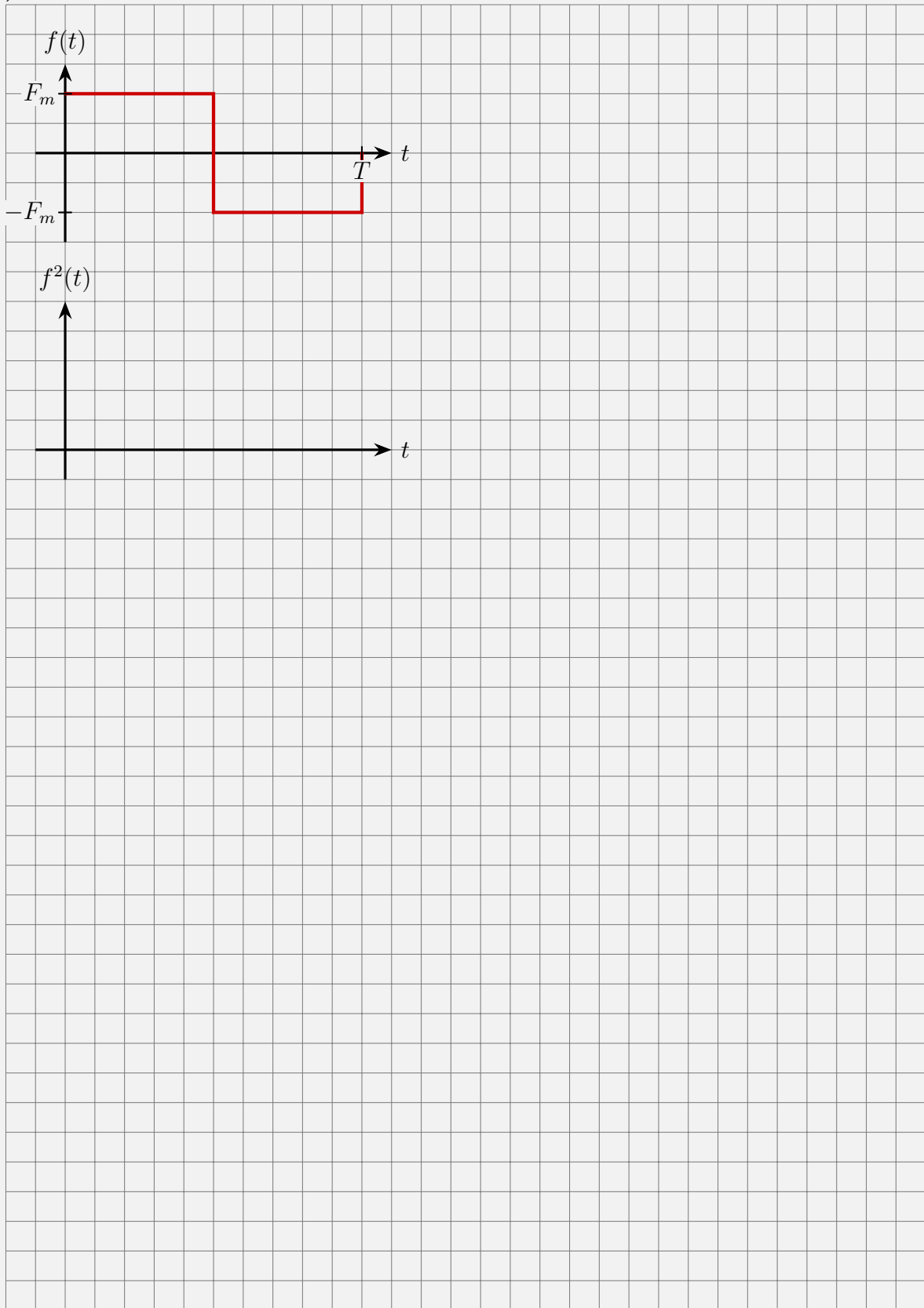
b)



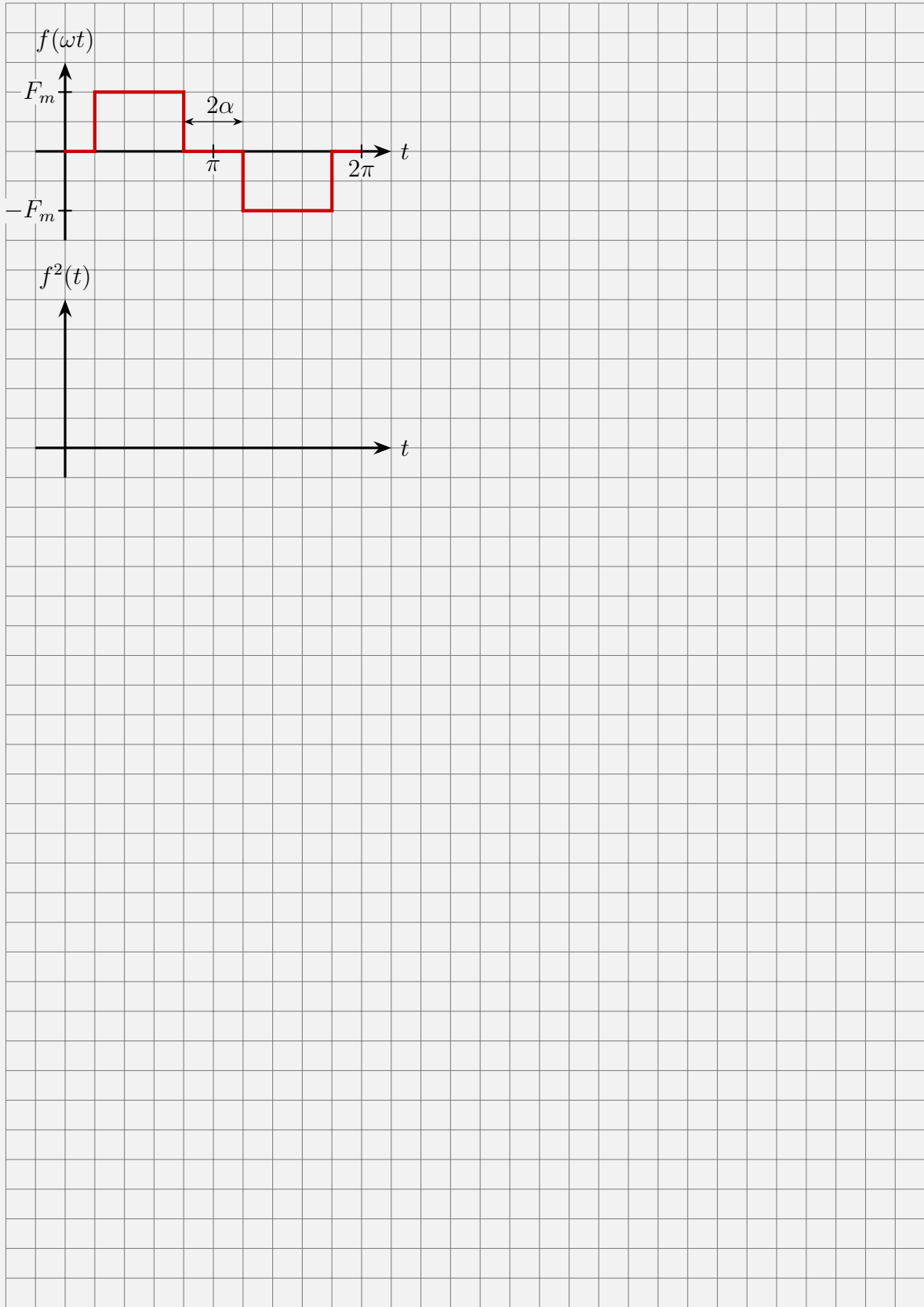
c)



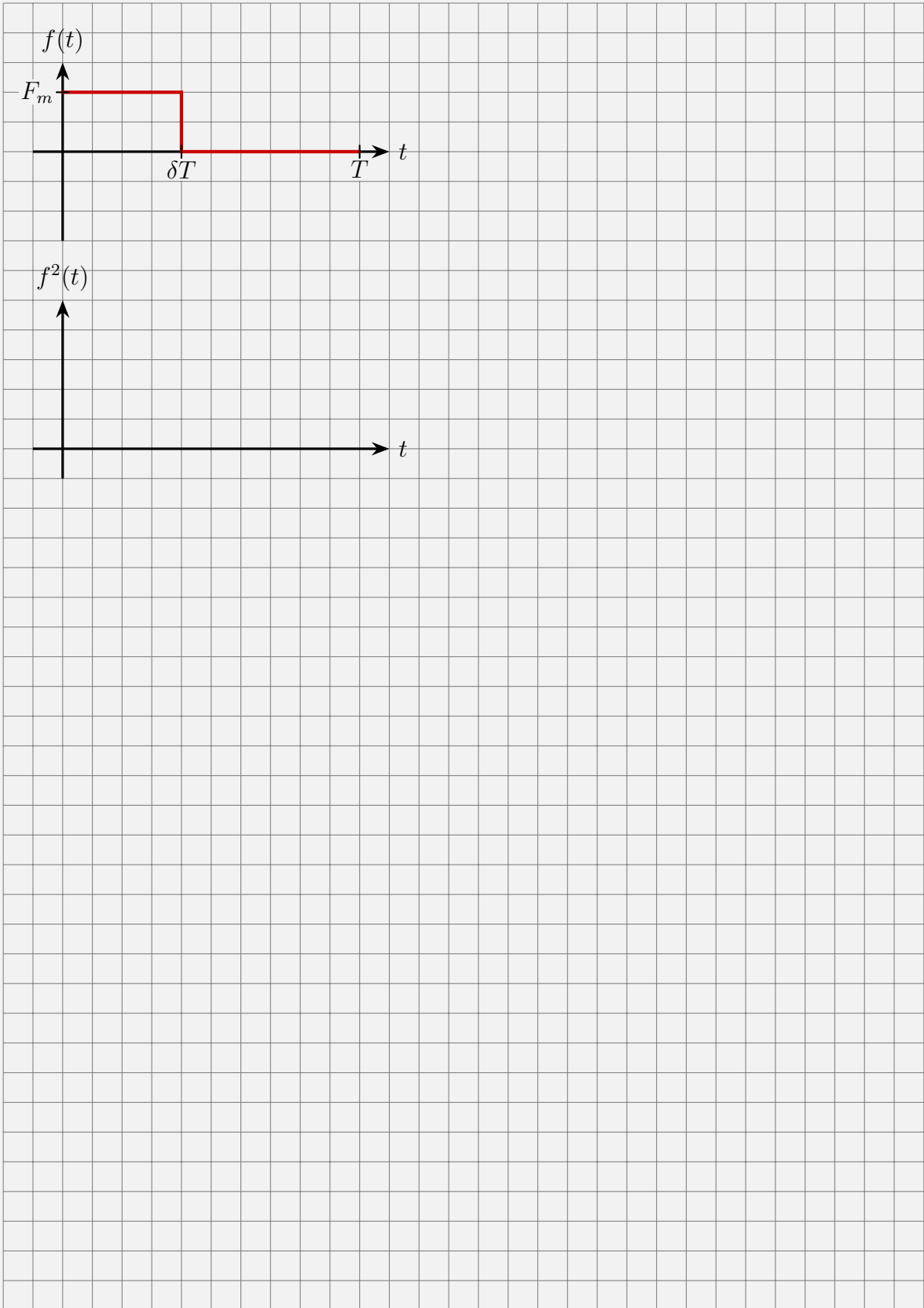
d)



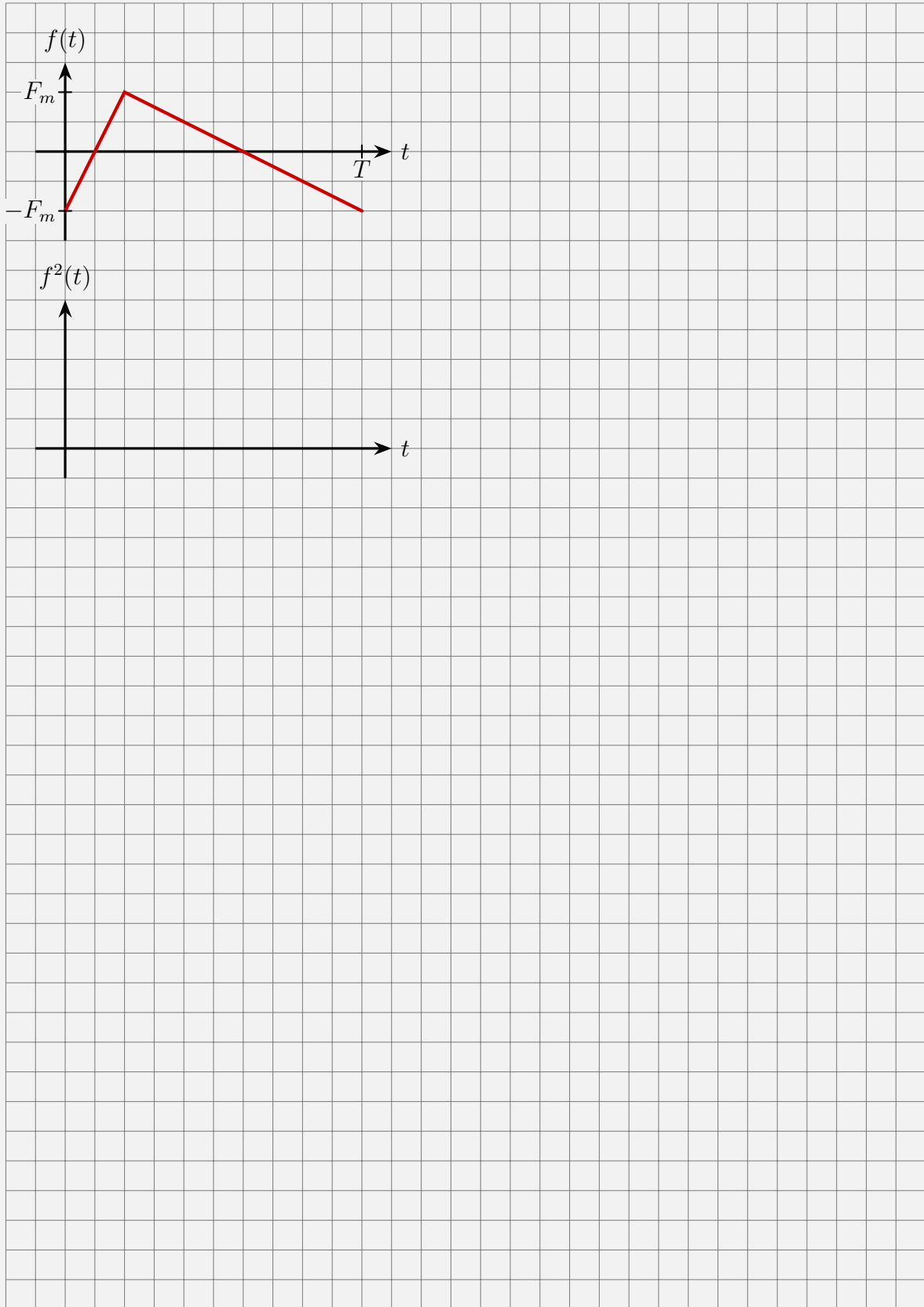
e)



f)

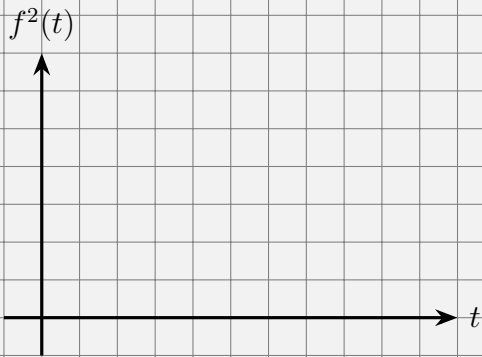
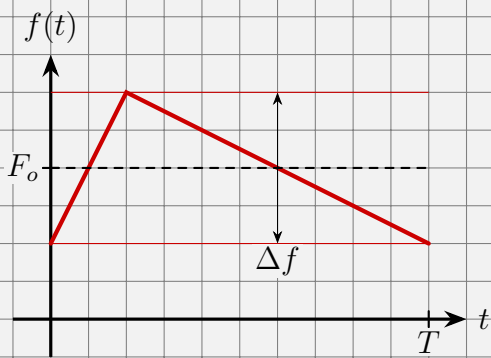


g)

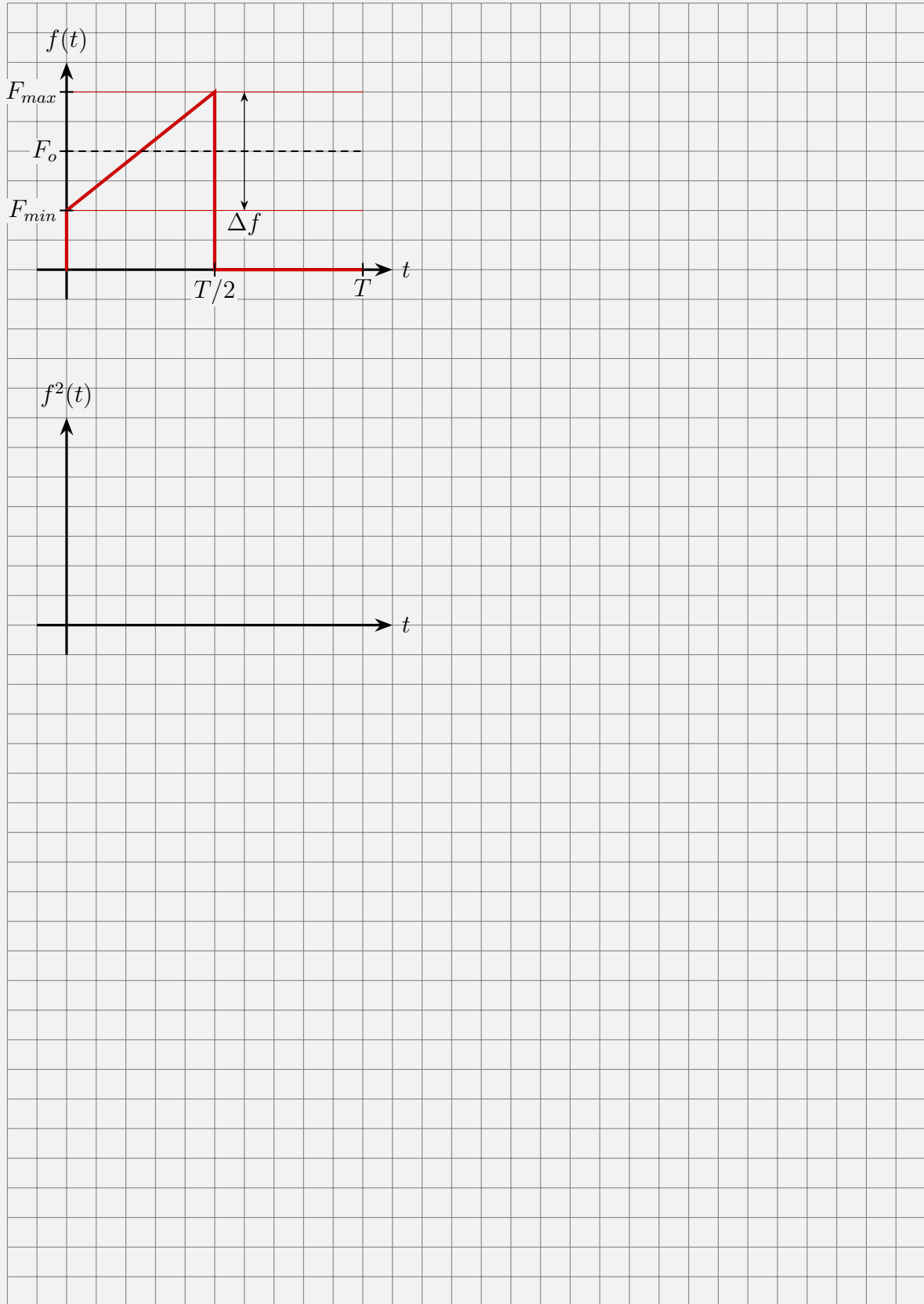


1.1

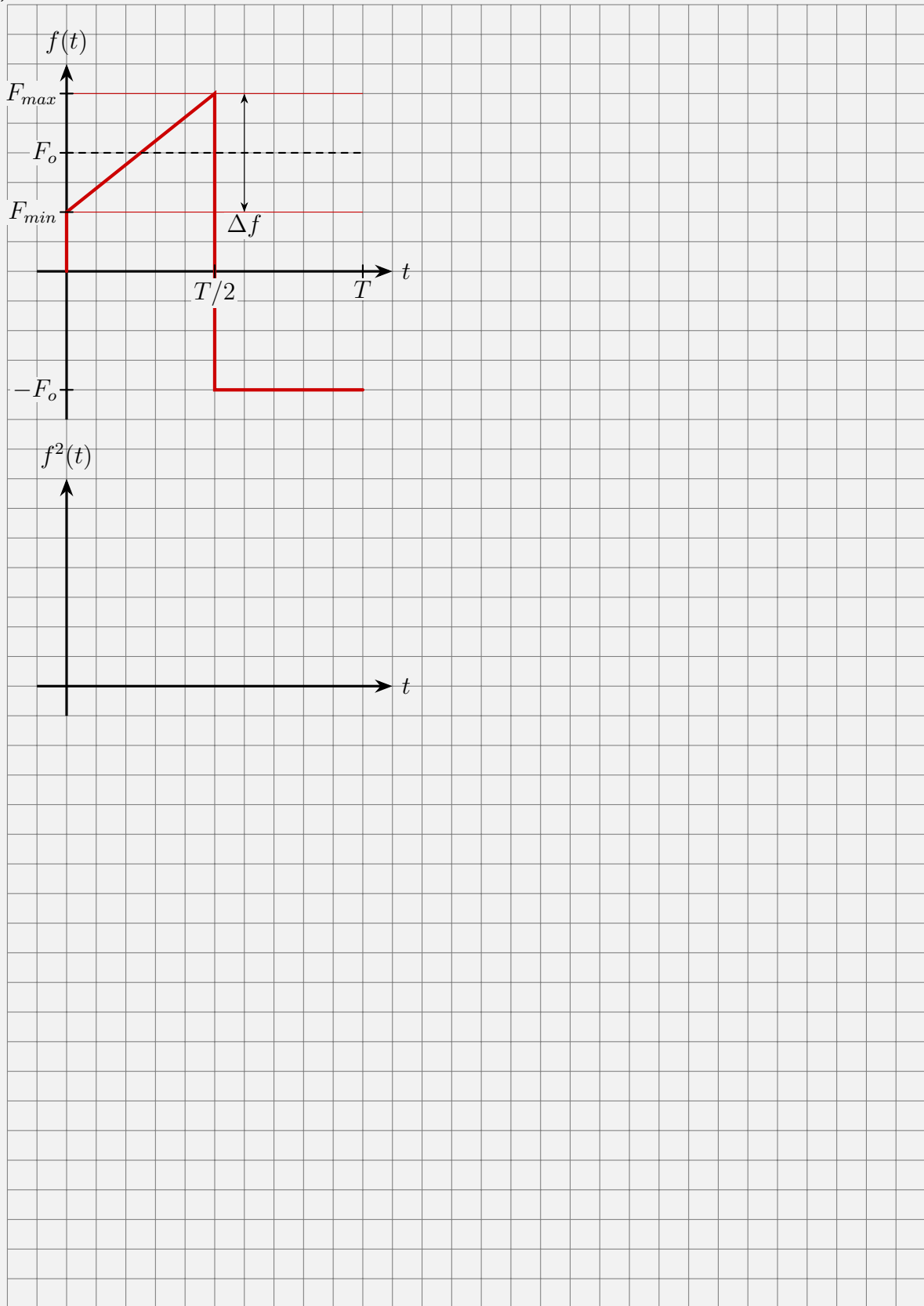
h)



i)



j)



Inductors and capacitor in periodic steady-state

Energy stored in an inductor and in a capacitor are state functions of their current and voltage respectively. Therefore, the stored energy is a periodic quantity and the average power at their port results equal to zero. As a consequence, for an inductor the next two conditions hold at the same time:

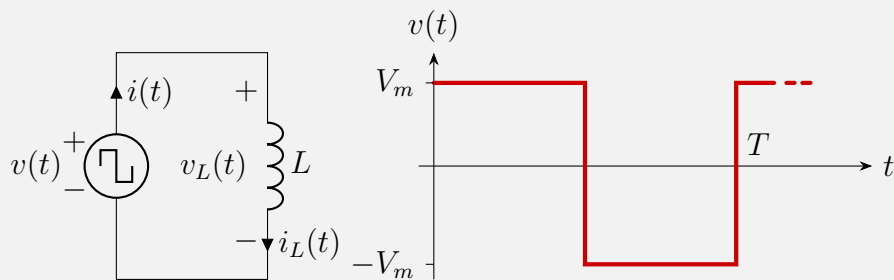
- the average voltage across an inductor V_L is zero;
- the current in the inductor is periodic;

In the same way for a capacitor we can state alternatively:

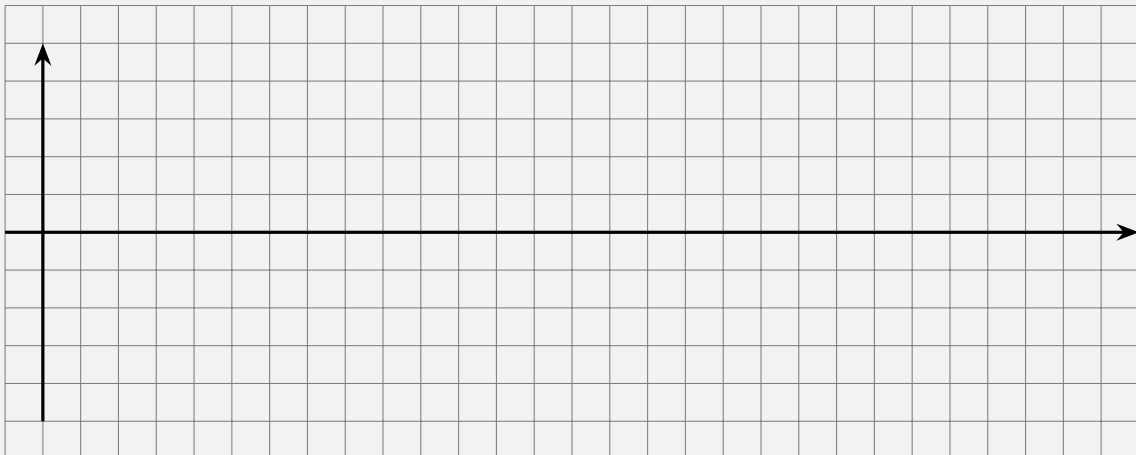
- the average current in a capacitor I_c is zero;
- the voltage across a capacitor is periodic;

Exercise 1.2: inductor with square-wave voltage.

Study the behavior of the following circuit where an inductor is supplied by a square-wave voltage. Use $L = 0.1$ H, $V_m = 1$ V and $T = 0.1$ s.



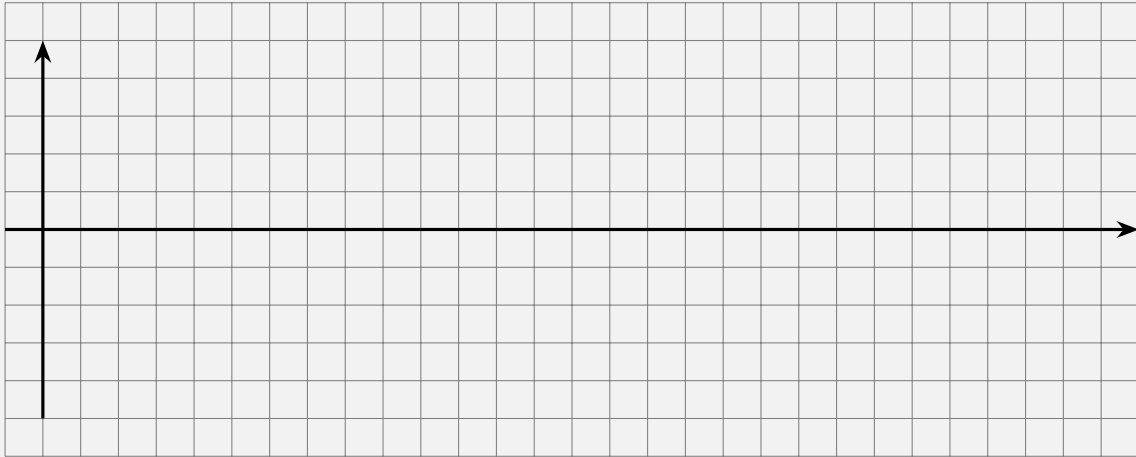
a) Draw two periods of the voltage and current of the inductor.



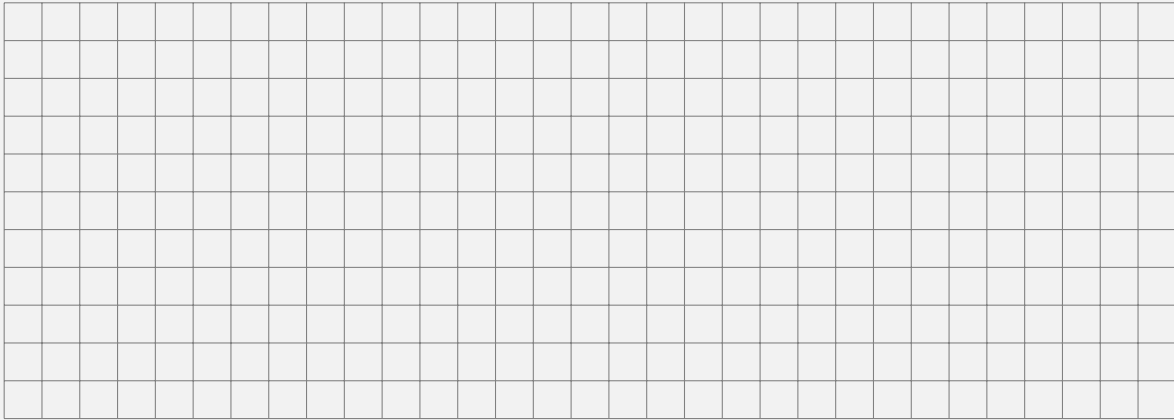
Q
1.2

1.2

b) Considering two periods, draw the instantaneous power and the stored energy for the inductor. Is there any relation between the two waveforms?



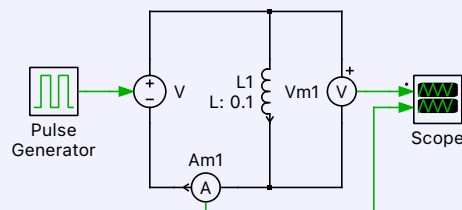
c) Compute the RMS value of the inductor current and the average value of the power.



1.1

Lab. 1.1: inductor with square-wave voltage (periodic_L)

Build a PLECS model to simulate the Exercise 1.2. Consider to start the exercise with the following model (that has to be completed/modified):

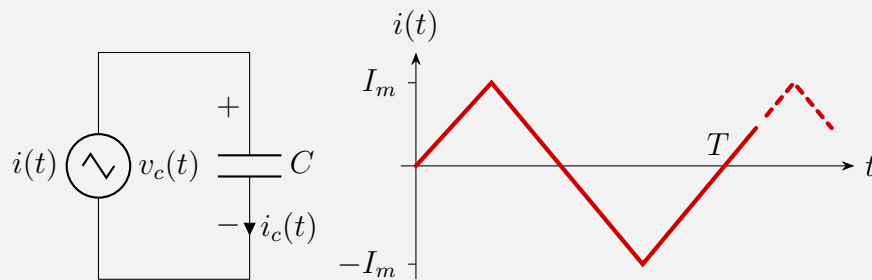


- Plot on a scope the voltage and current waveforms of the inductor.
- Plot on a second scope the instantaneous power and the magnetic energy stored in the inductor.
- Check on the previous waveforms that with a positive instantaneous power the stored energy increases while it decreases with a negative instantaneous power.

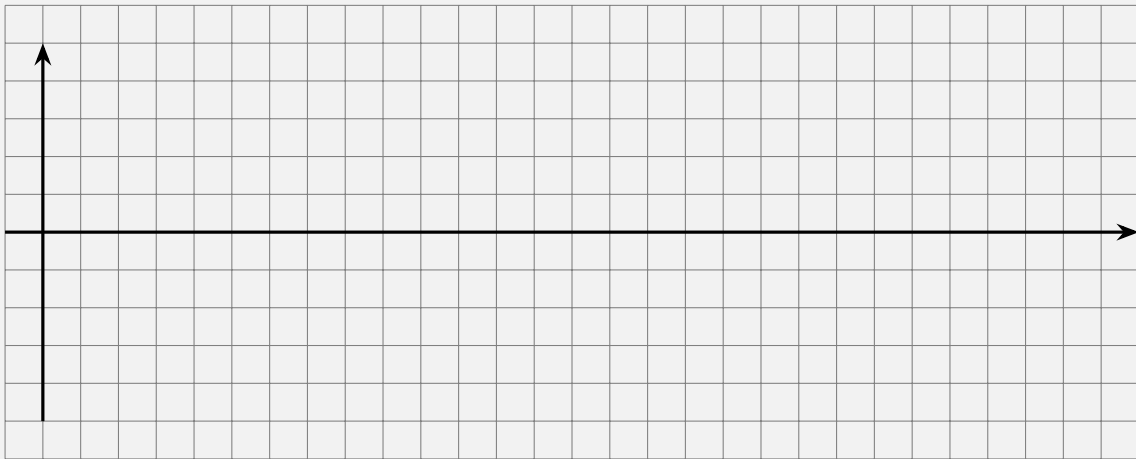
Exercise 1.3: capacitor with triangular-wave current.

1.3

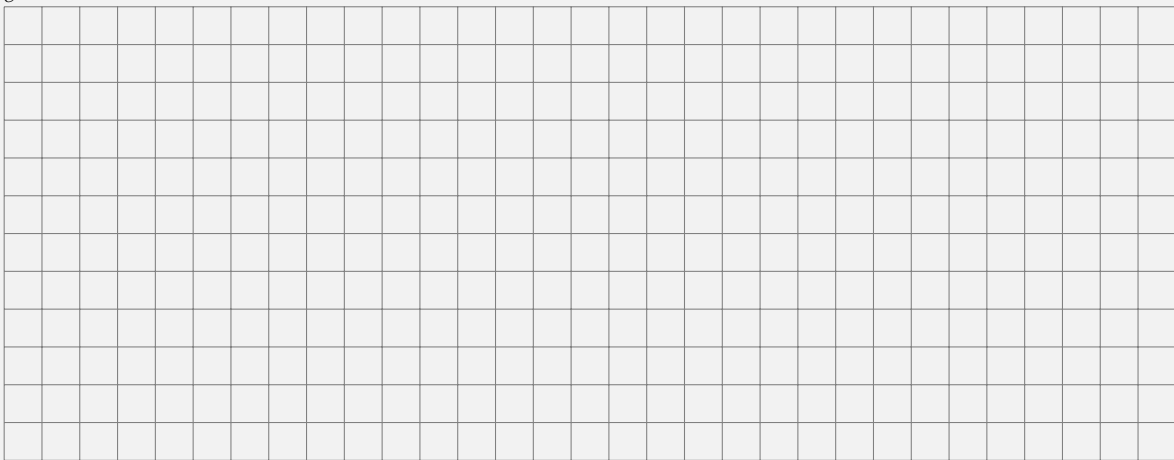
Study the behavior of the following circuit where a capacitor is supplied by a triangular-wave current generator. Use $C = 100 \mu\text{F}$, $I_m = \pm 1 \text{ A}$ and $T = 0.1 \text{ s}$.



a) Draw two periods of the voltage and current of the capacitor.



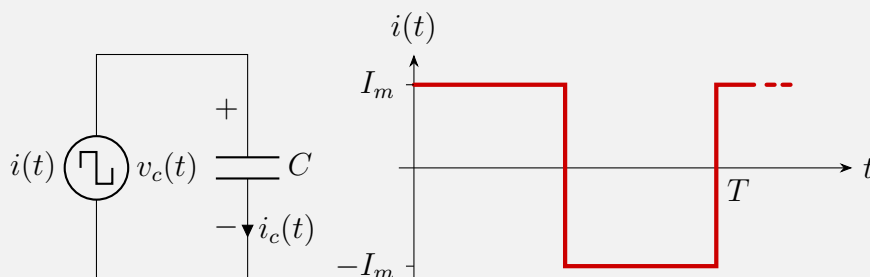
b) Compute the RMS value of the capacitor current $I_{c,rms}$ and the average value of voltage V_c .



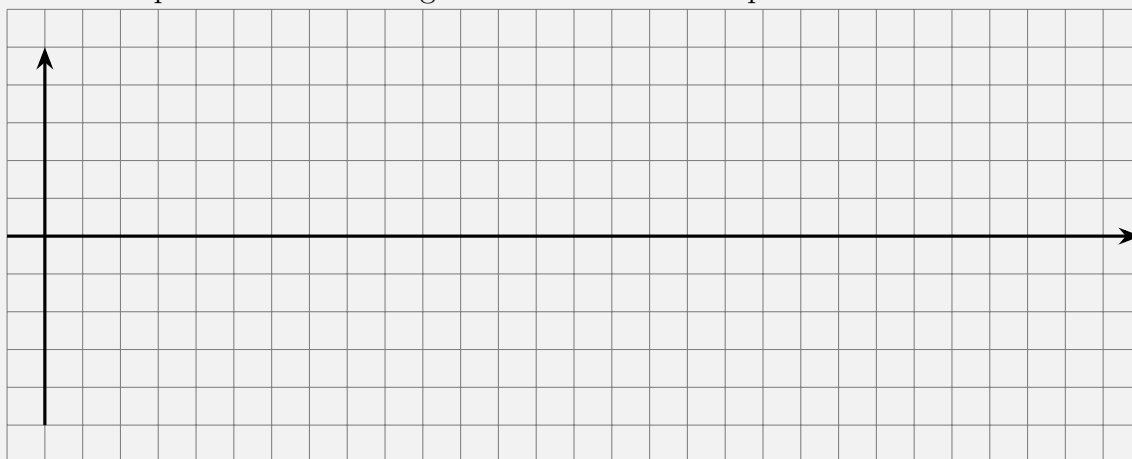
1.4

Exercise 1.4: capacitor with square-wave current.

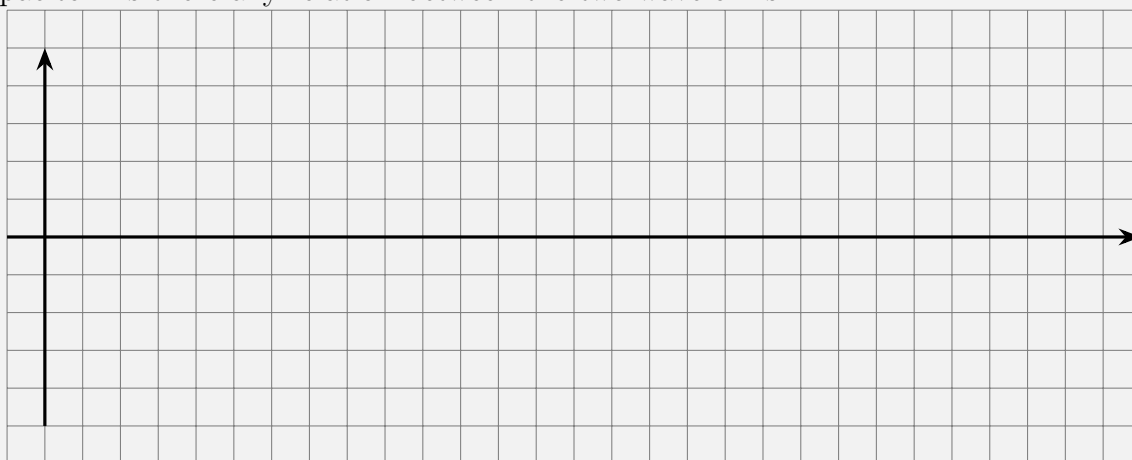
Study the behavior of the following circuit where a capacitor is supplied by a square-wave current generator. Use $C = 100 \mu\text{F}$, $I_m = \pm 0.5 \text{ A}$ and $T = 0.1 \text{ s}$.



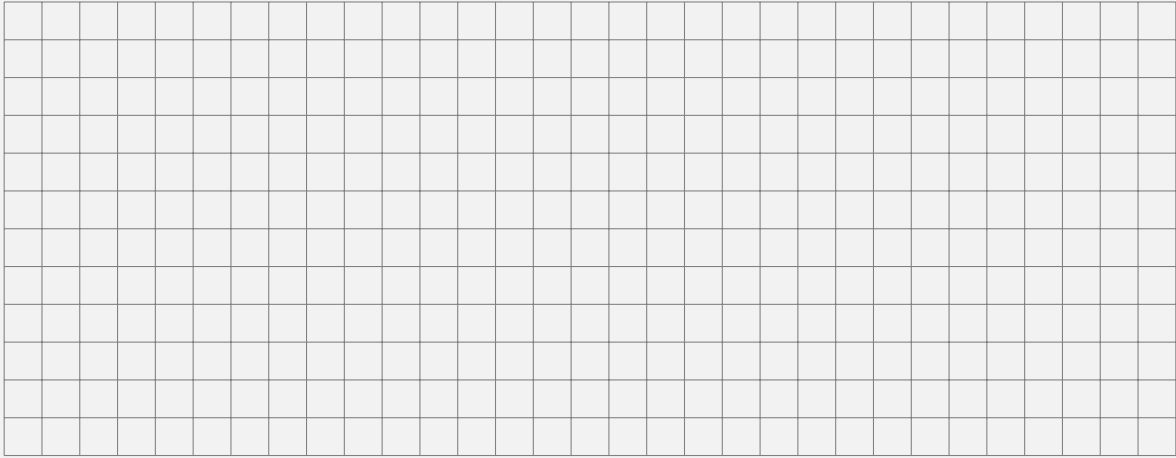
a) Draw two periods of the voltage and current of the capacitor.



b) Considering two periods, draw the instantaneous power and the stored energy for the capacitor. Is there any relation between the two waveforms?



c) Compute the RMS value of the capacitor voltage and the average value of the power.



1.4

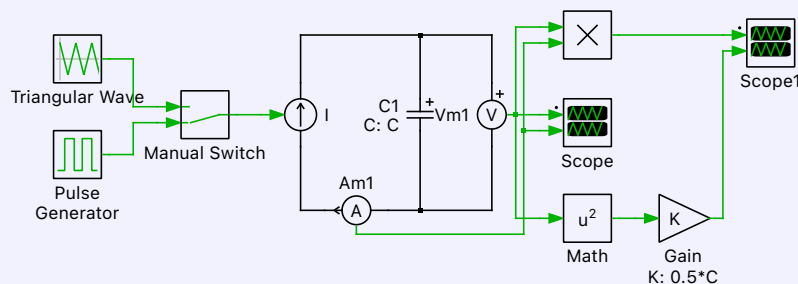
Lab. 1.2: analyze a capacitor with periodic current waveform (periodic_c)

1.2

Build a PLECS model to simulate the behavior of a capacitor $C=100 \mu\text{F}$ supplied by two different periodic current generators:

- a triangular wave at 10 Hz with peak value $\pm 1 \text{ A}$.
- A square wave at 10 Hz with peak value $\pm 0.5 \text{ A}$.

The exercise can be developed considering the following model as a starting point where the manual switch can be used to change the generator:



- Plot on a scope the voltage and current waveforms of the capacitor.
- Plot on a second scope the instantaneous power and the electrostatic energy stored in the capacitor.
- Check on the previous waveforms that with a positive instantaneous power the stored energy increases while it decreases with a negative instantaneous power.

Periodic generator with linear load

When a periodic generator is connected to a linear load, the circuit can be solved exploiting the superposition principle. The sources are considered by means of their Fourier series expansion and each harmonic is considered separately. Then, the responses of all harmonics are added together to achieve the total response of the circuit. If the voltage and current at one port are expressed as the following series:

$$v(t) = V_o + \sum_{n=1}^{\infty} V_n \cos(n\omega t + \theta_n)$$

$$i(t) = I_o + \sum_{n=1}^{\infty} I_n \cos(n\omega t + \phi_n)$$

the power at the port can be computed as:

$$P = \sum_{n=0}^{\infty} P_n = V_o I_o + \sum_{n=1}^{\infty} \frac{V_n I_n}{2} \cos(\theta_n - \phi_n) \quad (1.2)$$

(1.2) is particularly useful to compute power at the generators ports. For a resistor it is more practical to compute the rms value of the current and then compute:

$$P = R \cdot I_{rms}^2 = R \cdot \left[I_o^2 + \sum_{n=1}^{\infty} \frac{I_n^2}{2} \right] \quad (1.3)$$

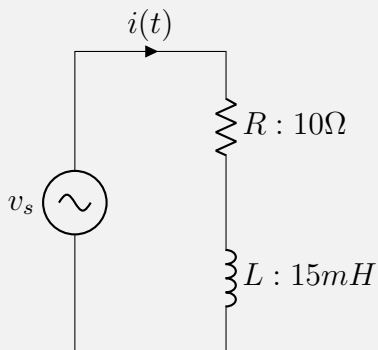
When in the circuit there is a voltage generator of constant value E , its power can be computed as:

$$P = E \cdot I_o$$

1.5

Exercise 1.5: periodic generator with linear load.

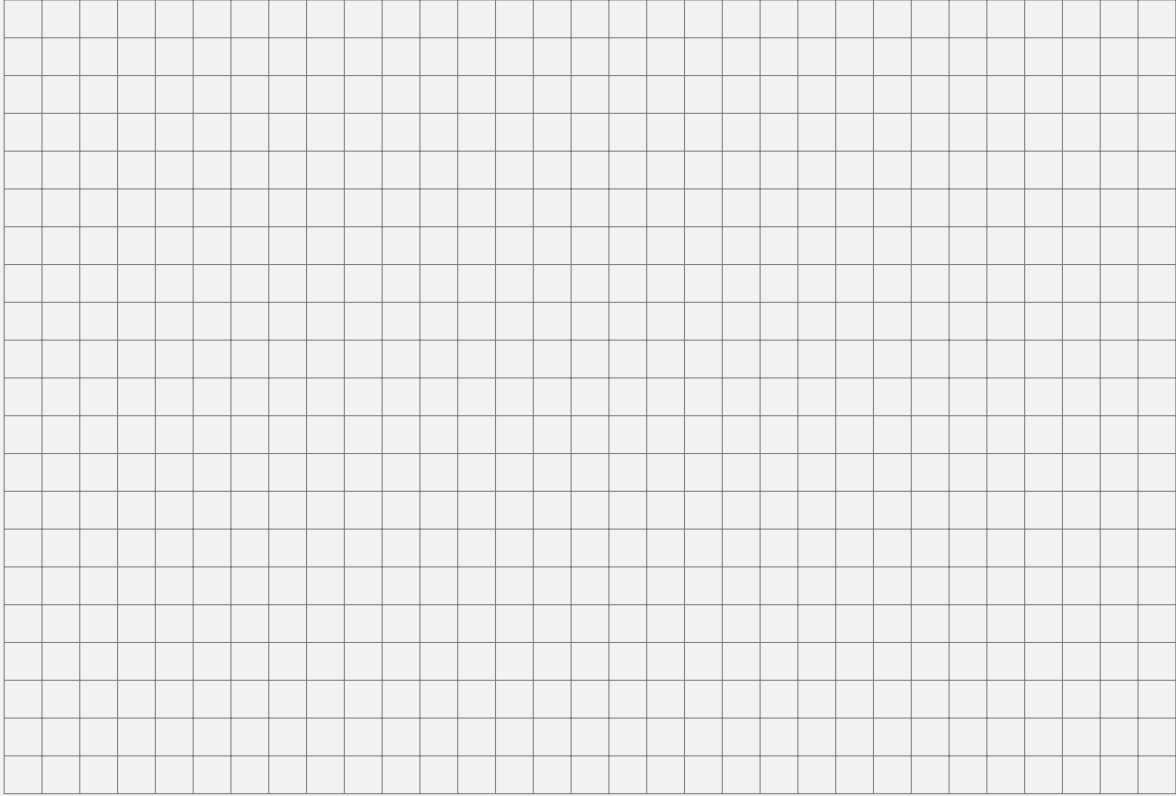
Study the behavior of the following circuit in which:



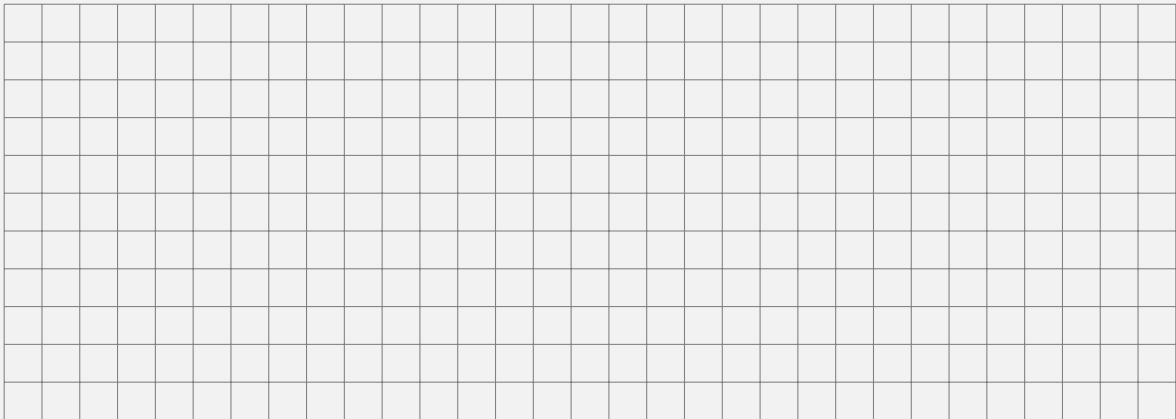
$$v_s = V_0 + V_1 \cdot \sin(\omega_1 t) + V_2 \cdot \sin(\omega_2 t - \frac{\pi}{4})$$

$$= 50 + 325 \cdot \sin(2\pi 50 t) + 150 \cdot \sin\left(4\pi 50 t - \frac{\pi}{4}\right)$$

a) Compute the expression of the current $i(t)$.



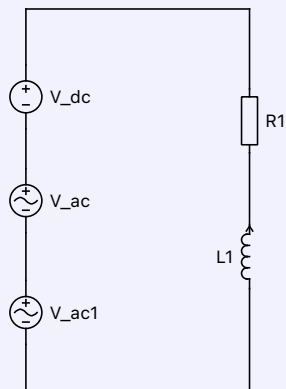
b) Compute analytically the RMS value of the current.



c) Compute analytically the average power of all the bipoles of the circuit.

d) Check that the conservation of power is verified.

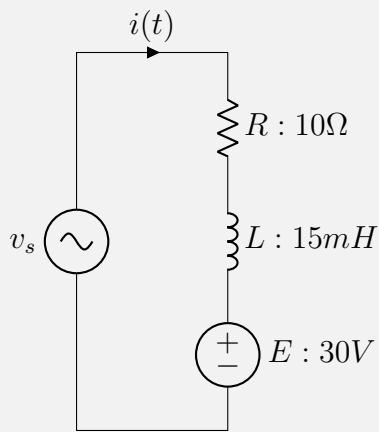
Lab. 1.3: periodic generator with linear load.



Exercise 1.6: periodic generator with RLE load.

1.6

Try to solve again the Exercise 1.5 considering in the load, besides the resistor R and the inductor L , also a dc generator of value $E = 30\text{ V}$.



$$v_s = 50 + 325 \cdot \sin(2\pi 50t) + 150 \cdot \sin\left(4\pi 50t - \frac{\pi}{4}\right)$$

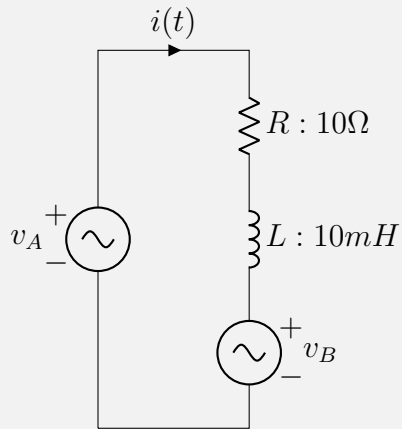
- a) Compute the current response of the circuit $i(t)$ and evaluate the power on each bipole of the circuit.



1.7

Exercise 1.7: two periodic generators with RL load.

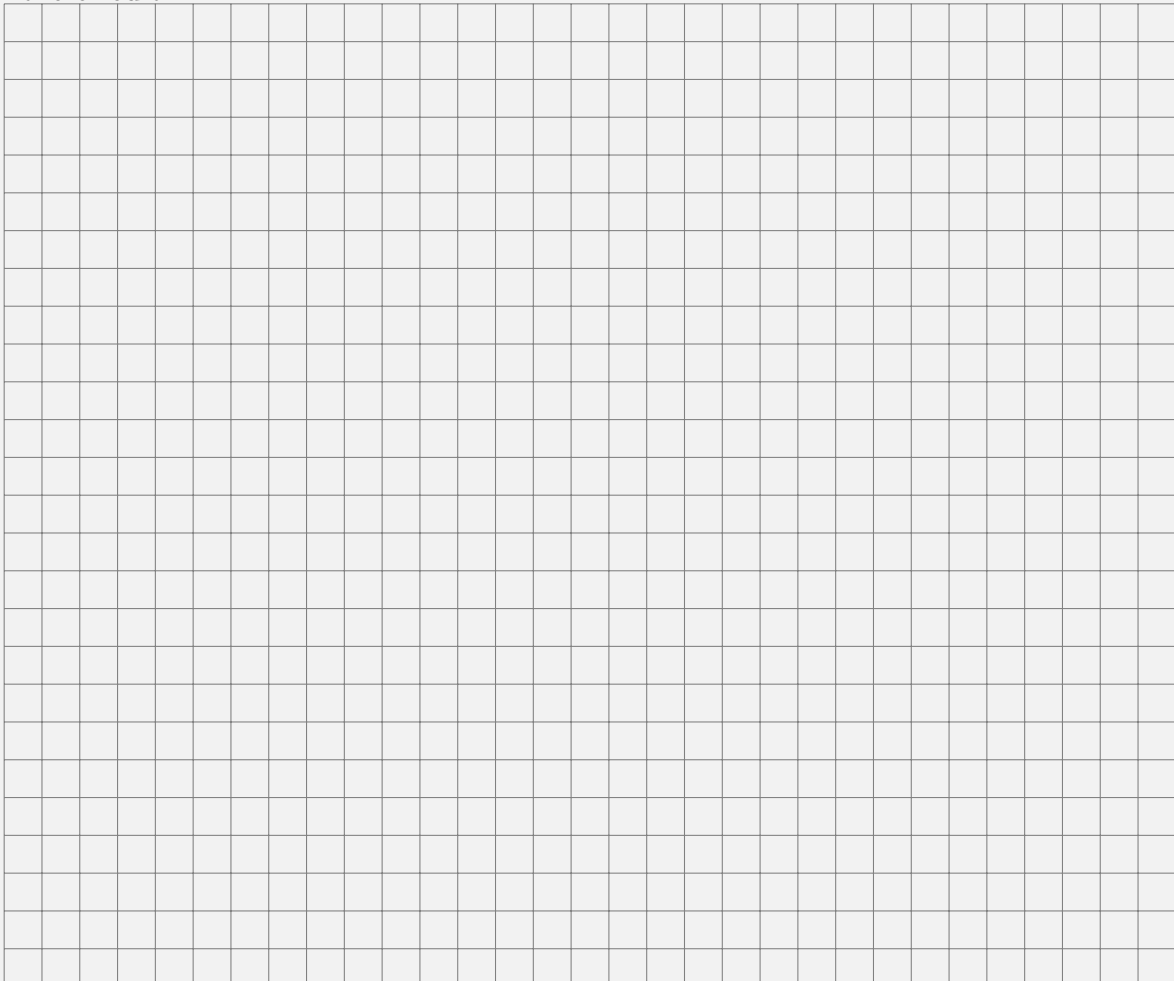
Consider the following circuit where two periodic generators are connected with a resistor R and an inductor L .



$$v_A = 150 + 73 \cdot \sin(268 \cdot t) + 68 \cdot \sin\left(536 \cdot t - \frac{\pi}{4}\right)$$

$$v_B = 50 + 73 \cdot \sin\left(268 \cdot t + \frac{\pi}{2}\right)$$

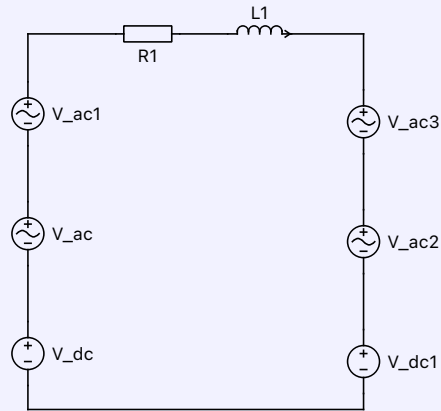
- a) Compute the current response of the circuit $i(t)$ and evaluate the power on each bipole of the circuit.



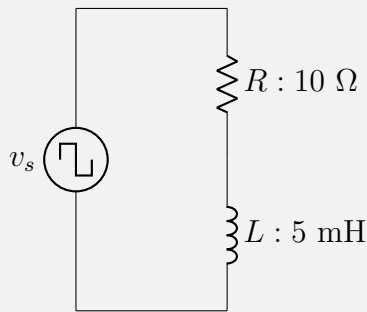
Lab. 1.4: two periodic generators with linear load.

▶
1.4

The Exercises 1.6 and 1.7 can be easily simulated in PLECS extending the Model 1.3. As an example, the following PLECS model allows to simulate all the circuits considered in the previous exercises adjusting properly the generators parameters. Solve the two circuits using PLECS and compare the simulation results with the analytical calculations.



Exercise 1.8: square-wave generator with linear load.



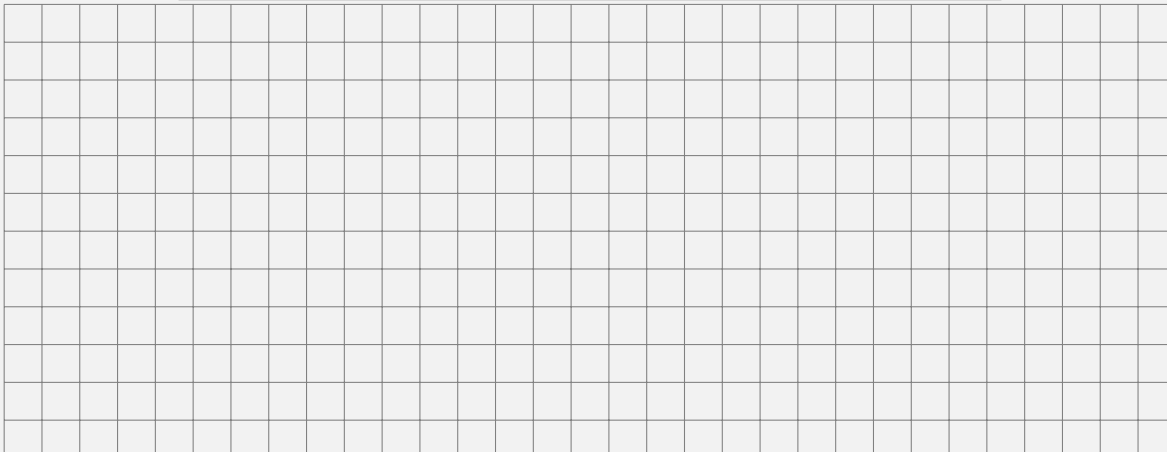
A square-wave voltage generator with duty cycle 50% and a peak value of $V_m = \pm 100 \text{ V}$ has a frequency $f = 50 \text{ Hz}$ and supply a linear load formed by a resistor of 10Ω in series with an inductor of 5 mH . The Fourier series for the square-wave source is:

$$v_s(t) = \sum_{n \text{ odd}} V_n \sin(n\omega t)$$

where: $V_n = \frac{4V_m}{\pi n}$ and $\omega = 2\pi f$.

a) Compute analytically the rms value of the voltage, the rms value of the current, the average power of the generator and the resistor. After the computations, fill the following table with all the values:

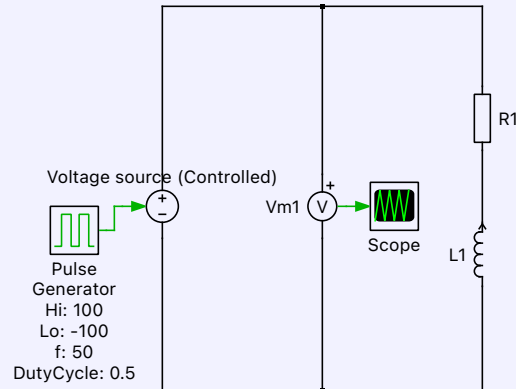
n	V_n	f_n	X_n	Z_n	I_n	ϕ_n	P_n
1							
3							
5							
7							
9							
11							
13							
15							
17							
19							



Lab. 1.5: square-wave generator with linear load.

▶
1.5

Simulate in PLECS the circuit of Exercise 1.8 and check the computations. For the simulation consider the following model as starting point where the square-wave generator is realized with a controlled voltage source:



Check that the actual voltage of the generator is as expected. i.e. ± 100 V at 50 Hz. Use the Fourier tool in the PLECS scopes to check the computations of the current harmonics. Compute the average power in the load.

Inductive current and energy recovery

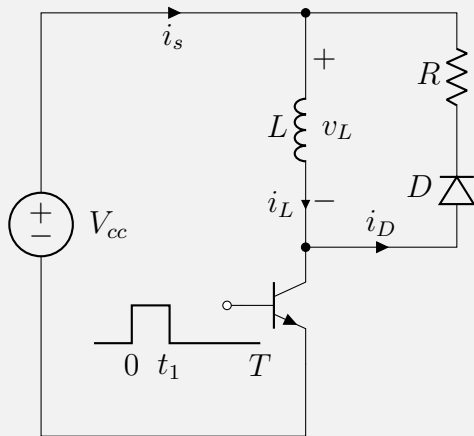
In many applications, inductors need to be energized and de-energized. This happens anytime we try to open a carrying current circuit with an electronic switch such as a transistor. For an inductor L , the stored energy is computed as:

$$w_m(t) = \frac{1}{2} Li(t)^2$$

and hence a sudden change in the current produce an over-voltage across the transistor which would be damaged. Therefore, the circuit must be designed to safely discharge the stored energy w_m , protecting the transistor from damage when it turns off. The efficiency of the circuit can be enhanced if this stored energy is redirected back to the source. In this section circuits characterized by a stored magnetic energy and a current commutation are considered.

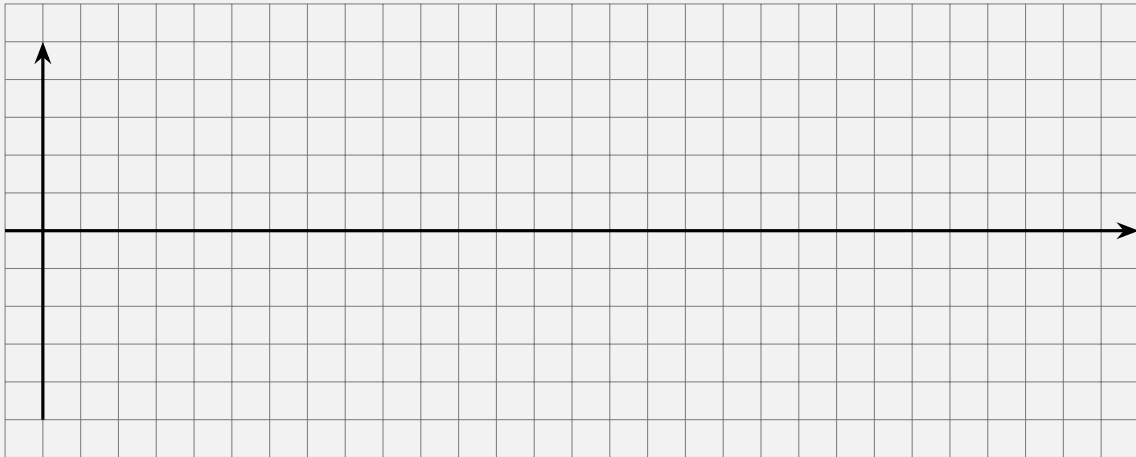
1.9

Exercise 1.9: commutation of an inductive current.

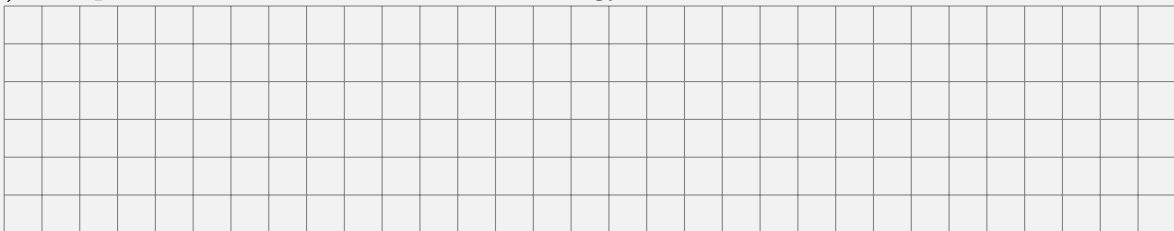


The circuit on the left is driven periodically operating the transistor with $T=0.1$ s and $t_1=10$ ms. The other parameters of the circuit are: $V_{cc}=100$ V; $L=100$ mH; $R=10$ Ω . Consider an ideal diode.

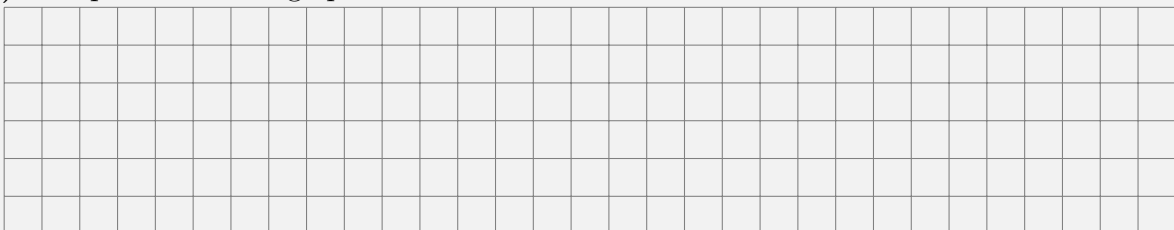
- a) Draw the graph of the inductor current $i_L(t)$ and compute its maximum value.



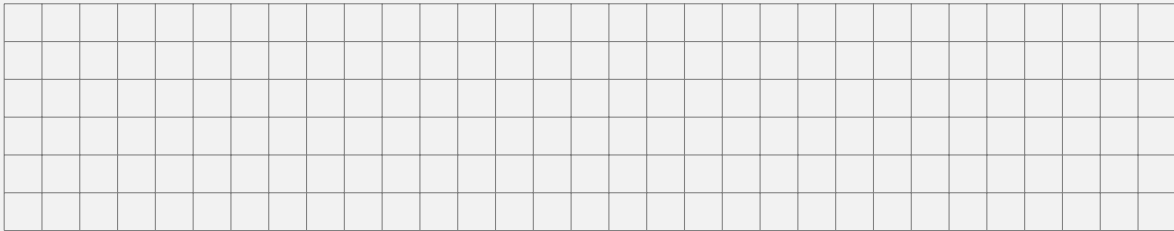
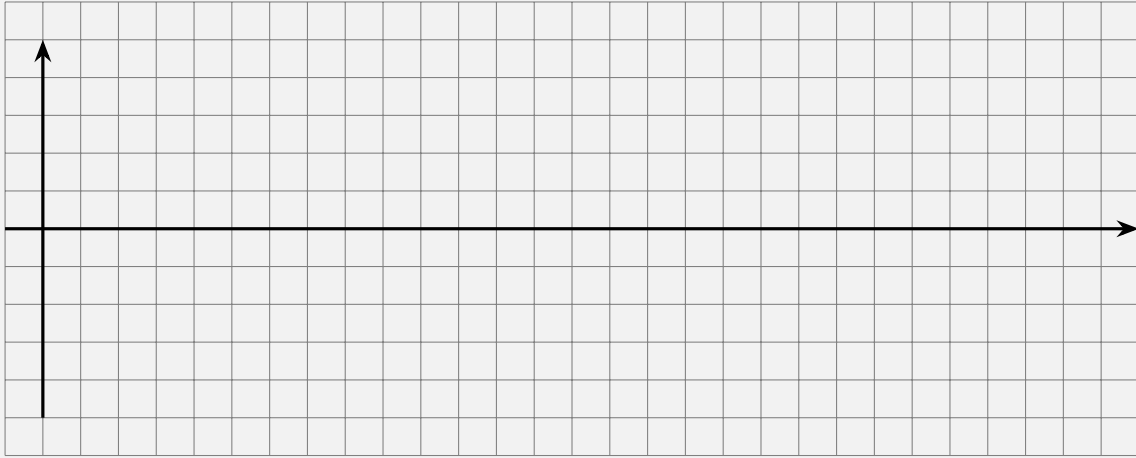
- b) Compute the maximum value of the energy stored in the inductor.



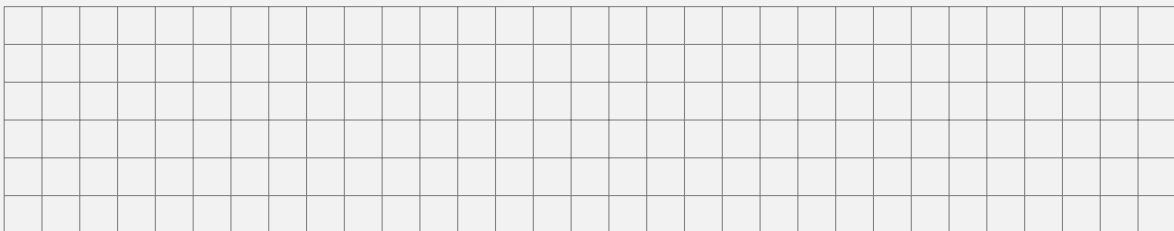
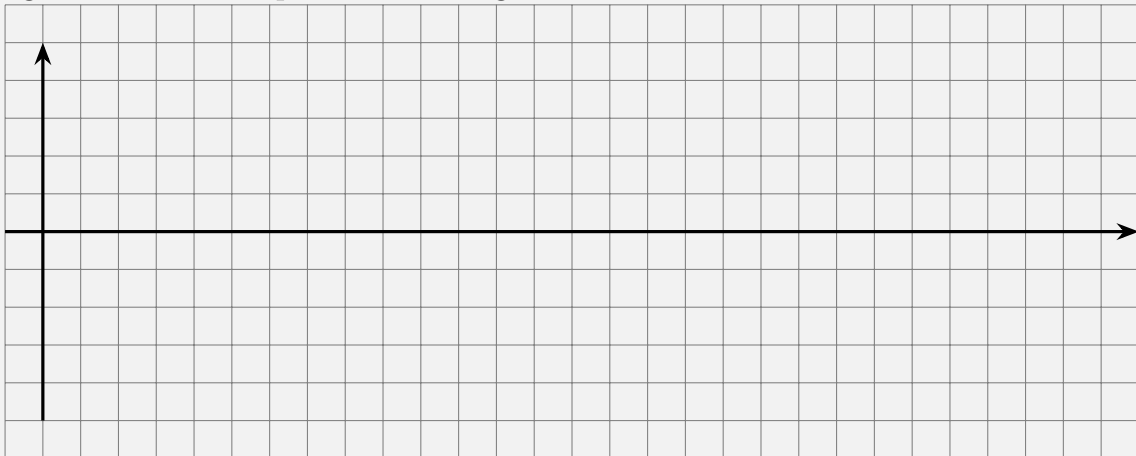
- c) Compute the average power absorbed in the resistor R .



d) Draw the graph of the current in the diode $i_D(t)$ and compute its rms value.

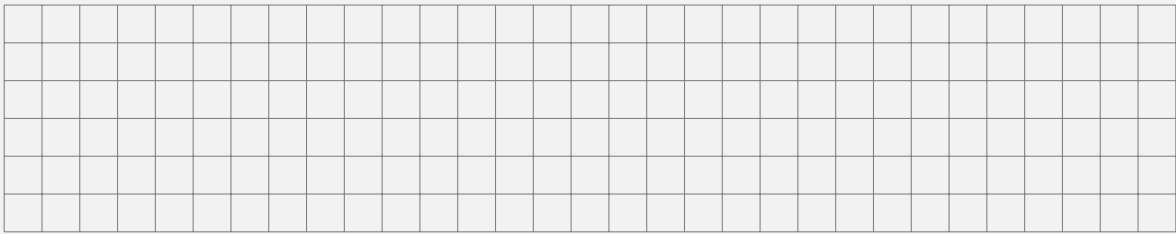
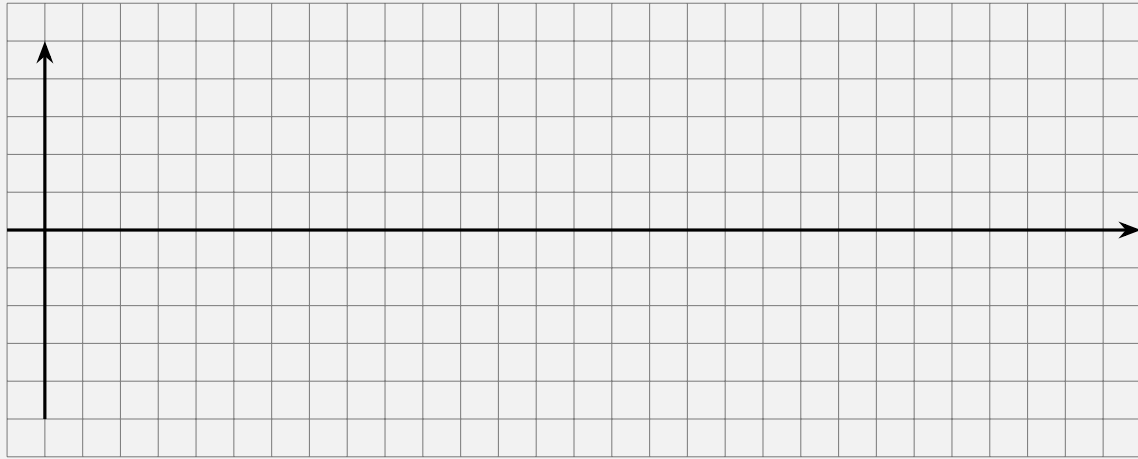


e) Draw the graph for the current i_s and for the instantaneous power $p_s(t) = V_{cc} \cdot i_s$ of the generator and compute their average values.



1.9

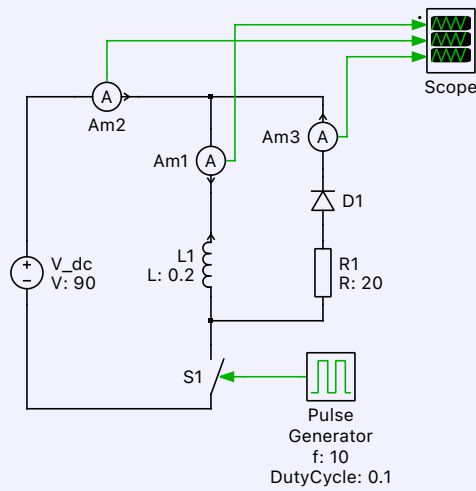
f) Draw the graph for the current i_s and for the instantaneous power $p(t) = V_{cc} \cdot i_s$ of the generator when an energy recovery (with two diode) is adopted. Compute also the average values for both the current and the power.



1.6

Lab. 1.6: commutation of an inductive current.

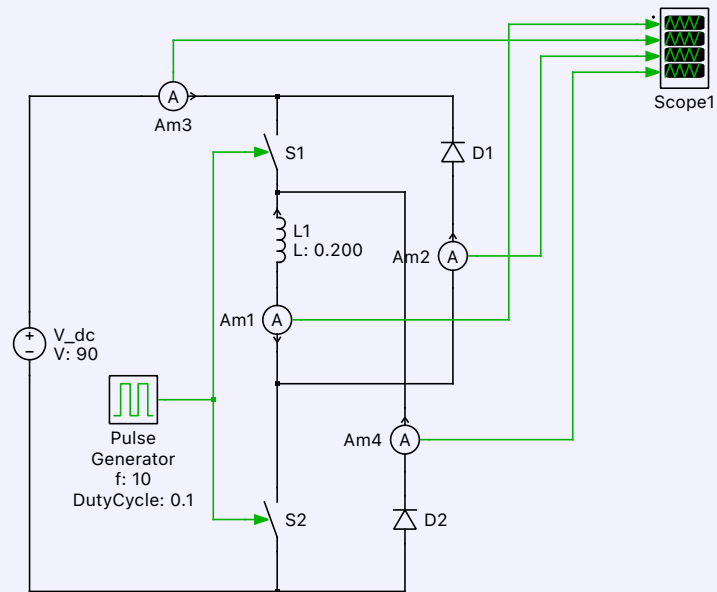
Simulate in PLECS the circuit of Exercise 1.9 and check the computations. For the simulation consider the following model as starting point:



Lab. 1.7: energy recovery simulation.

1.7

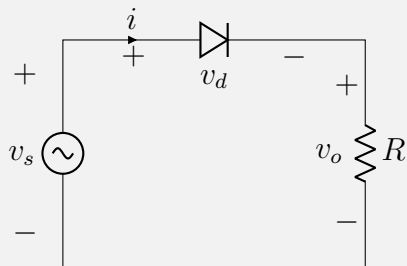
Modify the circuit of Simulation 1.6 to adopt an energy recovery for the magnetic energy stored in the inductor. For the simulation consider the following model as starting point:



CHAPTER 2

Half-wave rectifiers

Exercise 2.1: half-wave rectifier with resistive load.

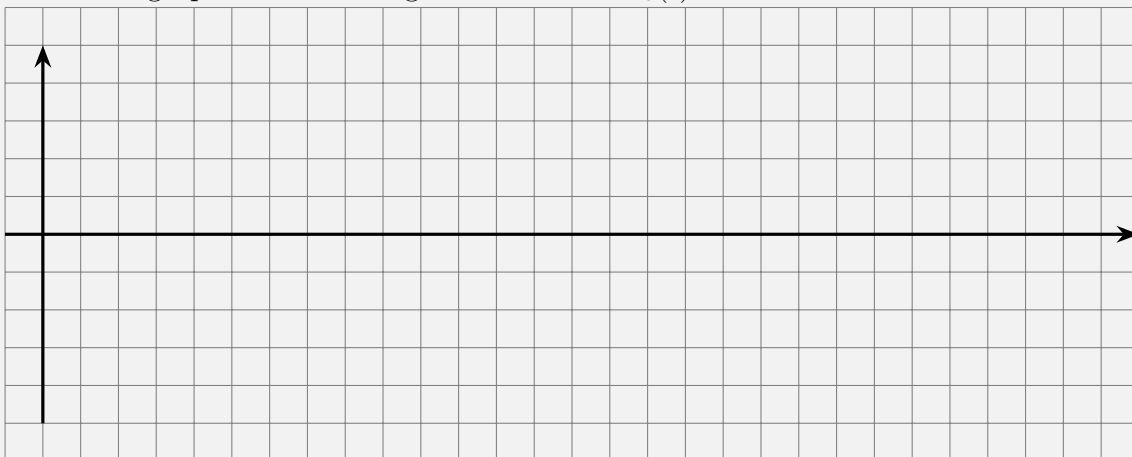


The half-wave rectifier in the figure supplies a resistor $R=15\ \Omega$. The generator parameters are:

$$v_s = \sqrt{2} \cdot 200 \sin(2\pi ft)$$

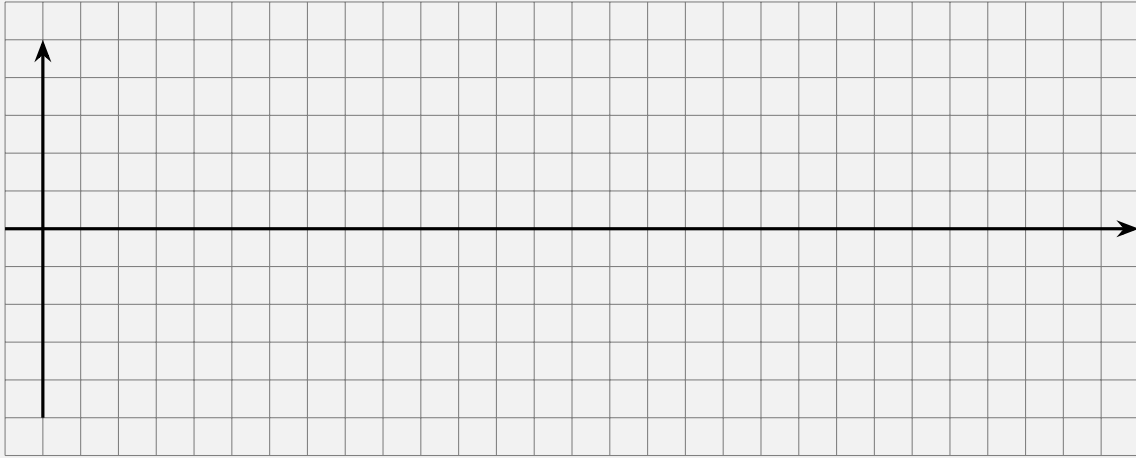
$$f = 50\text{ Hz}$$

a) Draw the graph for the voltage of the source $v_s(t)$.

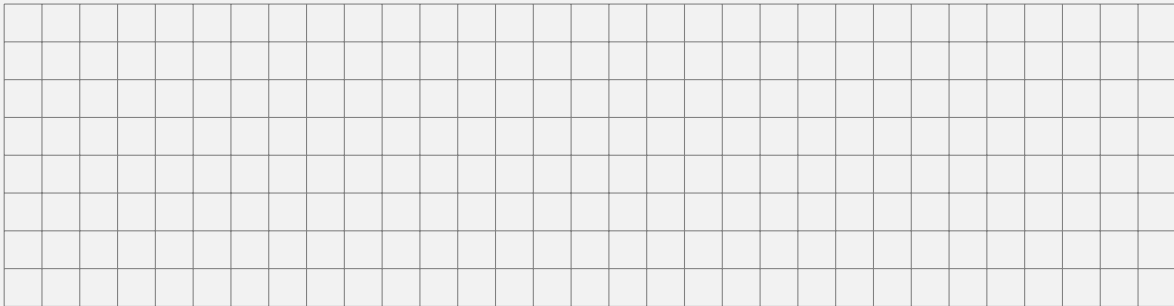


2.1

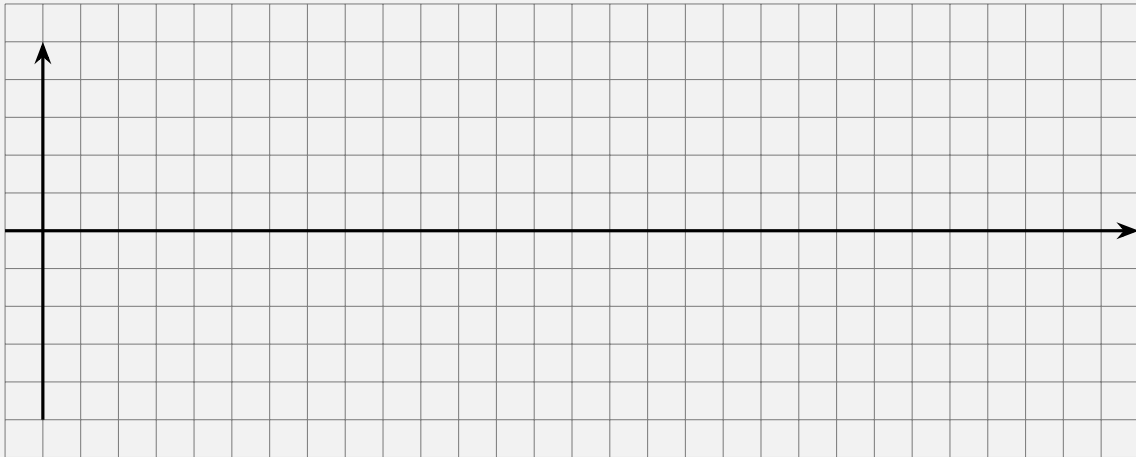
b) Draw the graph for the output voltage $v_o(t)$.



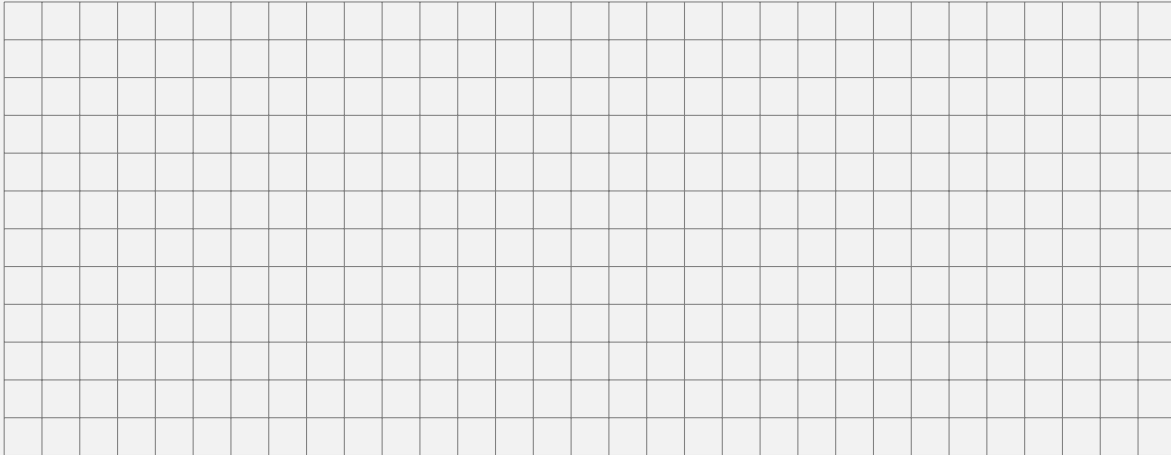
c) Compute the average and rms value of the output voltage $v_o(t)$.



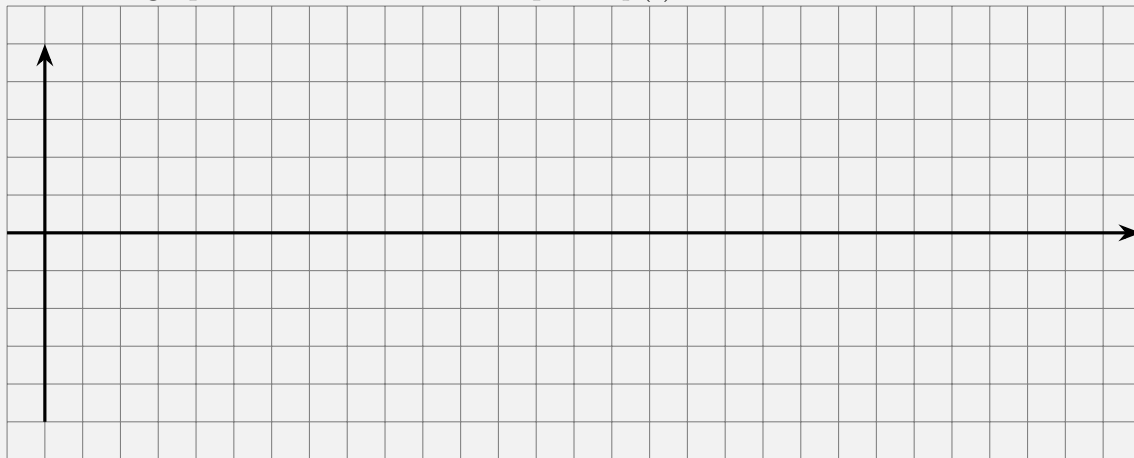
d) Draw the graph for the circuit current $i(t)$.



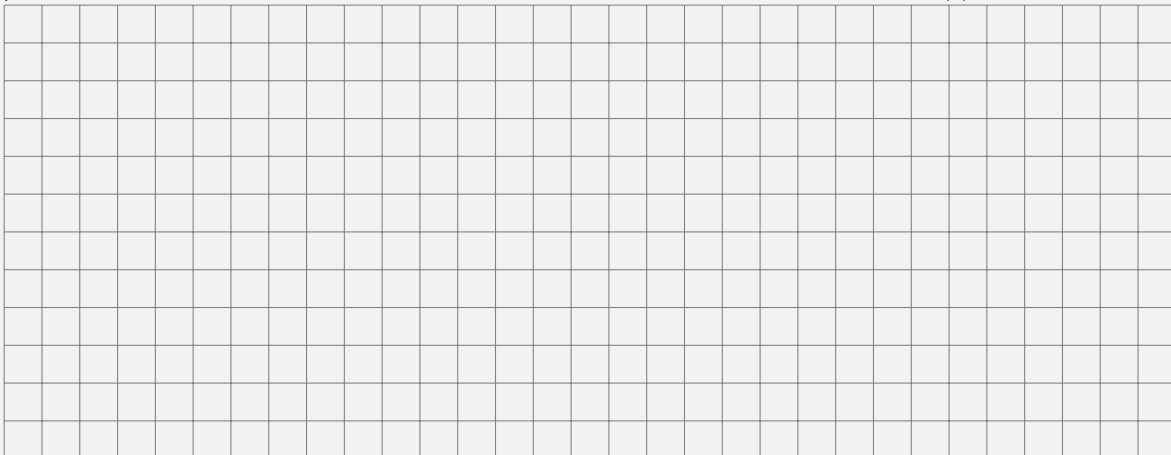
e) Compute the average and rms value of the circuit current $i(t)$.



f) Draw the graph for the instantaneous power $p(t)$ on the resistor.



g) Compute the average power and the power factor for the source $v_s(t)$.

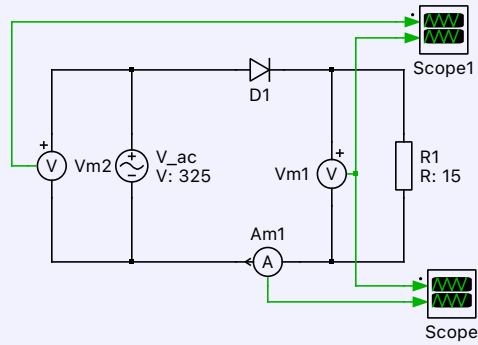




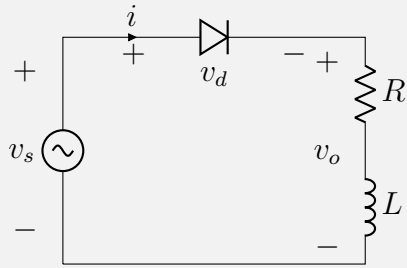
2.1

Lab. 2.1: half-wave rectifier with resistive load.

Build a PLECS model to simulate the Exercise 2.1 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.2: half-wave rectifier with resistive and inductive load.

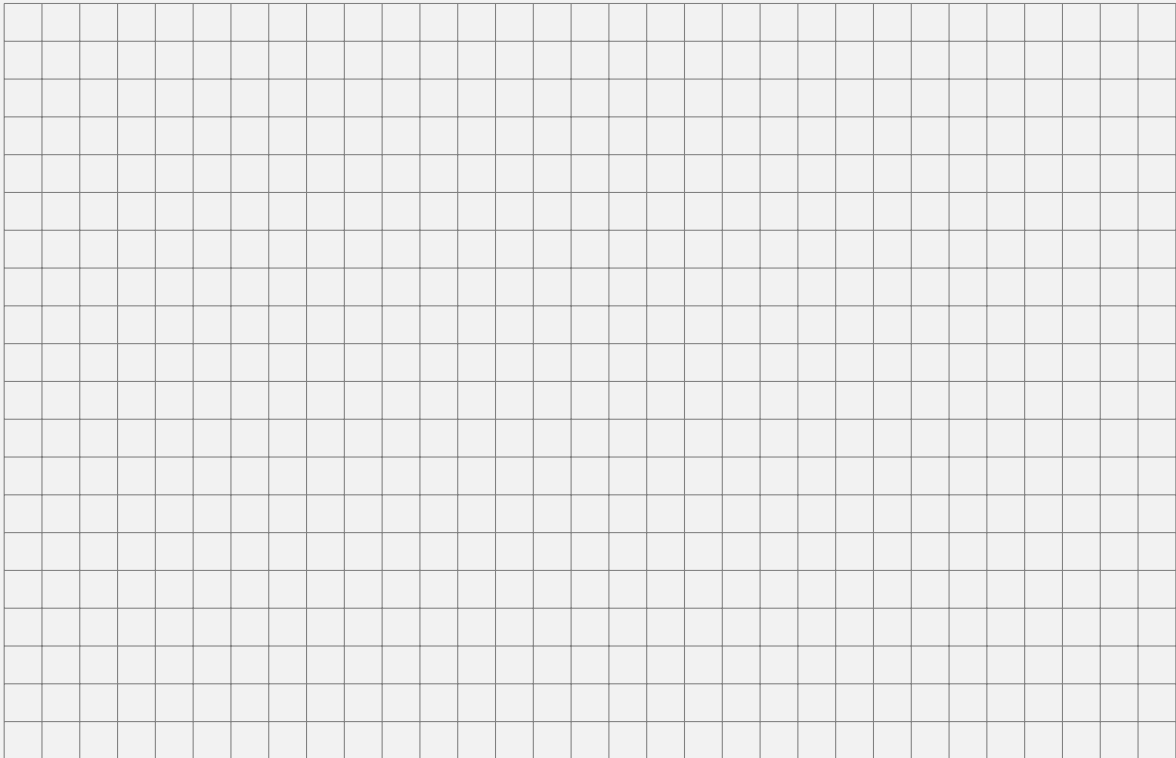


The half-wave rectifier in the figure supplies a resistor $R=30\ \Omega$ and an inductor $L=0.150\ \text{H}$. The generator parameters are:

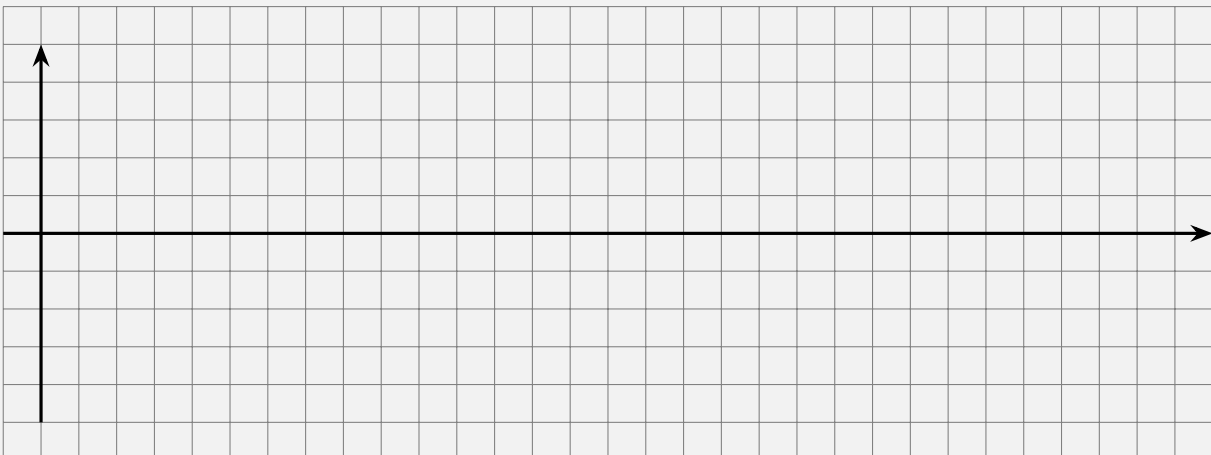
$$v_s = 100 \sin(2\pi ft)$$

$$f = 50\ \text{Hz}$$

- a) Compute the extinction angle β and write the expression for the current $i(t)$.

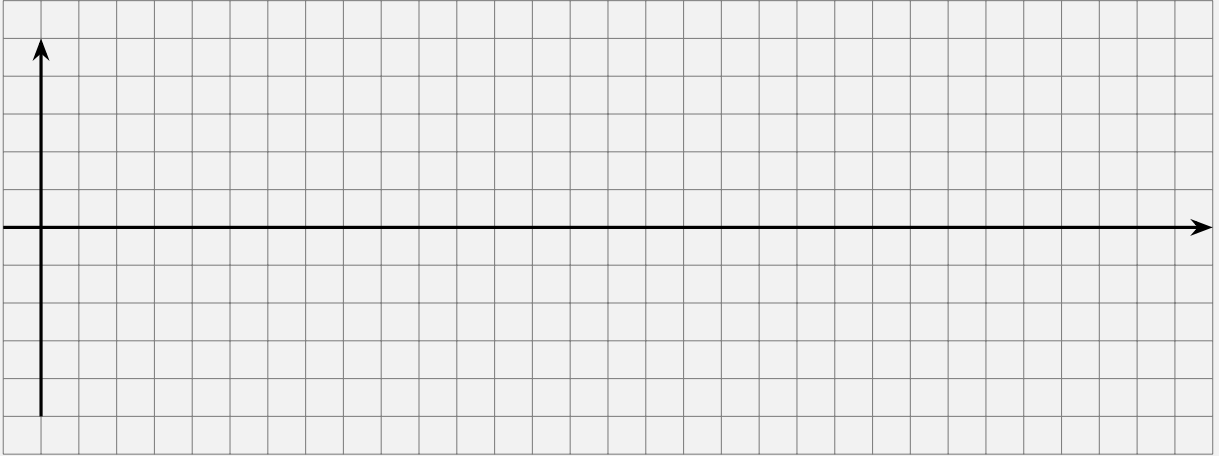


- b) Draw the graph for the voltage for the source $v_s(t)$ and sketch the current $i(t)$.

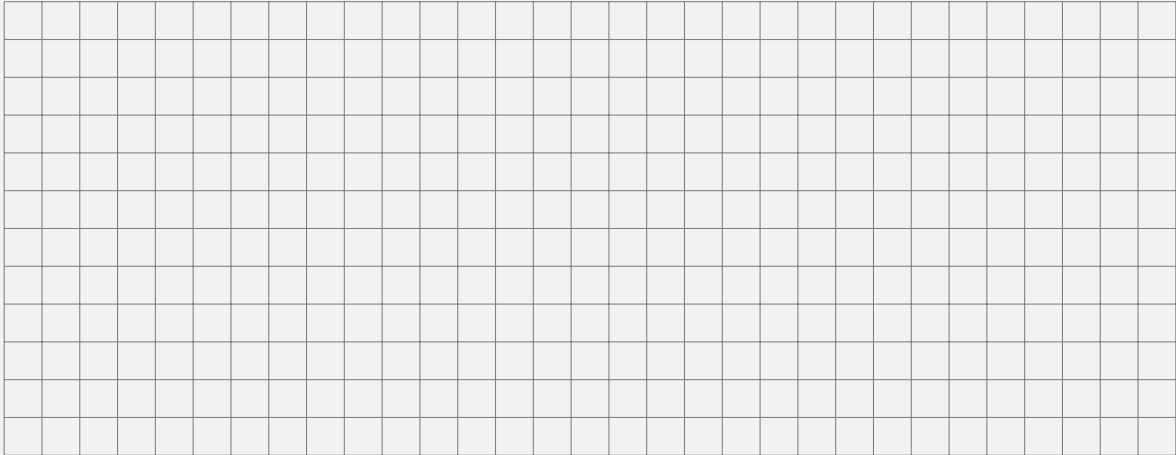


2.2

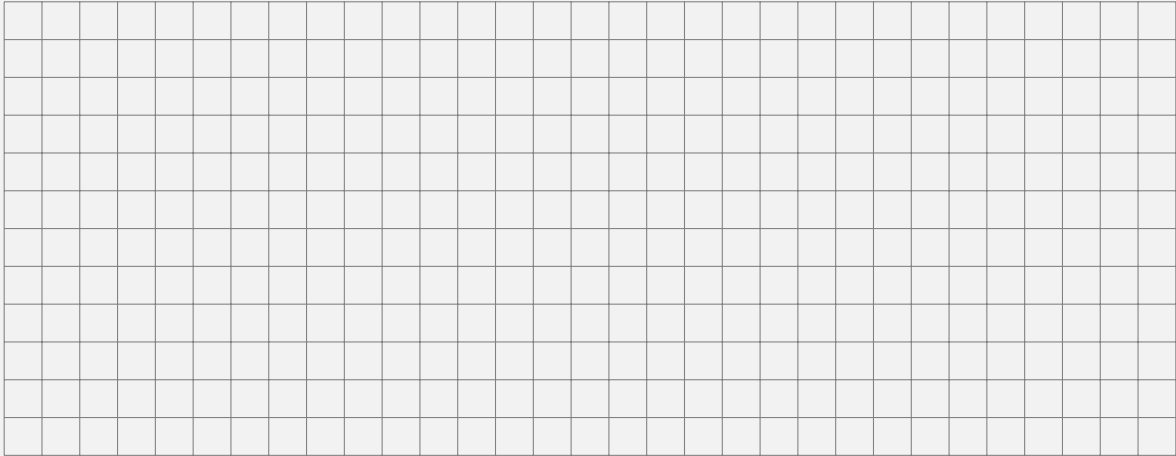
c) Draw the graph for the output voltage $v_o(t)$.



d) Compute the average value of the output voltage $v_o(t)$.



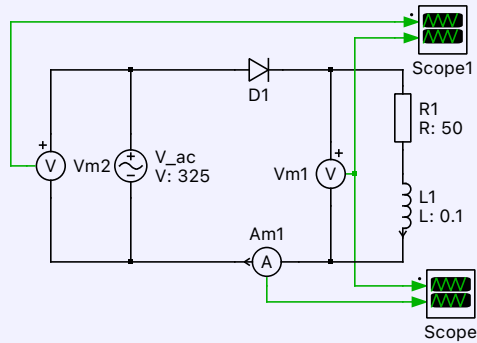
e) Compute the average value of the circuit current $i(t)$.



Lab. 2.2: half-wave rectifier with resistive and inductive load.

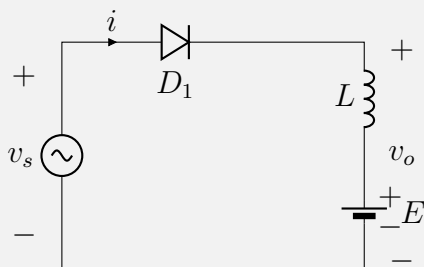
2.2

Build a PLECS model to simulate the Exercise 2.2 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.3: half-wave rectifier with inductive and battery load.

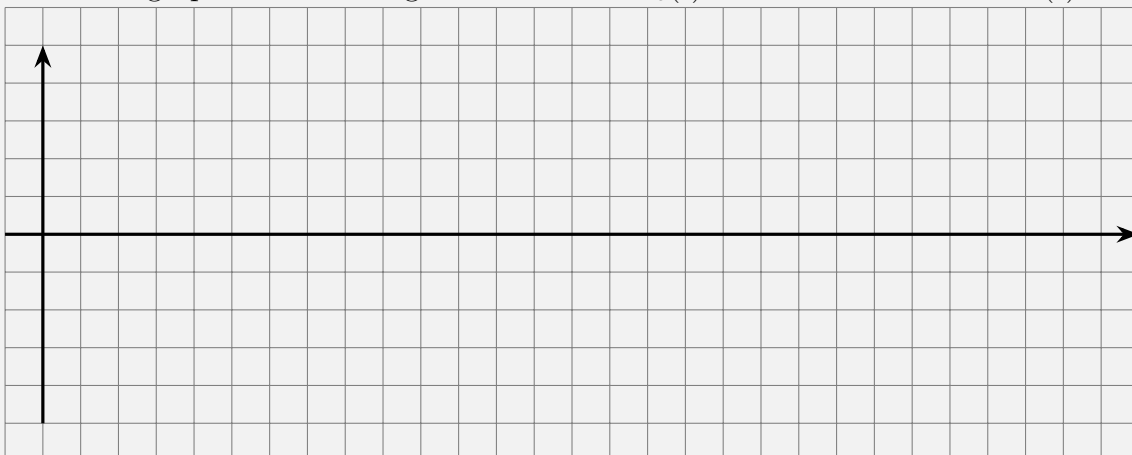
2.3



The half-wave rectifier in the figure supplies an inductor $L=0.01$ H in series with a battery $E=100$ V. The generator parameters are:

$$v_s = V_m \sin(2\pi ft) = 325 \cdot \sin(2\pi 50 \cdot t)$$

- a) Draw the graph for the voltage for the source $v_s(t)$ and sketch the current $i(t)$.



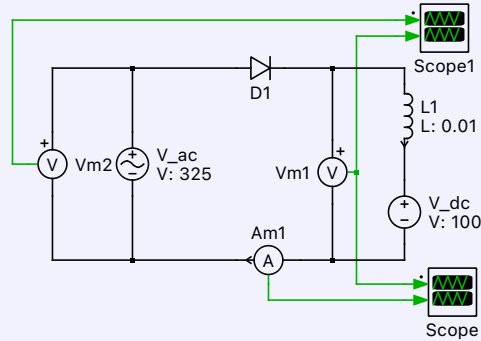
b) write an expression for the current in the circuit $i(t)$.

[illegible][illegible][illegible]

Lab. 2.3: half-wave rectifier with inductive and battery load.

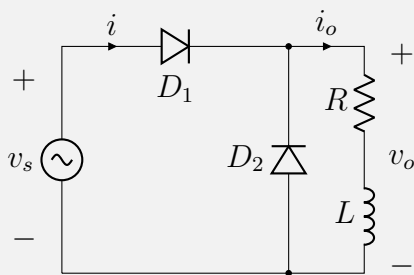
2.3

Build a PLECS model to simulate the Exercise 2.3 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.4: half-wave rectifier with free-wheeling diode.

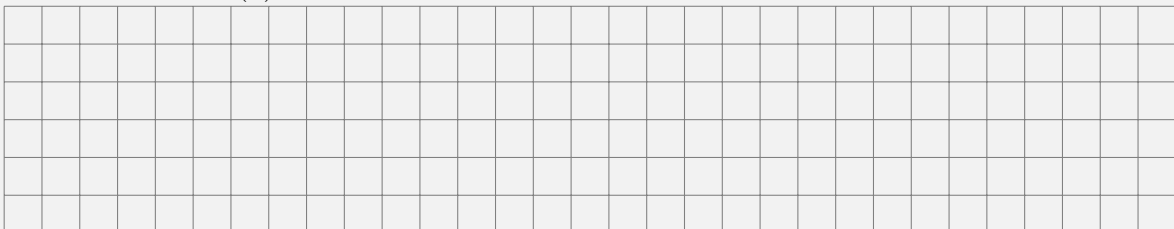
2.4



The half-wave rectifier in the figure supplies a resistor $R=5\ \Omega$ and an inductor $L=0.030\ \text{H}$. The generator parameters are:

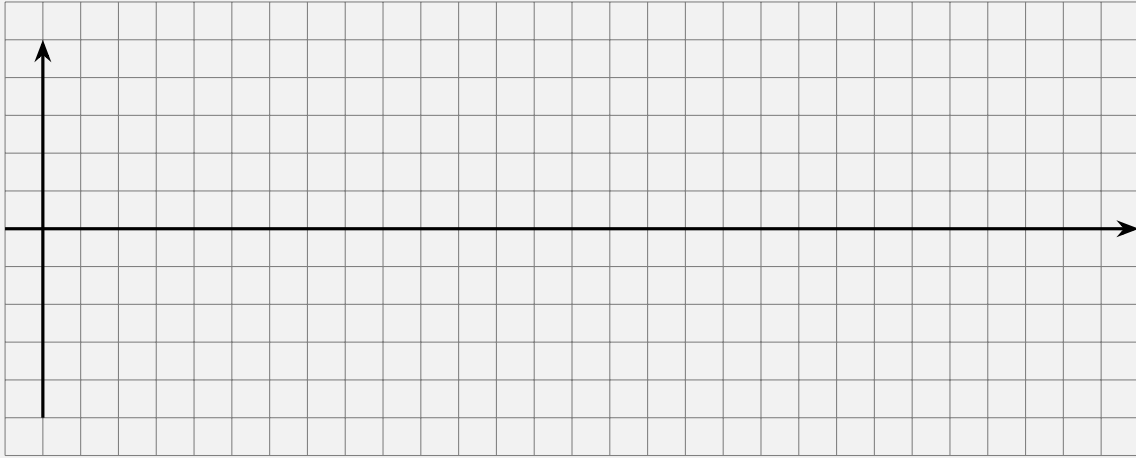
$$v_s = V_m \sin(2\pi ft) = 200 \cdot \sin(2\pi \cdot 50 \cdot t)$$

- a) Estimate the time necessary for the circuit to reach steady-state starting from a zero initial condition $i_o(0) = 0$.

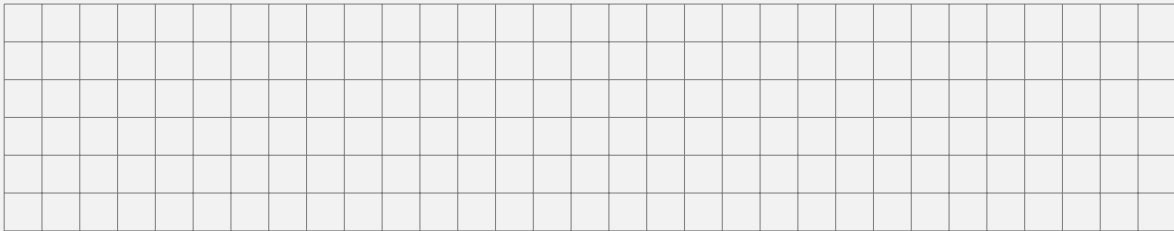


2.4

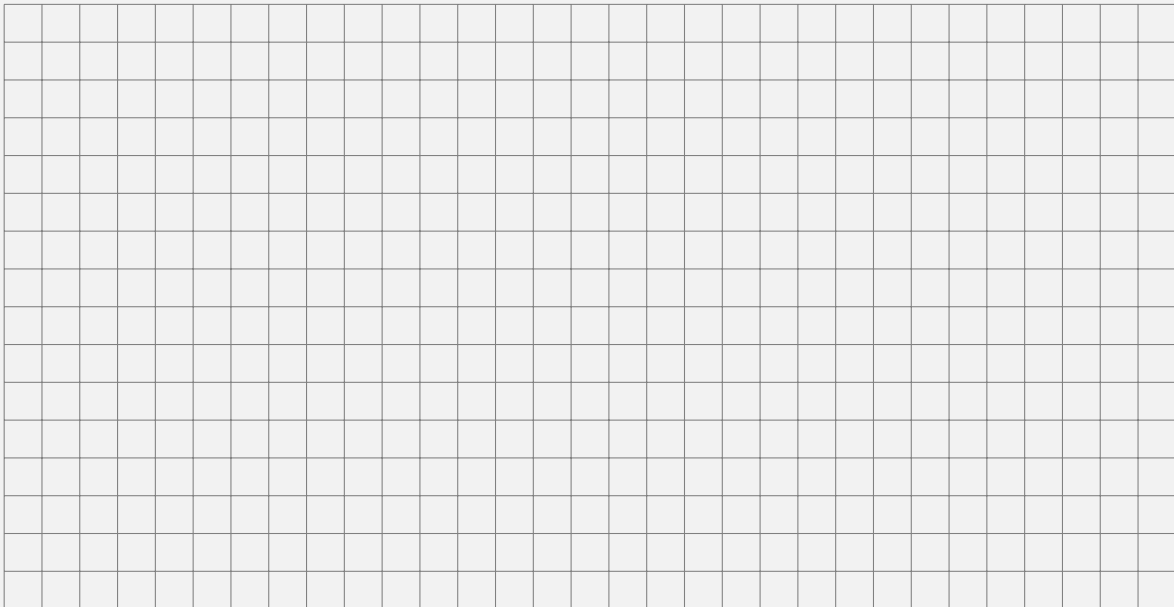
b) Draw the graph for the output voltage $v_o(t)$.



c) Compute the average voltage and current in the load.

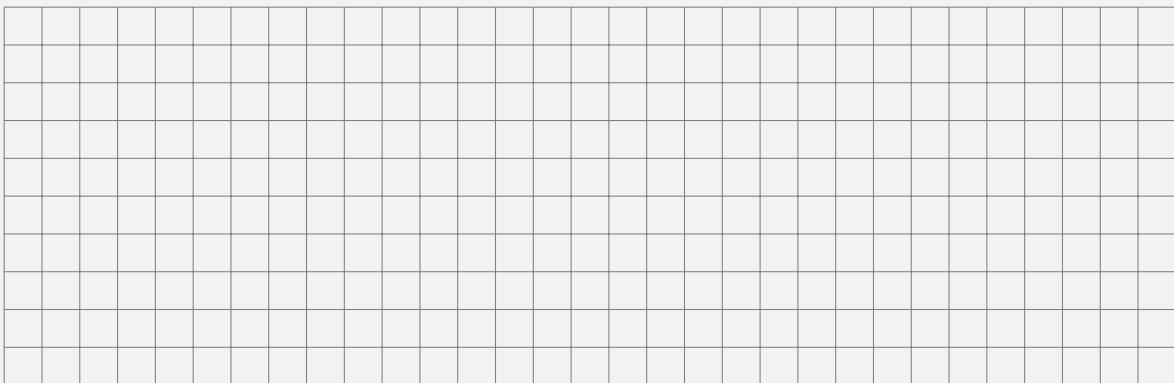
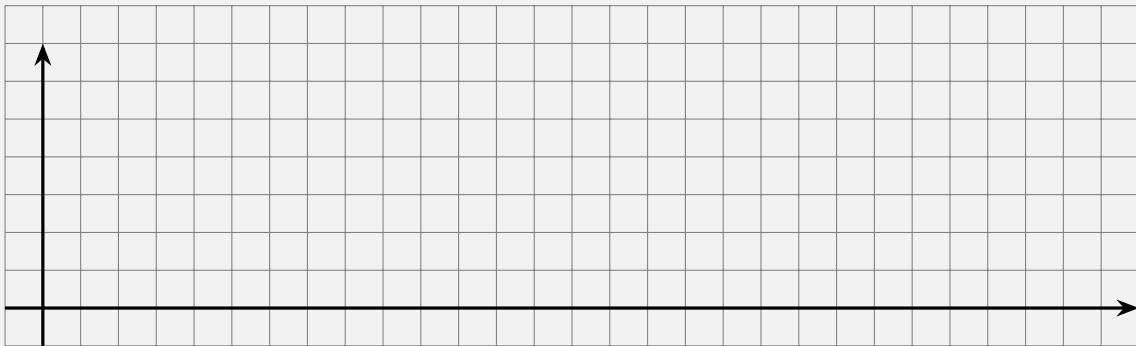
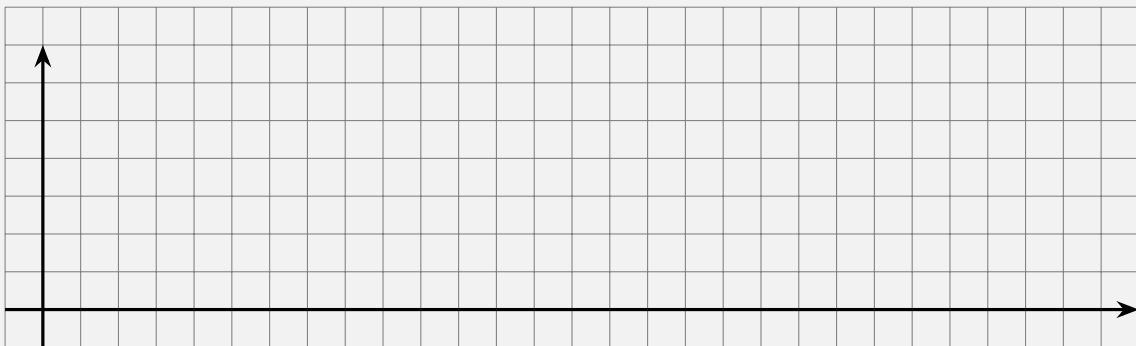
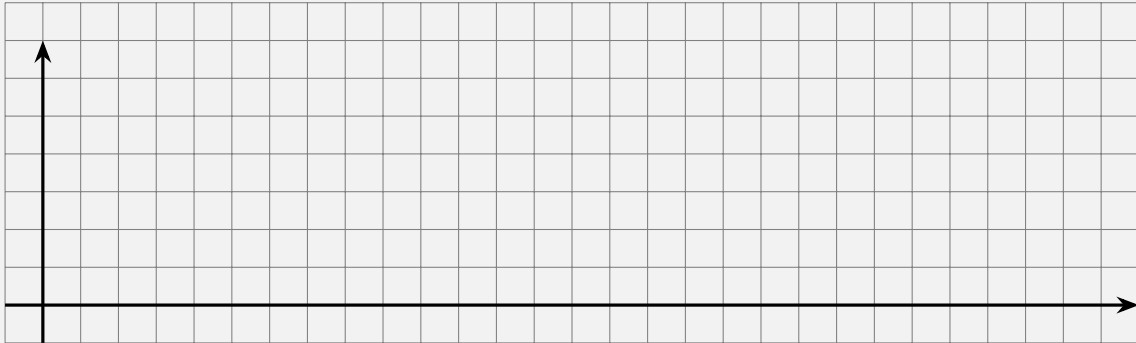


d) Compute the power delivered to the load.



2.5

d) Assuming an infinite inductance, draw the graph for the generator current i_s , in the two diodes i_{D1} and i_{D2} , and compute the power factor of v_s .

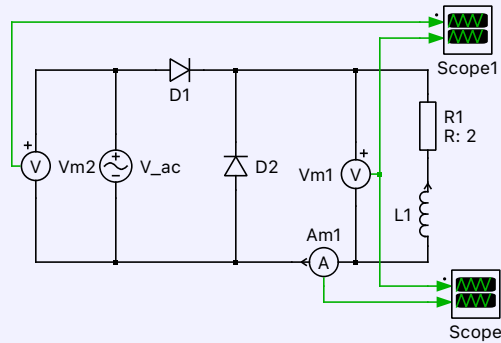


Lab. 2.4: half-wave rectifier with free-wheeling diode.



2.4

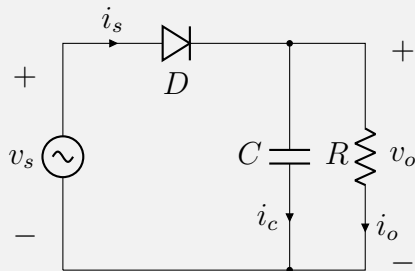
Build a PLECS model to simulate the Exercise 2.4 and 2.5. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.6: half-wave rectifier with capacitor filter.



2.6



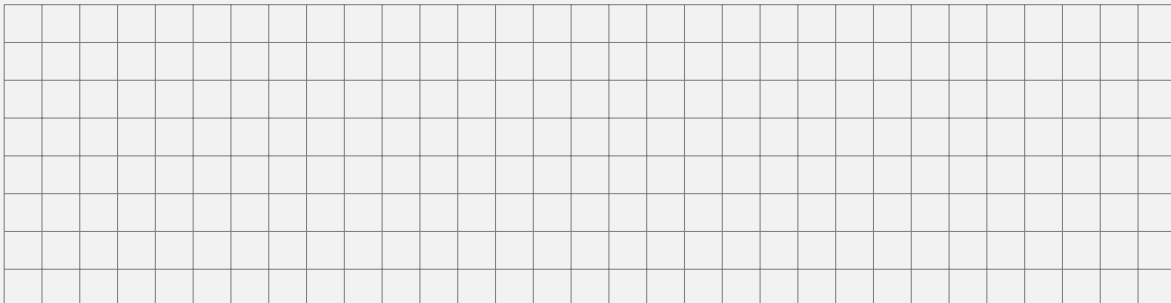
$$v_s = V_m \cdot \sin(2\pi ft)$$

$$V_m = 325$$

$$f = 50Hz$$

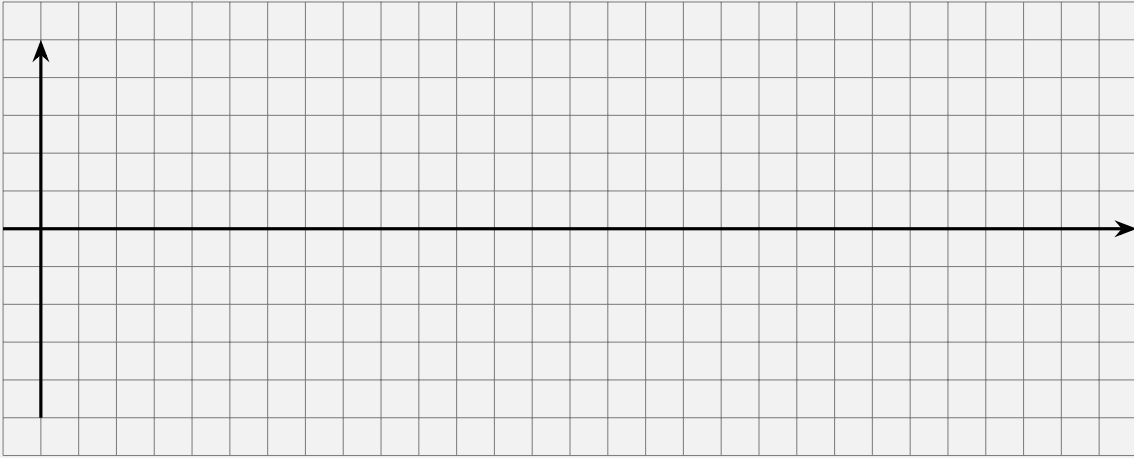
$$R = 100\Omega$$

a) Compute the capacitor value to achieve an output voltage ripple $\Delta v_o \leq 0.1V_m$

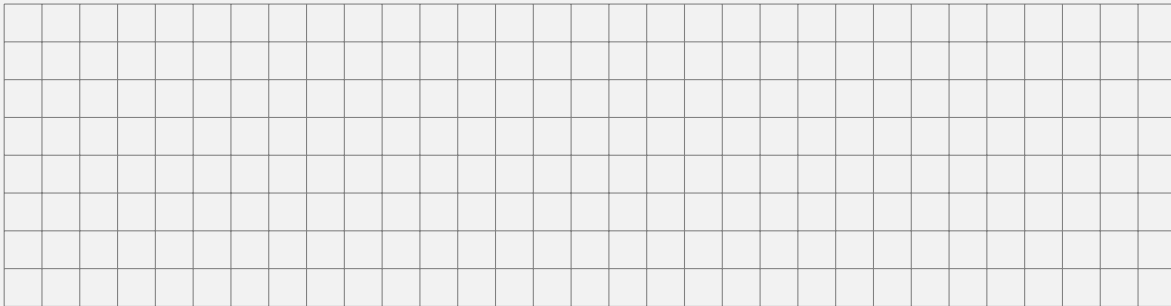


2.6

b) Sketch an approximated graph for the output voltage $v_o(t)$ highlighting Δv_o .



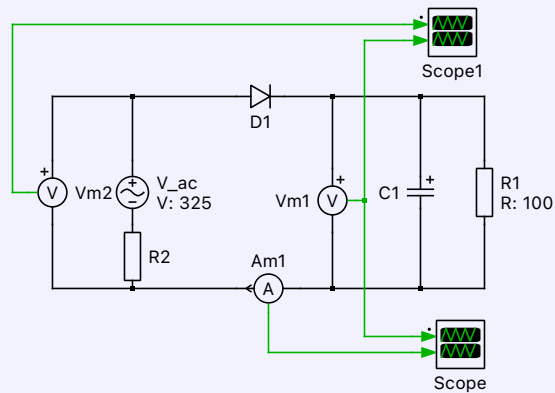
c) Compute an estimation for the average output voltage V_o



2.5

Lab. 2.5: half-wave rectifier with capacitor filter.

Build a PLECS model to simulate the Exercise 2.6 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Q 2.7



$$f = 50\text{Hz}$$

a) Compute the phase angle α to adopt for controlling the rectifier to obtain and average output voltage $V_o=50$ V.

[illegible][illegible][illegible]



Exercise 2.8: controlled half-wave rectifier with specified power.

Considering again the rectifier of Exercise 2.7

- a) Compute the angle α to achieve an output power $P = 200$ W.

This image shows a full page of blank graph paper. The background is a solid light gray color. Overlaid on this background is a precise grid of thin, dark gray horizontal and vertical lines. These lines intersect to form a series of small, identical squares across the entire surface of the page, providing a guide for drawing or writing.

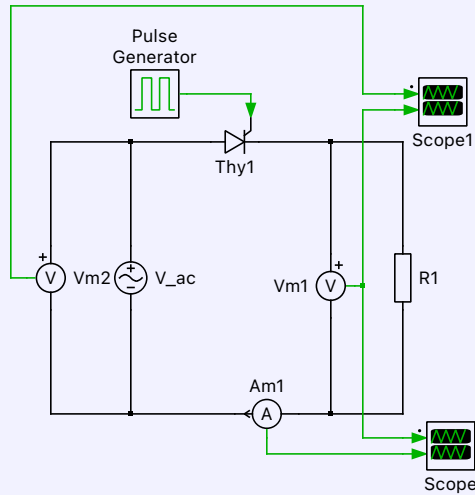
- b)** Is it possible to obtain a power $P = 250$ W in the resistor? Motivate the answer.

[illegible]

Lab. 2.6: controlled half-wave rectifier.

▶
2.6

Build a PLECS model to simulate the Exercise 2.7 and 2.8. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:

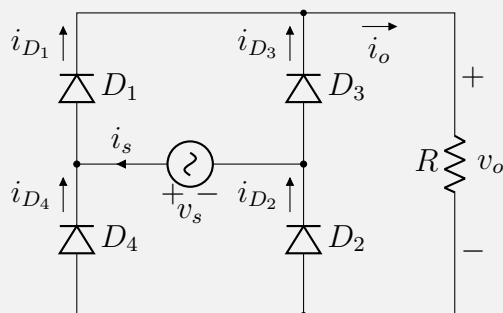


Before performing the computations in PLECS check that the output voltage waveform is as expected, in particular check if a proper phase delays (with the correct unit of measurements) has been set for the Pulse Generator block driving the SCR.

CHAPTER 3

Full-wave rectifiers

Exercise 3.1: full-wave rectifier with resistive load.



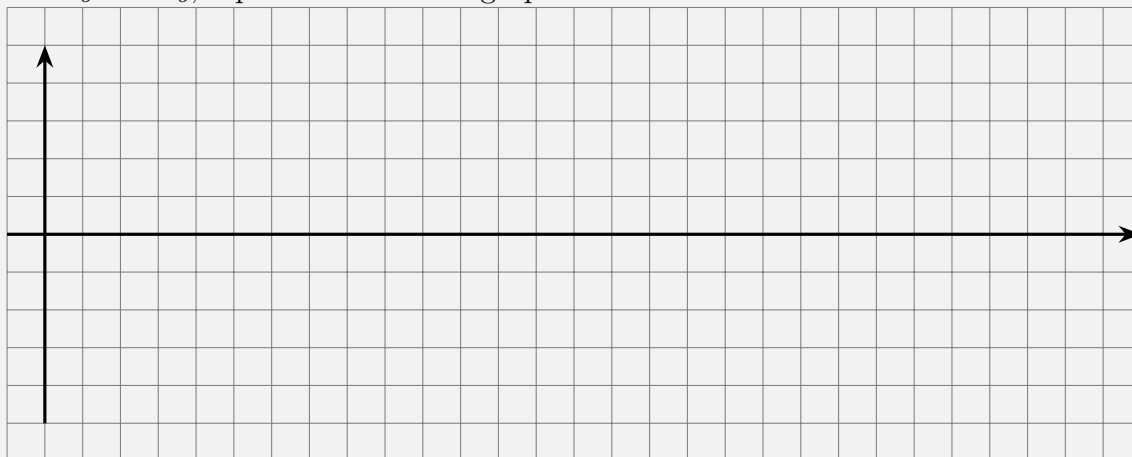
Given the rectifier with the following parameters:

$$v_s = \sqrt{2} \cdot 230 \sin(2\pi ft)$$

$$f = 50 \text{ Hz}$$

$$R = 100 \, \Omega$$

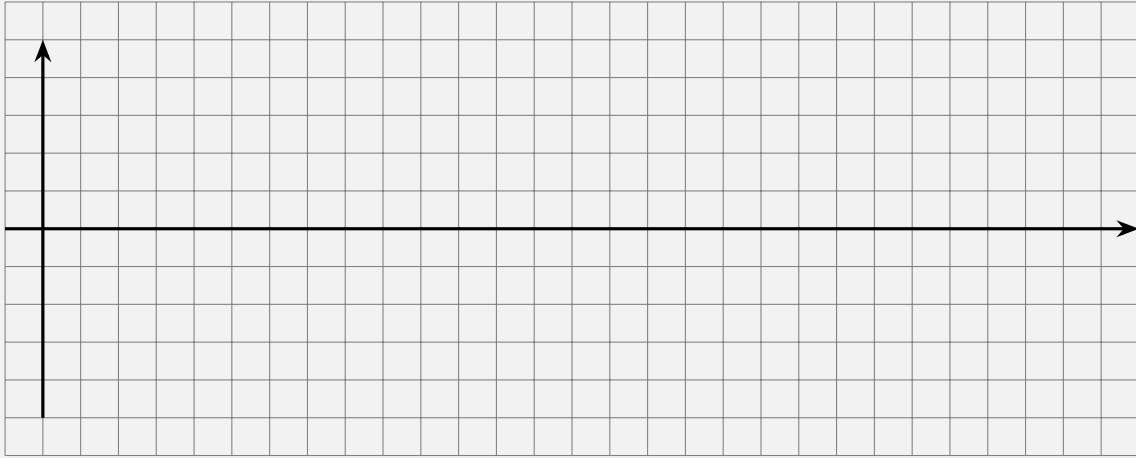
- a) Draw the graphs for the output voltage $v_o(t)$ and current $i_o(t)$. Compute their mean values V_o and I_o , report them on the graph.



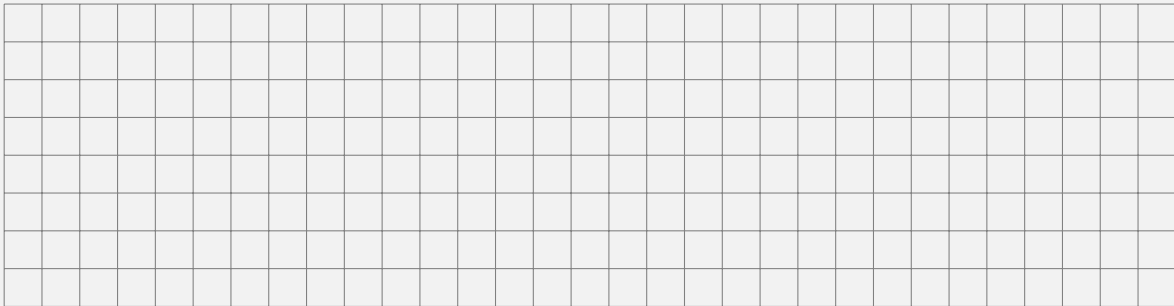
3.1

3.1

b) Draw the graphs for the input voltage $v_s(t)$ and current $i_s(t)$.



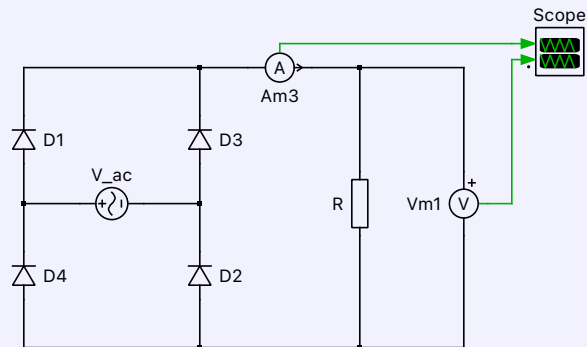
c) Compute the average power and the power factor for the source $v_s(t)$.



3.1

Lab. 3.1: full-wave rectifier with resistive load.

Build a PLECS model to simulate the Exercise 3.1 and 2.8. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:

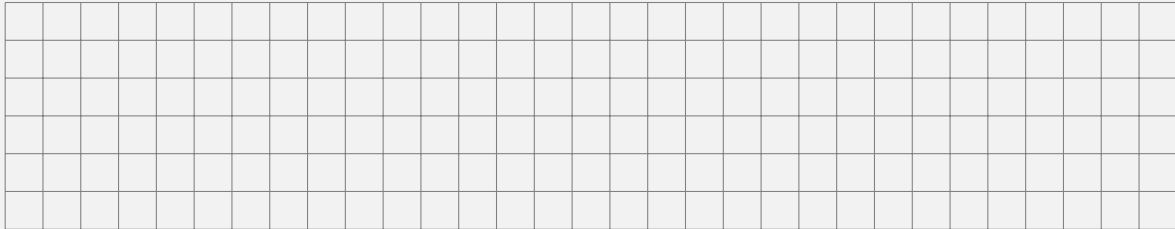


Exercise 3.2: full-wave rectifier with resistive load and capacitive filter.

3.2

Consider again the rectifier of Exercise 3.1.

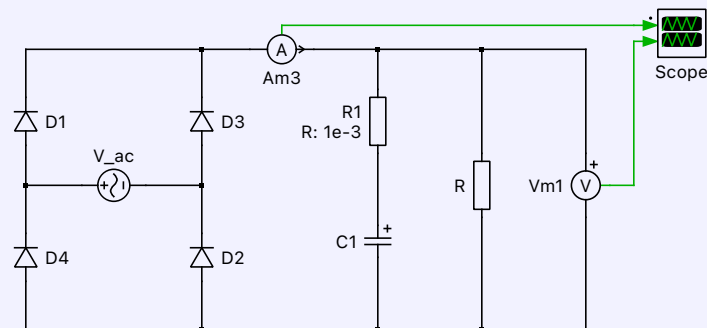
- a) Compute a capacitor to filter the output voltage in order to achieve a ripple $\Delta v_o \leq 0.1V_m$



Lab. 3.2: full-wave rectifier with capacitive filter.

3.2

Build a PLECS model to simulate the Exercise 3.2. Consider the following PLECS model to start the simulation:



Note the small (negligible) resistor in series with the capacitor. This is necessary to obtain a solvable circuit, i.e. a circuit with no parallel connection between capacitors and voltage source.

c) Compute the power delivered to the load.

[illegible]

d) Compute the power factor for the generator v_s .

A full-page sheet of white graph paper with a light gray grid pattern. The grid consists of small, uniform squares covering the entire area. There are no margins, text, or other markings on the page.

e) Considering an infinite inductance in the load, compute the average and rms current in the diodes and the power factor for v_s .

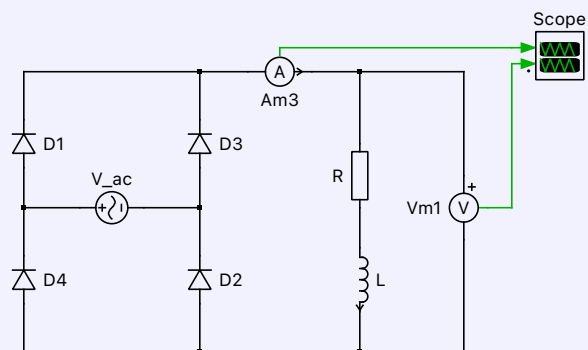
[illegible]



3.3

Lab. 3.3: full-wave rectifier with RL load.

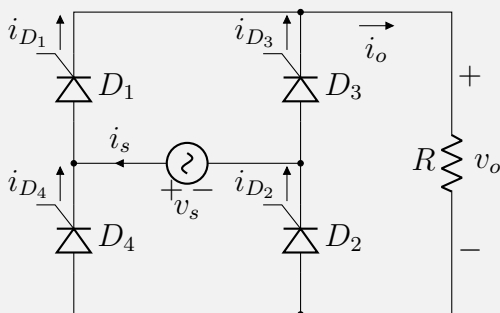
Build a PLECS model to simulate the Exercise 3.3. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Lab. 3.4: full-wave rectifier with RL load and source inductance.

Note that a resistor of adequate value ($100\ \Omega$) is considered in parallel with the source inductance to achieve a solvable model. This is an approximation of the considered model, nevertheless the achieved results are acceptable.

Exercise 3.5: controlled full-wave rectifier with resistive load.



$$v_s = 300 \sin(2\pi ft)$$

$$f = 50\text{Hz}$$

$$R = 10 \, \Omega$$

$\alpha = 60^\circ$ phase angle for scr

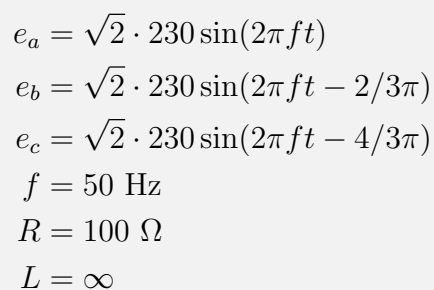
a) Compute the average current in the load.

[illegible]

b) Compute the power absorbed by the load.

[illegible]

Exercise 3.6: full-wave three-phase rectifier.



- [illegible]

- [illegible]

-
- A blank coordinate plane with a grid. The x-axis and y-axis are shown with arrows at their ends. The grid consists of 20 squares by 10 squares.

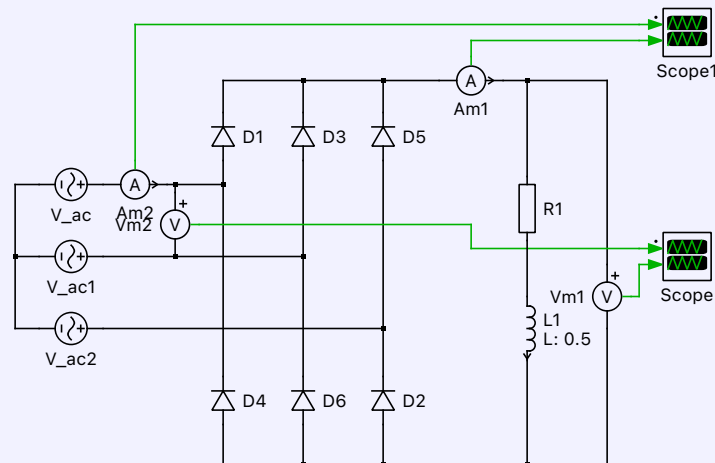
d) Compute the power and the power factor for the generator e_a .



3.6

Lab. 3.6: full-wave three-phase rectifier.

Build a PLECS model to simulate the Exercise 3.6. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



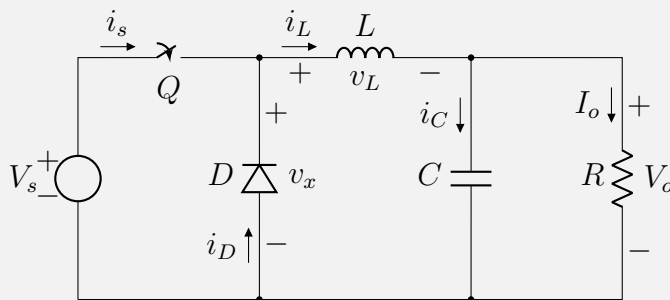
Note that for the inductance an adequate value has been set in the simulation to consider a “large enough” value.

3.6

CHAPTER 4

dc-dc converter

Exercise 4.1: design of a buck converter.



Given the dc-dc converter with the following parameters:

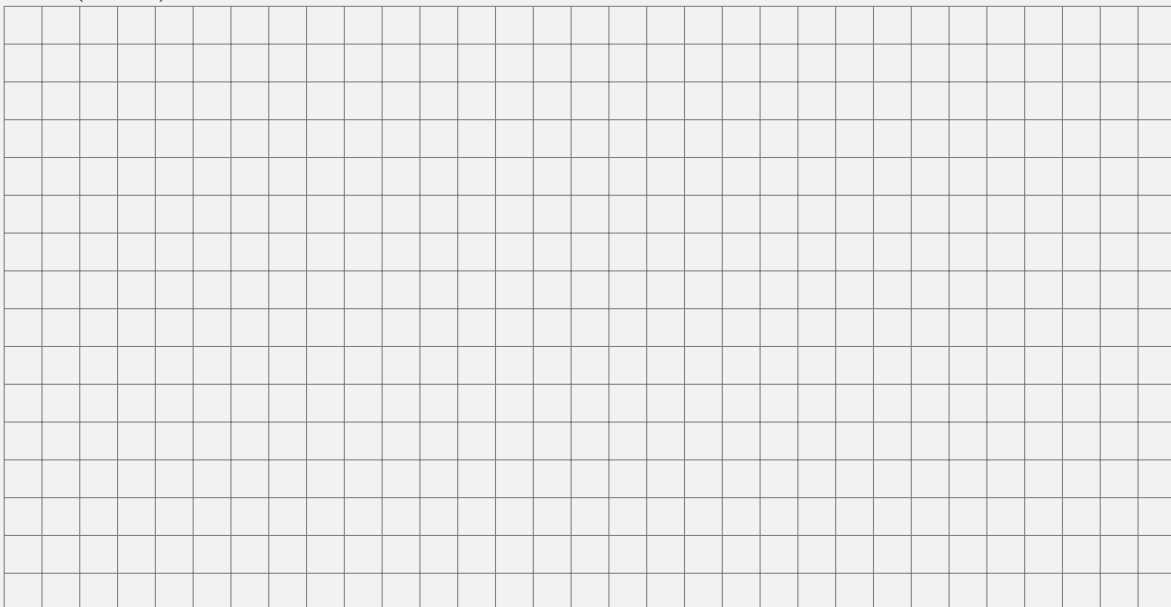
$$V_s = 60 \text{ V}$$

$$V_o = 15 \text{ V}$$

$$R = 8 \text{ } \Omega$$

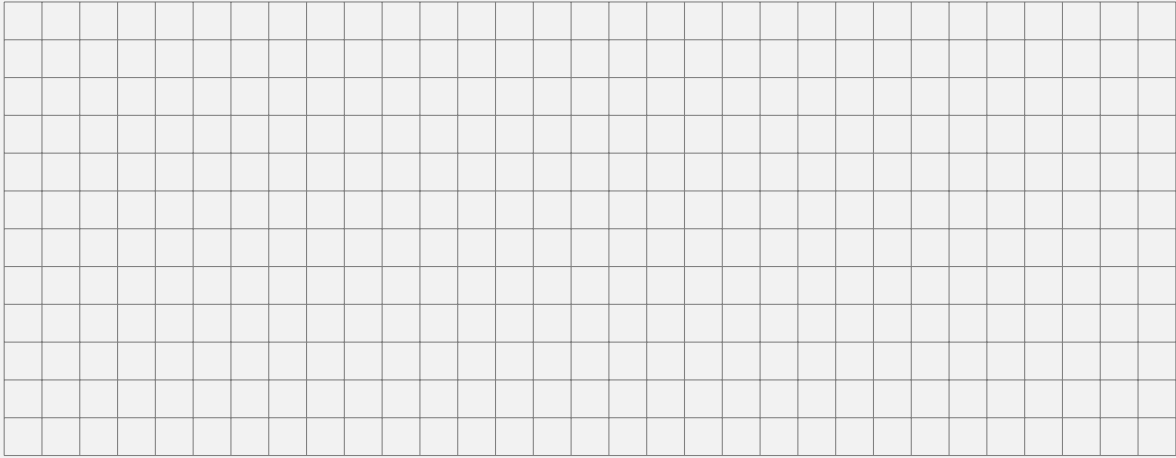
$$f = 20.0 \text{ kHz}$$

a) Determine the values of L that ensure the converter operates in Continuous Conduction Mode (CCM).

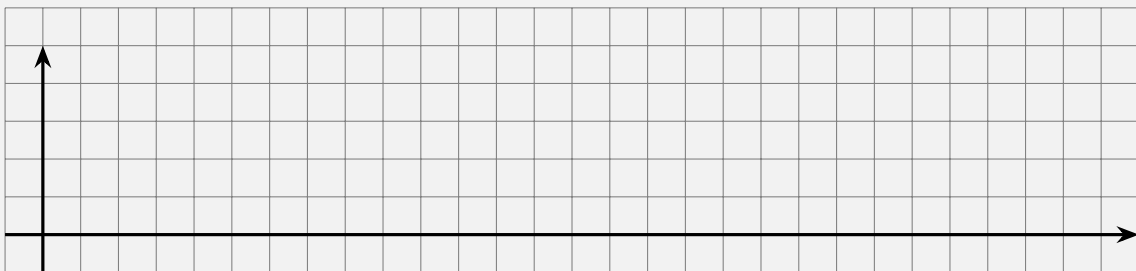
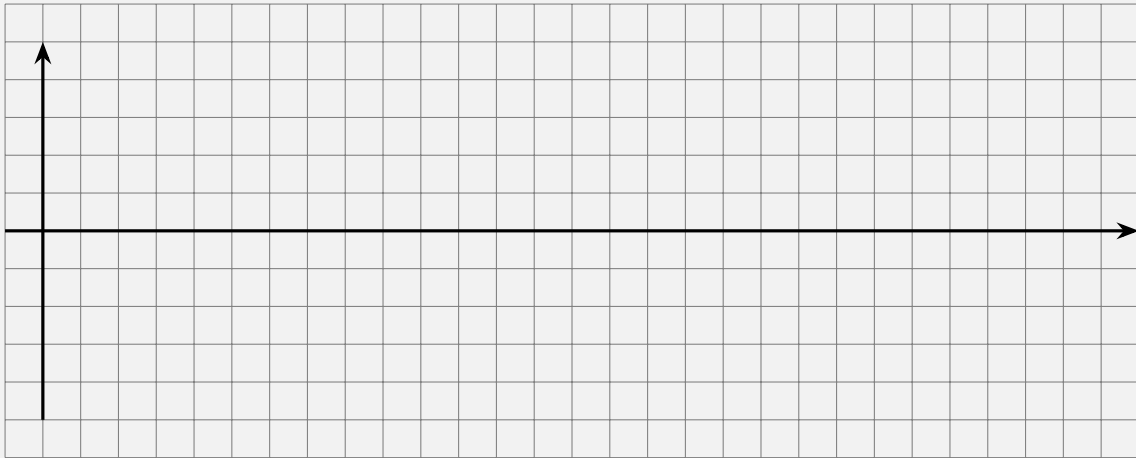


4.1

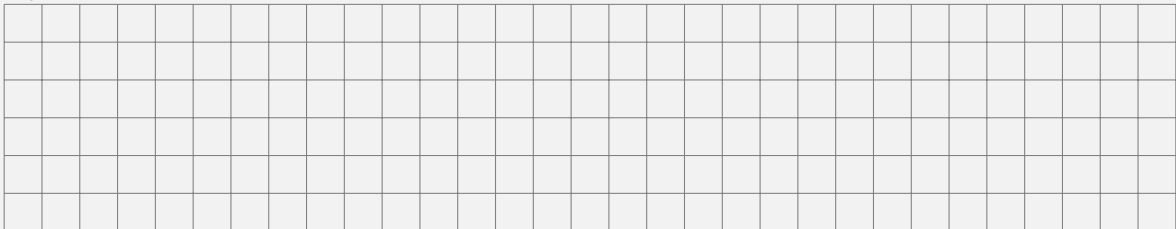
b) Compute the minimum, the maximum and the rms value of the inductor current i_L .



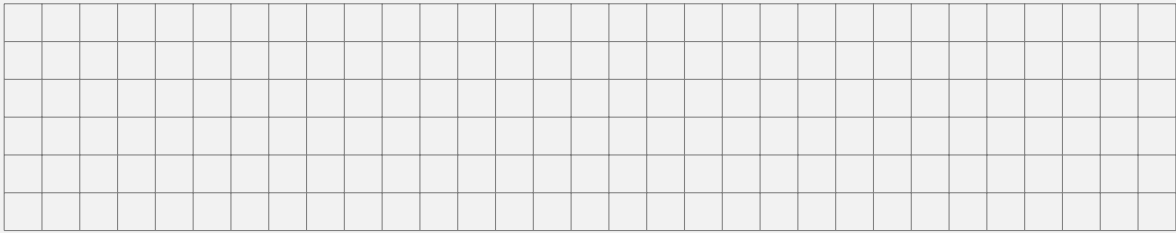
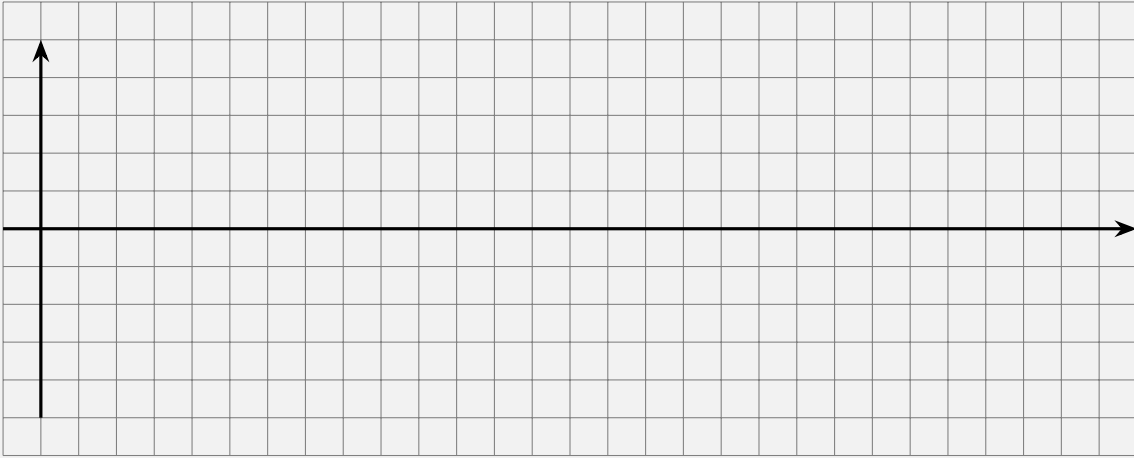
c) Plot the voltage and current waveforms for the inductor.



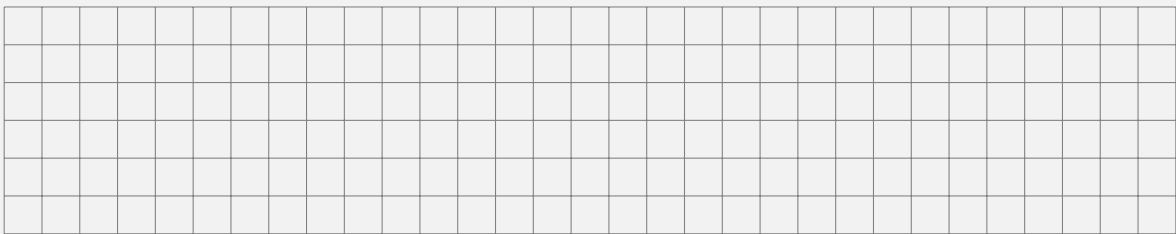
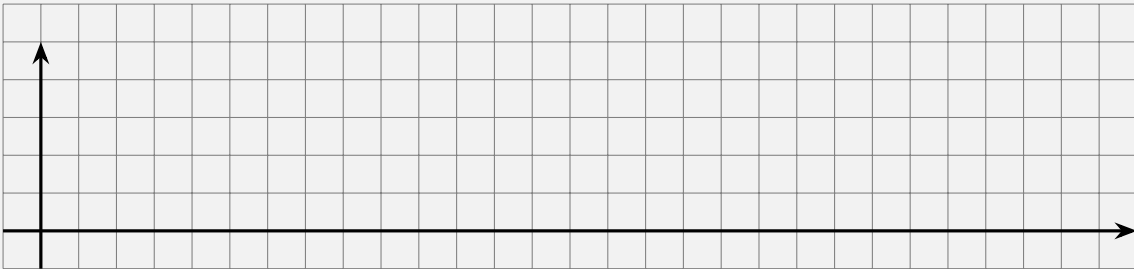
d) Determine the value C that ensure the converter maintain an output voltage ripple $\Delta v_o \leq 5\%$.



e) Plot the current waveform for the capacitor. Compute the average and rms values.

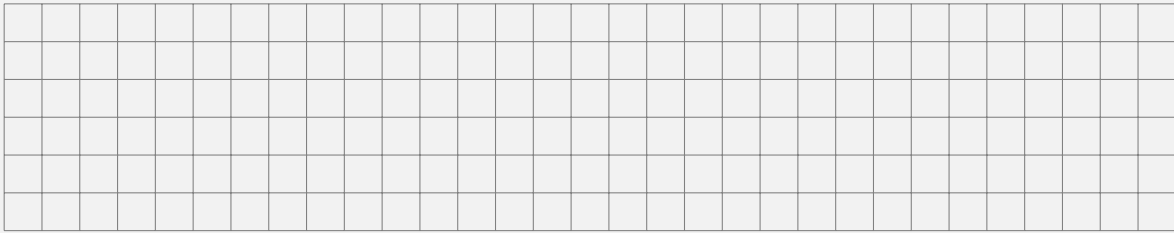


f) Draw the waveform for the switch current i_s and compute its average value.



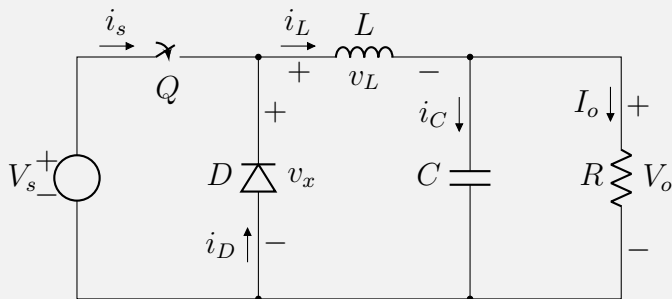
4.1

g) Draw the waveform for the diode current i_D and compute its average value.



4.2

Exercise 4.2: consider again the same converter of Exercise 4.1 with the additional requirement of a maximum current ripple in the inductor of $0.20 \cdot I_o$. How will change the converter design?



Given the dc-dc converter with the following parameters:

$$V_s = 60 \text{ V}$$

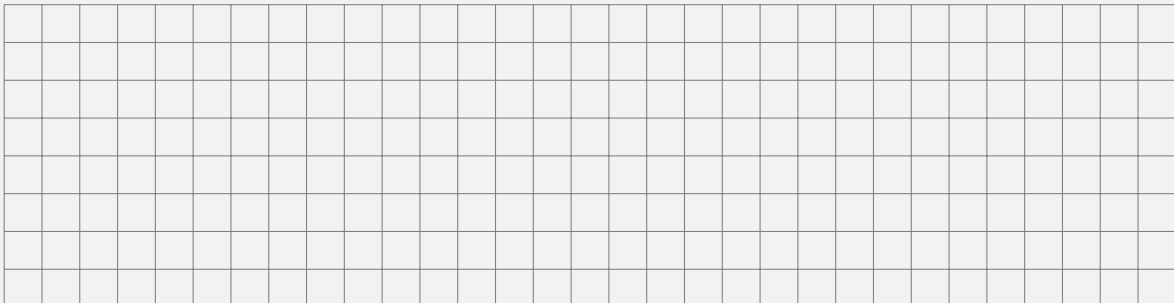
$$V_o = 15 \text{ V}$$

$$R = 8 \text{ } \Omega$$

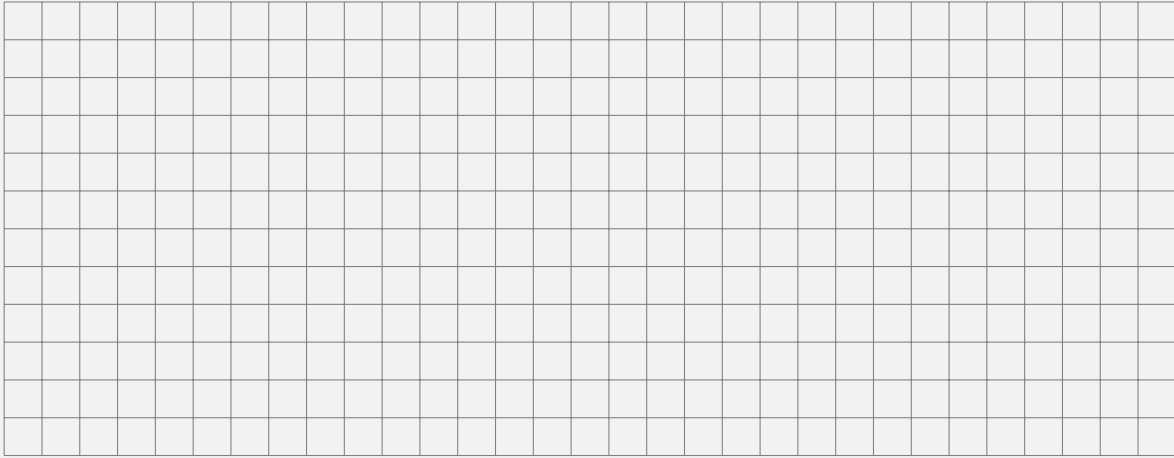
$$f = 20.0 \text{ kHz}$$

$$\Delta i_L \leq 0.20 \cdot I_o \%$$

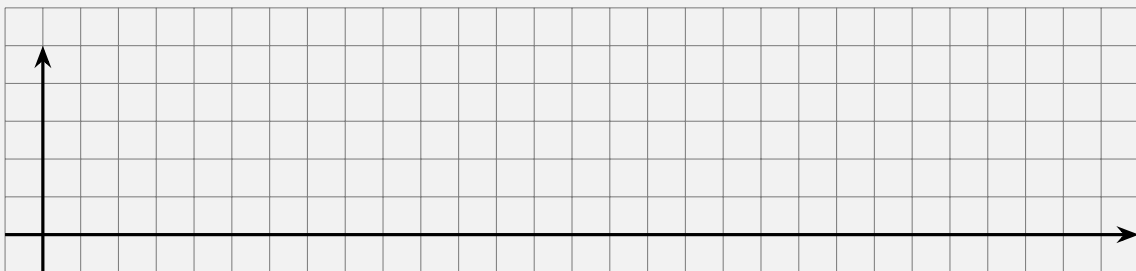
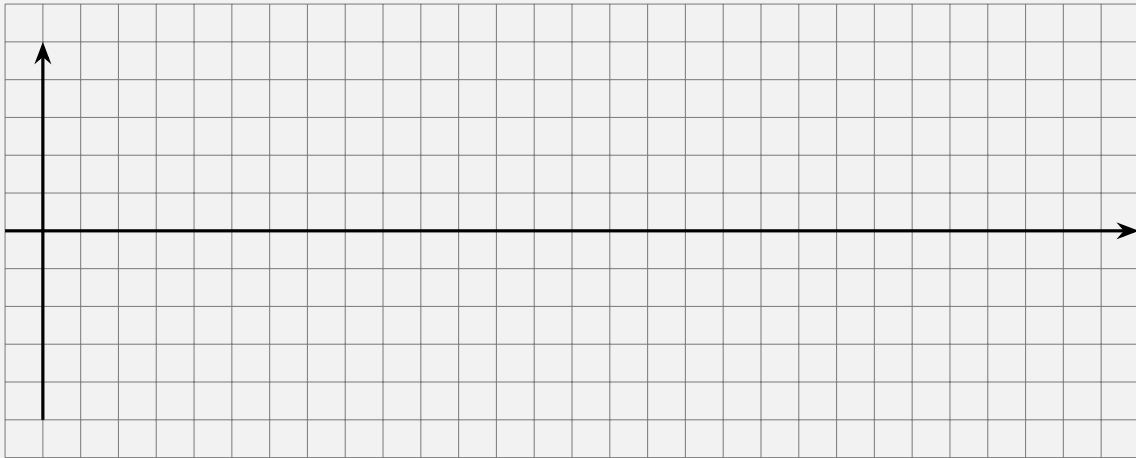
a) Determine the values of L that ensures the required current ripple Δi_L .



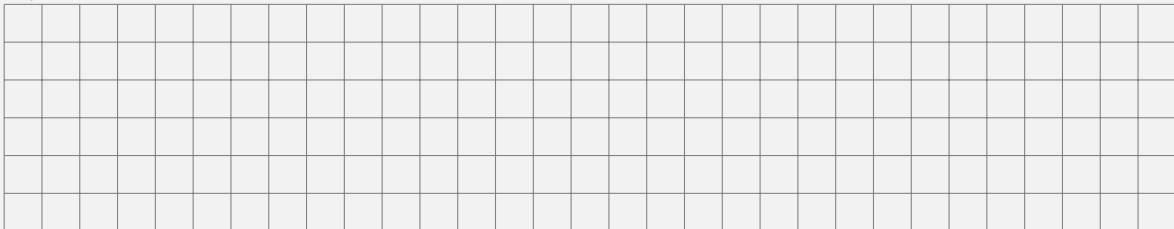
b) Compute the minimum, the maximum and the rms value of the inductor current i_L .



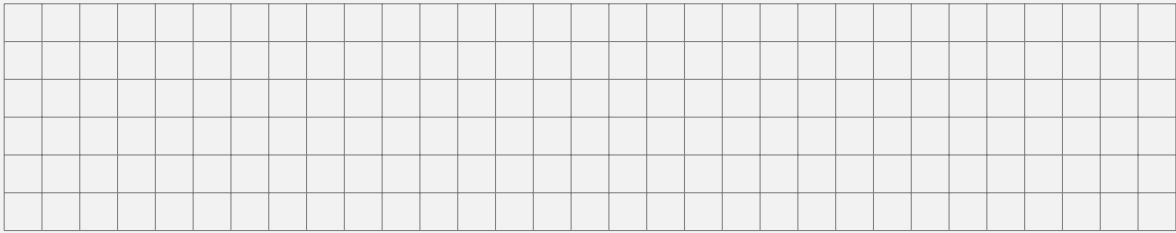
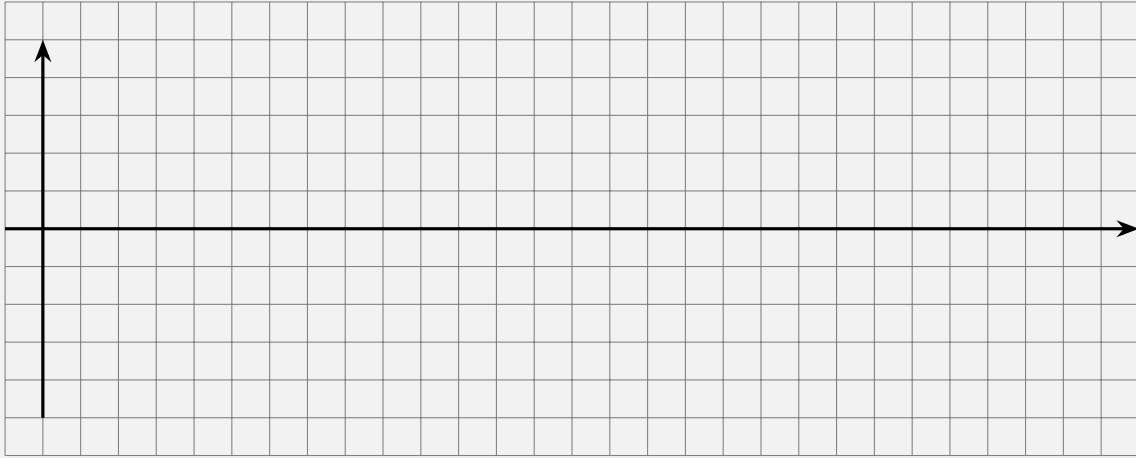
c) Plot the voltage and current waveforms for the inductor.



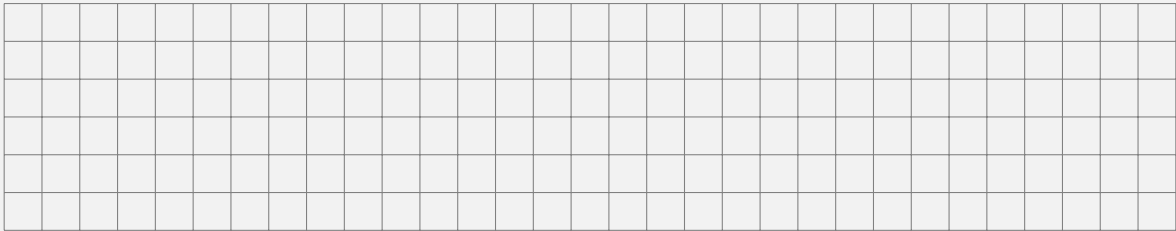
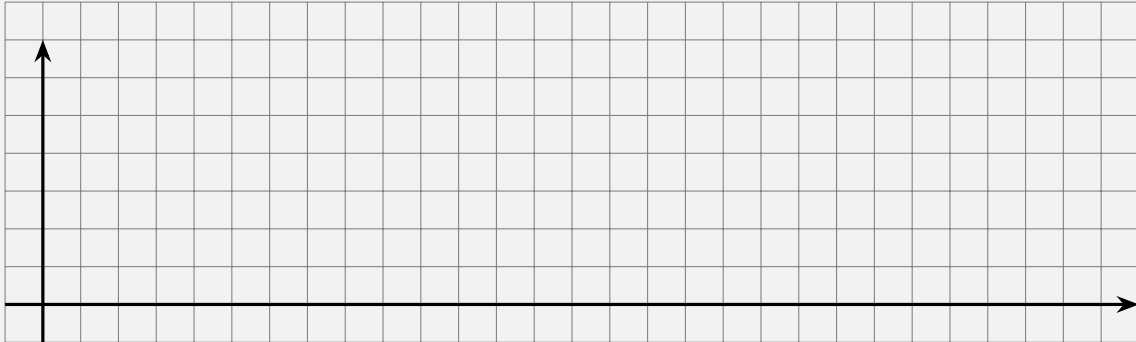
d) Determine the value C that ensure the converter maintain an output voltage ripple $\Delta v_o \leq 5\%$.



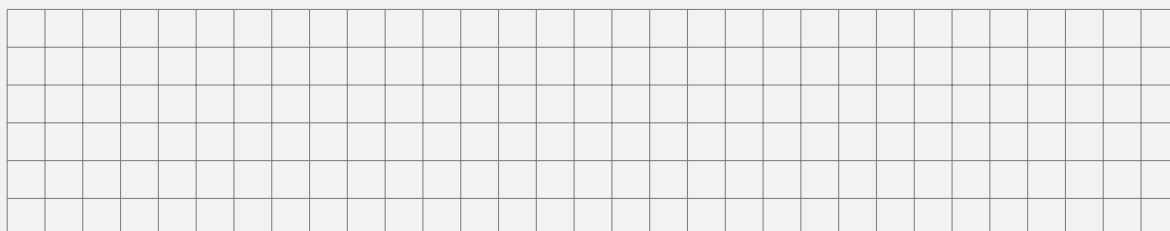
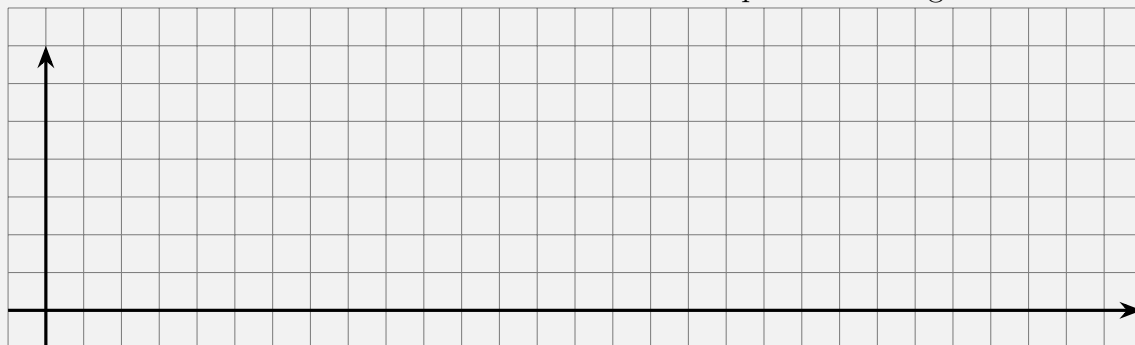
e) Plot the current waveform for the capacitor. Compute the average and rms values.



f) Draw the waveform for the switch current i_s and compute its average value.



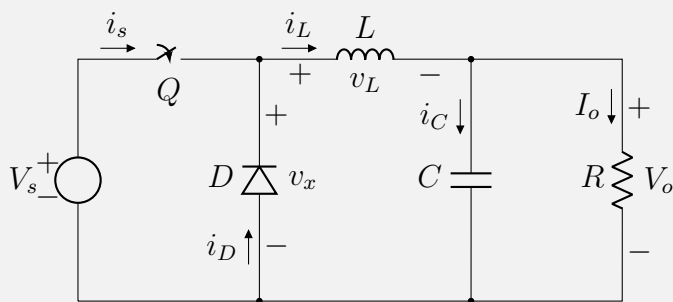
g) Draw the waveform for the diode current i_D and compute its average value.



Q
4.2

4.3

Exercise 4.3: dc-dc buck converter in DCM operation.



A buck converter has the following parameters:

$$V_s = 75 \text{ V}$$

$$V_o = 25 \text{ V}$$

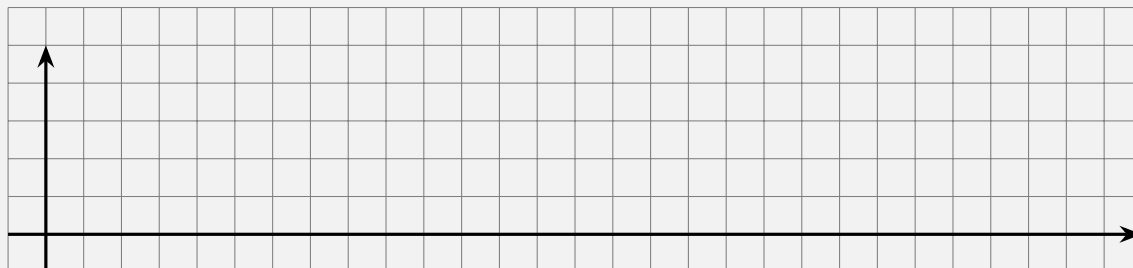
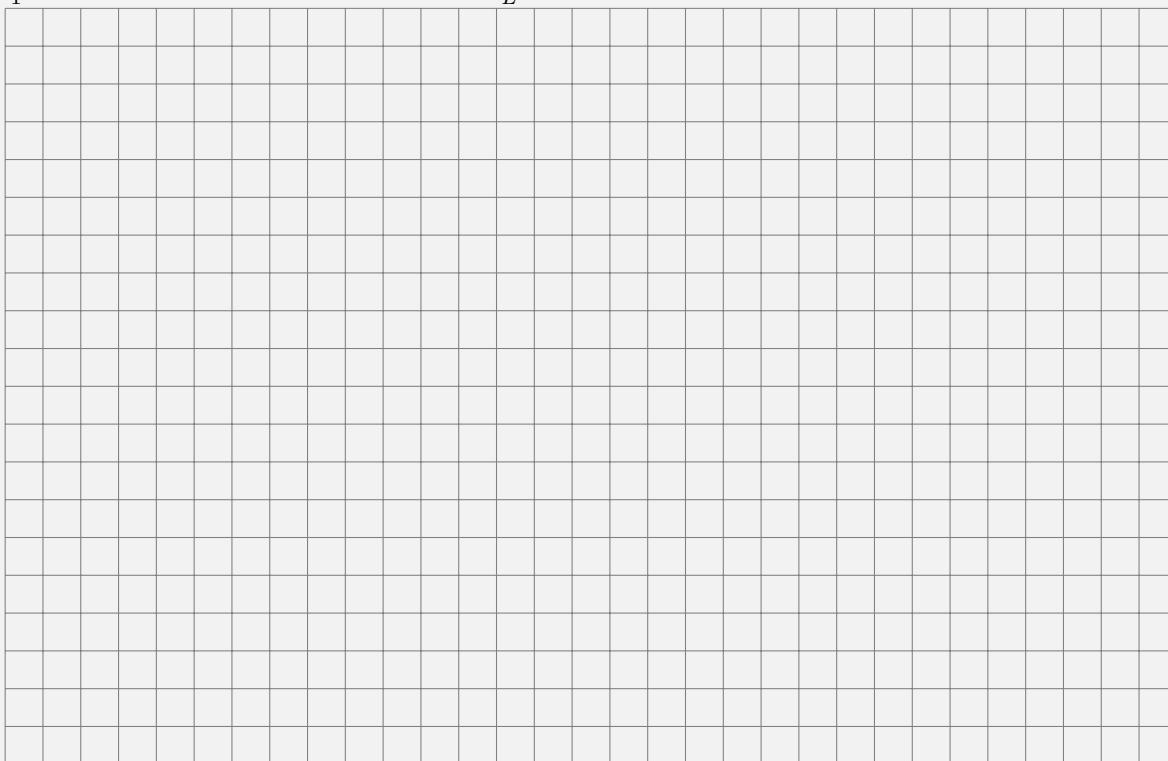
$$f_s = 10 \text{ kHz}$$

$$C = 100 \mu\text{F}$$

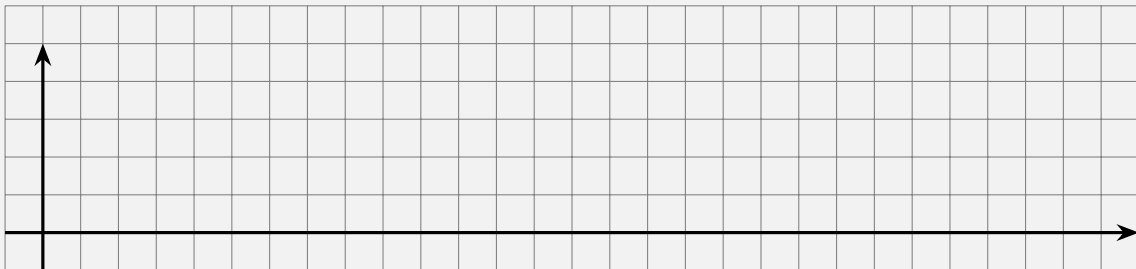
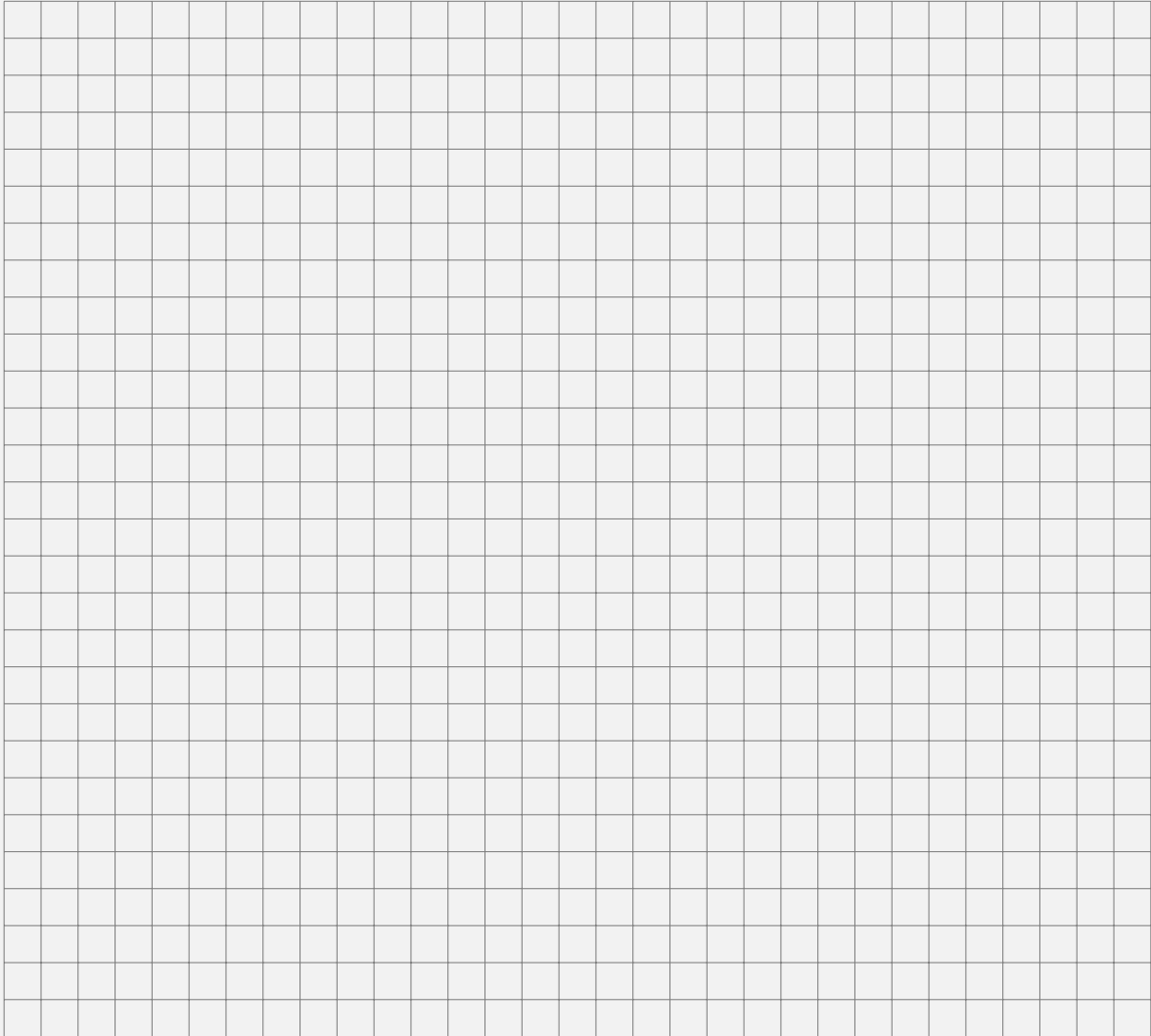
$$L = 500 \mu\text{H}$$

The converter operates with two different load, i.e. $R_1 = 5 \Omega$ and $R_2 = 20 \Omega$.

- a) Compute the duty cycle to obtain the desired output voltage $V_o = 25 \text{ V}$ with the load $R_1 = 5 \Omega$ and draw the waveform of i_L .



b) Compute the duty cycle to obtain the desired output voltage $V_o = 25\text{ V}$ with the load $R_L = 20\ \Omega$ and draw the waveform of i_L .



9
4.3

c) Compute the limit value of load resistor R_{lim} which allows the converter to operate in ccm with the given parameters.

A large grid of graph paper for calculations, consisting of 20 columns and 30 rows of small squares.

d) It has been found that to have $V_o=25$ V the duty cycles are:

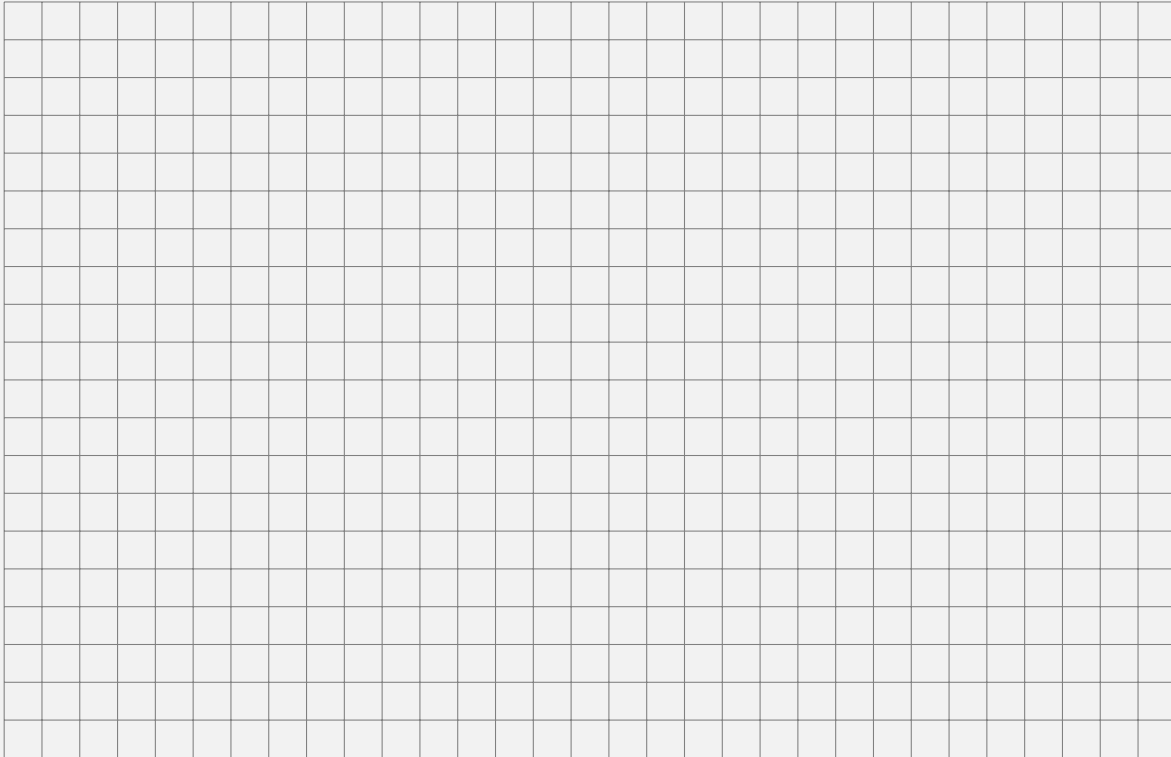
- $\delta = 1/3 = 0.333$ when $R_1 = 5 \Omega$
- $\delta = 0.288 < 1/3$ when $R_2 = 20 \Omega$

Which would be the output voltage with $R_2 = 20 \Omega$ maintaining $\delta = 1/3$?

A large grid of graph paper, consisting of 20 columns and 30 rows, provided for calculations.

4.3

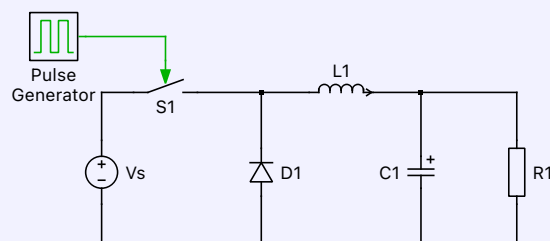
e) Compute the inductance to obtain ccm operation of the converter with $R_2=20\ \Omega$ load.



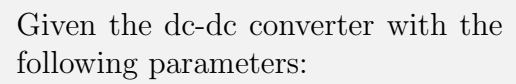
4.1

Lab. 4.1: simulation of dc-dc buck converter.

Build a PLECS model to simulate the Exercise 4.1, 4.2 and 4.3. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulations:



4.4

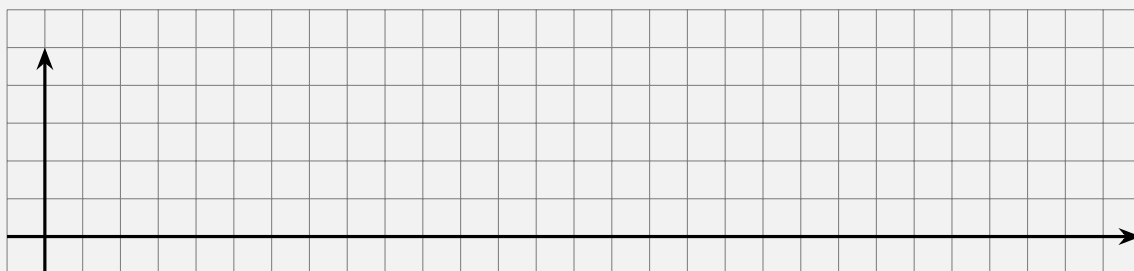
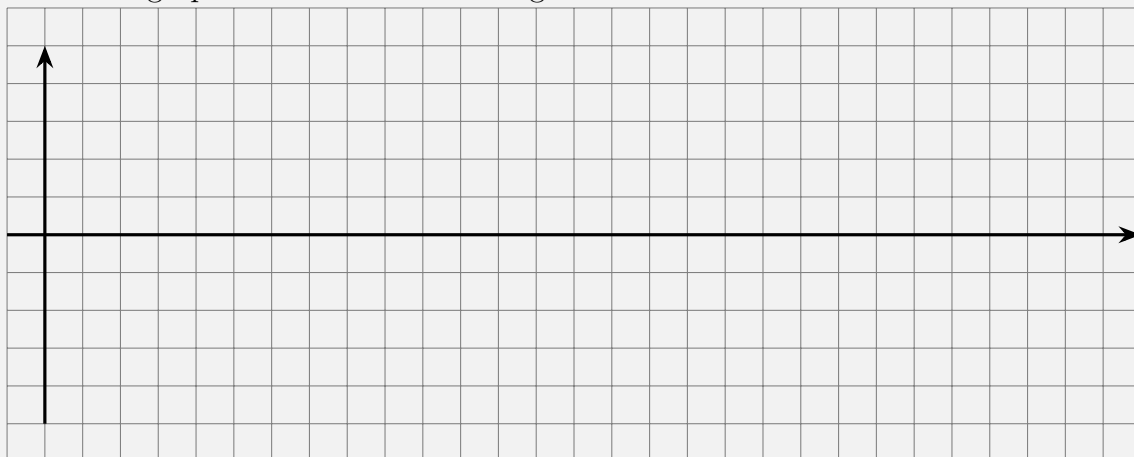


a) Compute the average output voltage V_o and the average output current I_o .

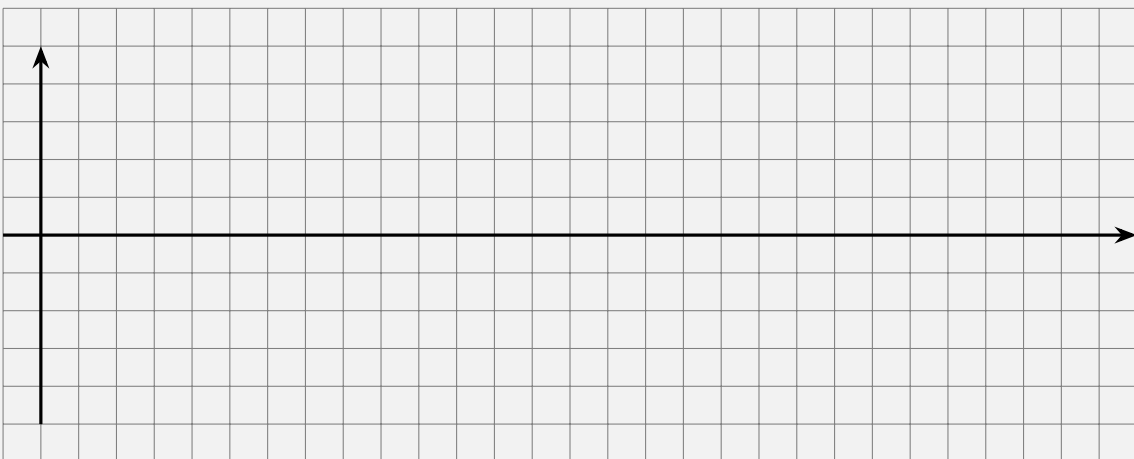
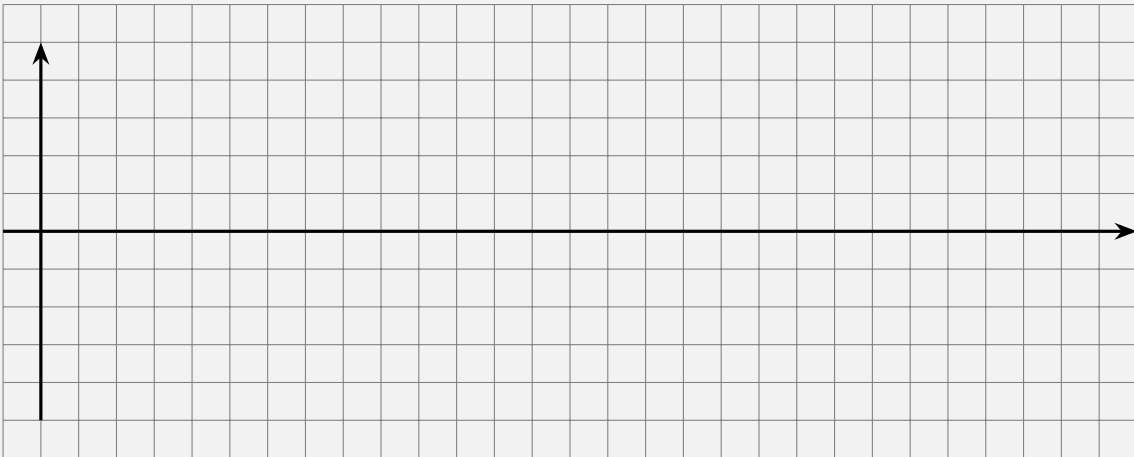
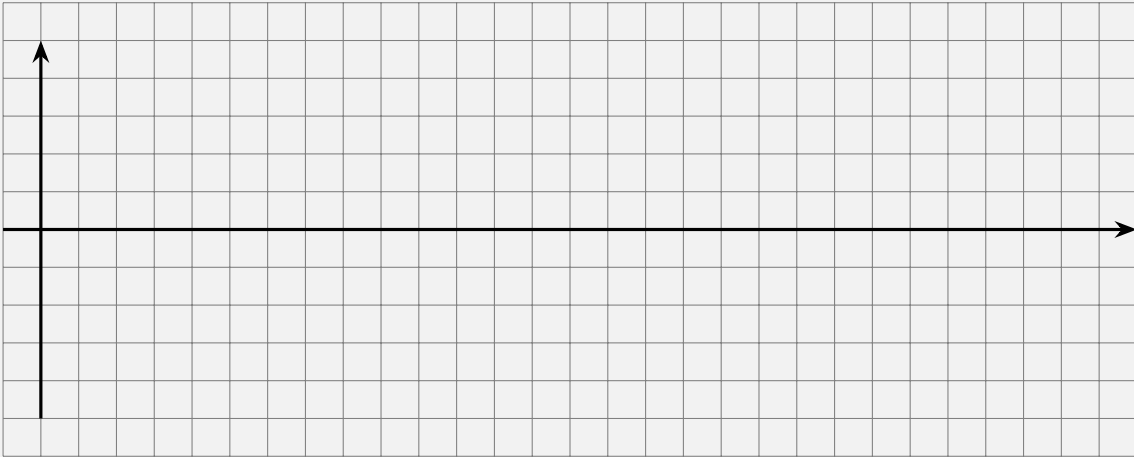
[illegible][illegible]

9
4.4

c) Draw the graph for the inductor voltage v_L and current i_L .



d) Draw the graphs and compute the average values for the switch current i_Q , the diode current i_D and the capacitor current i_C . Report the average values on the graphs.

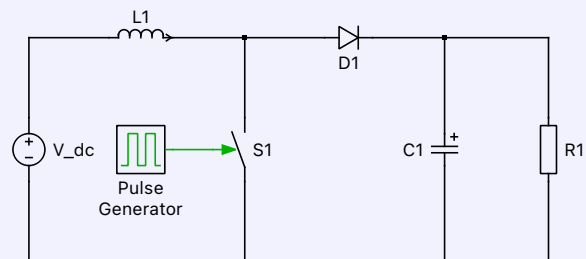




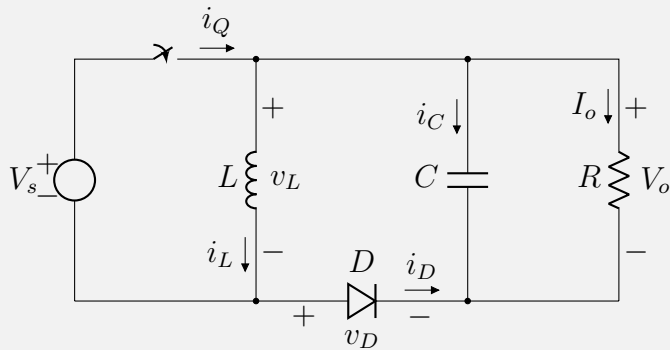
4.2

Lab. 4.2: simulation of dc-dc boost converter.

Build a PLECS model to simulate the Exercise 4.4 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulations:



Exercise 4.5: dc-dc buck-boost converter.



Given the dc-dc converter with the following parameters:

$$\begin{array}{ll} V_s = 12 \text{ V} & \delta = 0.6 \\ f = 200 \text{ kHz} & R = 10 \text{ } \Omega \\ L = 10 \text{ } \mu\text{H} & C = 20 \text{ } \mu\text{F} \end{array}$$

- a) Compute the voltage at the output.

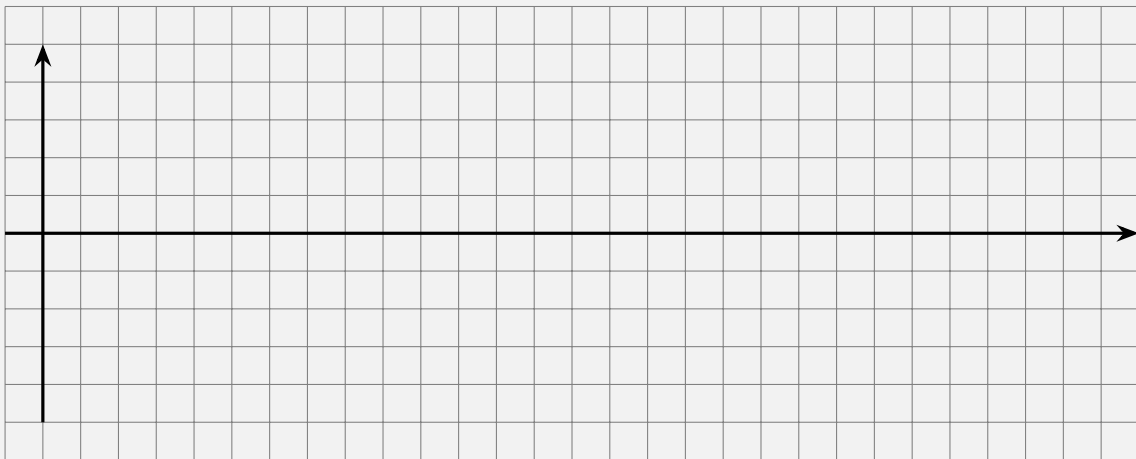
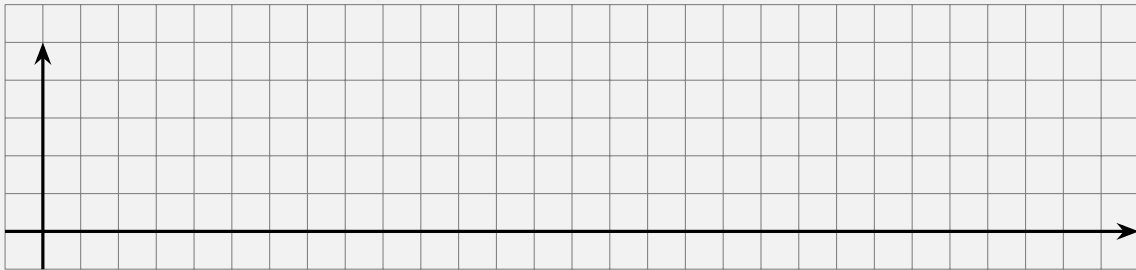
[illegible]

- b) Draw the graph for the inductor voltage v_L and current i_L .

A blank coordinate plane with a grid. The x-axis and y-axis are shown, intersecting at the origin. The grid extends 10 units in each direction from the origin.A blank coordinate grid with x and y axes. The grid is 20 units wide and 10 units high. The x-axis is horizontal and the y-axis is vertical, both with arrows at their ends. The grid lines are thin and gray, and the background is white.

4.5

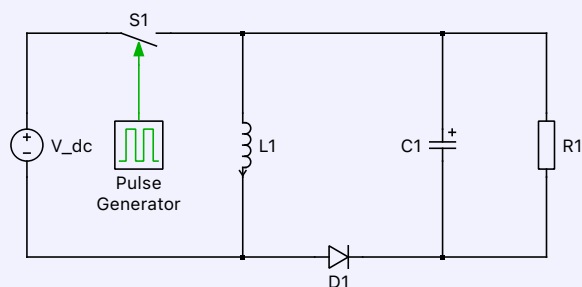
c) Draw the graphs and compute the average values for the switch current i_Q , the diode current i_D and the capacitor current i_C . Report the average values on the graphs.



4.3

Lab. 4.3: simulation of dc-dc buck-boost converter.

Build a PLECS model to simulate the Exercise 4.5 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulations:



Part II

Appendix

Brief introduction to PLECS

PLECS¹ is a software for the modeling and simulation of power conversion systems including, among the others, power sources, power converters and various type of loads. PLECS includes a comprehensive component library, which covers the electrical, as well as the magnetic, thermal, and mechanical aspects of power conversion systems and their controls. Power electronics circuits are captured with a schematic editor in a way that is familiar and intuitive for electrical engineers. Typical power electronics components such as diodes, switches, inductors and capacitors are placed on the circuit diagram and simply connected by drawing wires. Additional information along with a complete set of tutorials can be found at the webpage <http://www.plexim.com/>. To get an overview of the software's capabilities, it is possible to test several models running directly the simulations from the website. Alternatively, it is possible to download a demo version of the software to run the simulations on a local PC.

The best way to learn the software is to use it in performing computations. Therefore, in the following pages some preliminary examples are considered to introduce the fundamental of the software and run the first simulations.

Studying the transient of an RC circuit

The first electrical system we are going to model is a simple RC network, as shown in Fig. A.1. A capacitor is charged by a DC voltage source via a resistor. Various quantities will be considered during the charging transient.

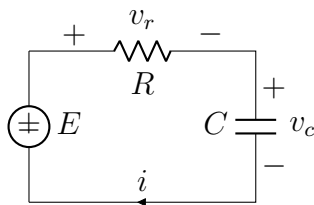


Fig. A.1: RC circuit.

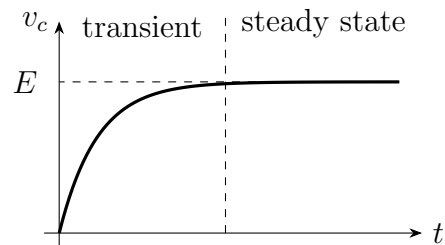


Fig. A.2: Voltage response of the RC network

Open PLECS program: the yellow schematic window that appears is the modeling interface. In the libraries there are various components from which it is possible to create the

¹PLECS is a registered trademark of Plexim GmbH.

circuits. It is possible to browse through the available libraries and see which components are available.

The Library Browser can always be pull up from the Window menu of the schematic window or pressing **Ctrl-L** (for Mac users, use **Cmd** instead of **Ctrl** for all shortcuts from here on).

Components

The components needed to build the circuit should be dragged and dropped into the yellow model window from the Library Browser. To duplicate an existing component already present in the model window, hold down the **Ctrl** key, then drag and drop the component to a new location, or use the right mouse button. It is also possible to select the component and use **Copy** and **Paste**.

The electrical components required to create the RC network are located in the Electrical domain library, specifically under the Sources, Meters and Passive Components sub-libraries. The Scope component can be found in the System library. Instead of navigating through the library it is also possible to search a component by typing the first letters of its name in the search bar.

Signals

In addition to the electrical connections (black wires) that are used to connect electrical components, PLECS makes also use of unidirectional signals. The signals are in green and have an arrowhead to indicate their direction. These signals carry non-electrical information like measurement values or triggering pulses for switches. Signals can be used in calculations or displayed in a scope. Electrical connections cannot be fed into a scope directly, it is always necessary to use a voltmeter or an ammeter to convert the electrical quantities into a signal first.



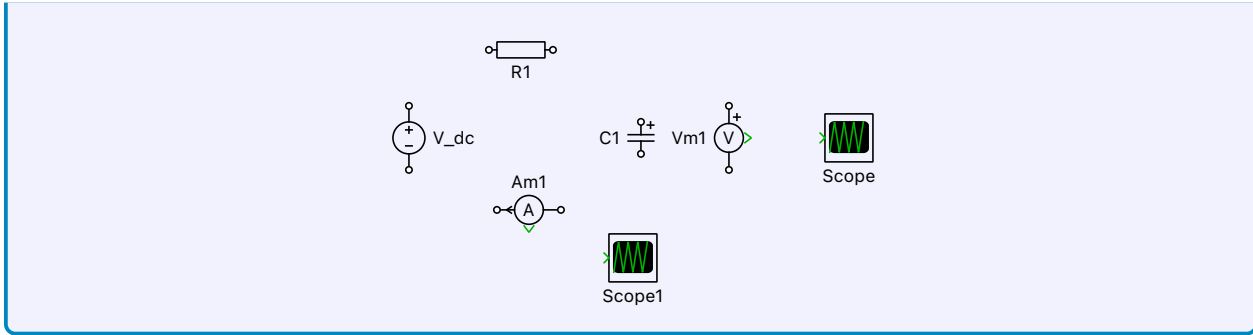
A.1

Lab. A.1: preparing the component for the RC circuit (RC_00)

Perform the following steps:

- drag and drop the Voltage Source DC, Inductor, Resistor, Capacitor, Voltmeter, Ammeter and Scope components from the library onto the schematic window.
- It is possible to move the required components with the left mouse button. To rotate selected components press **Ctrl-R**, to flip them horizontally press **Ctrl-F**, or to flip them vertically press **Ctrl-I**. All these functions can also be accessed with a right mouse click on the component in the Format menu.

When all the components have been inserted, the model is as reported in the following:



Connections

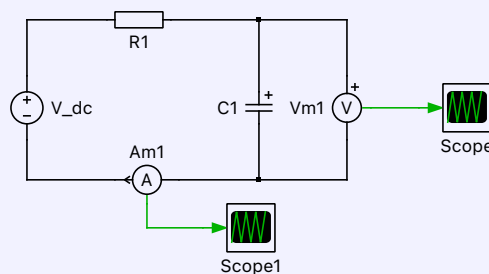
Each component in the current schematic has black circles or green arrowheads to represent terminals. Bringing the mouse pointer close to such a terminal, the pointer shape changes from an arrow to a cross. It is then possible to drag a connection to another component by holding the left mouse button down. When another terminal or an existing connection of the same connection type is approached, the pointer shape changes into a double cross. As soon as the mouse button is released, a new connection will be created. For creating a branch connection from an existing connection, place the mouse cursor over it, hold **Ctrl**, then click and drag the cursor in the desired direction where the branch is desired.

Lab. A.2: connect all the elements of the circuit (RC_01)

Perform the following steps:

- connect the components for completing the RC circuit.
- Connect the output terminal of the Voltmeter and the Ammeter to the input terminal of the scopes.

The complete circuit should appear as in the following model:



Component Parameters

At each component a unique name is assigned by default. When desired, the name can be changed by double-clicking on the component in the schematic. The specific name of the components does not affect the simulation but it could be useful to specify them for documentation purposes. The parameters of the components, on the other hand, are of great importance since they determine the system behavior and hence the simulation results. Each

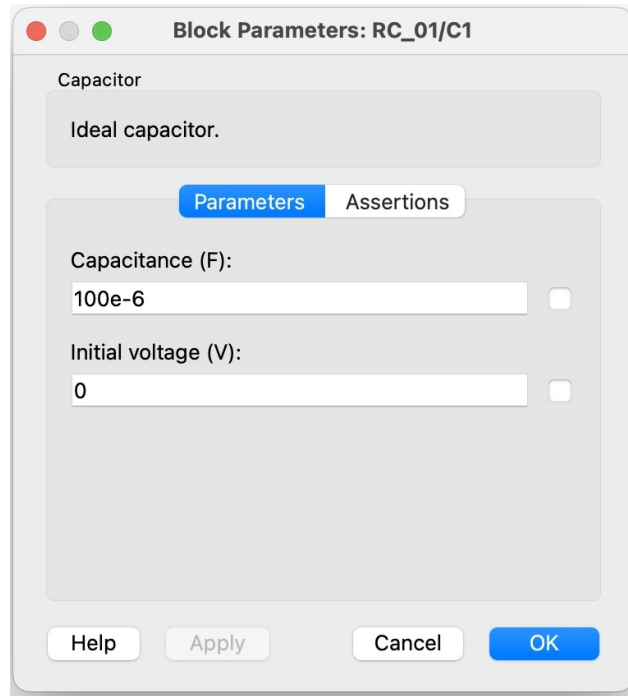


Fig. A.3: Dialog to set capacitor properties.

components has its own specific parameters. For example: the inductance of an inductor, the capacitance of a capacitor, or the voltage of a DC voltage source. Double-clicking on the component icon opens a dialog window in which the specific parameters of the component can be inserted. As an example, Fig. A.3 shows the dialog to setup the properties for a capacitor. The checkbox at the right of each parameter field allows to show (and edit) the specified value directly from the schematic window. Pressing the help button open the documentation for the selected component in which specific details and information can be found.

Units

PLECS does not understand units since letters can be used as custom variables. Therefore, the user must make sure that the variables are scaled correctly. In general, it is recommended to use the SI quantities. For example, use $10\text{e-}3$ or 0.01 , instead of 10 mH for an inductance value.



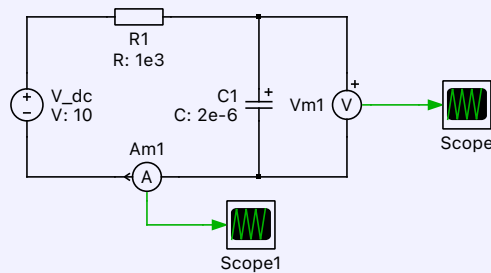
A.3

Lab. A.3: set proper values for the circuit parameters (RC_02)

Perform the following steps:

- set the DC voltage source at 10 V .
- Set the resistor at $1\text{ k}\Omega$.
- Set the capacitor at $2\text{ }\mu\text{F}$.

The complete circuit should appear as in the following model:



Simulation

Once the model has been prepared, before the simulation it is necessary to check/specify some more parameters such as the simulation time, solver type, solver options, model initialization commands etc. All these options are located in the Simulation Parameters dialog window from the Simulation menu. This dialog can also be opened using the keyboard shortcut **Ctrl-E**, see Fig. A.4. In the Initialization tab it is possible to define custom constants which can be then used to set component parameters. For example it would be possible to define $R=1000$ and $C=2e-6$ in the Initialization tab and then use the constant R and C to specify the model parameters.

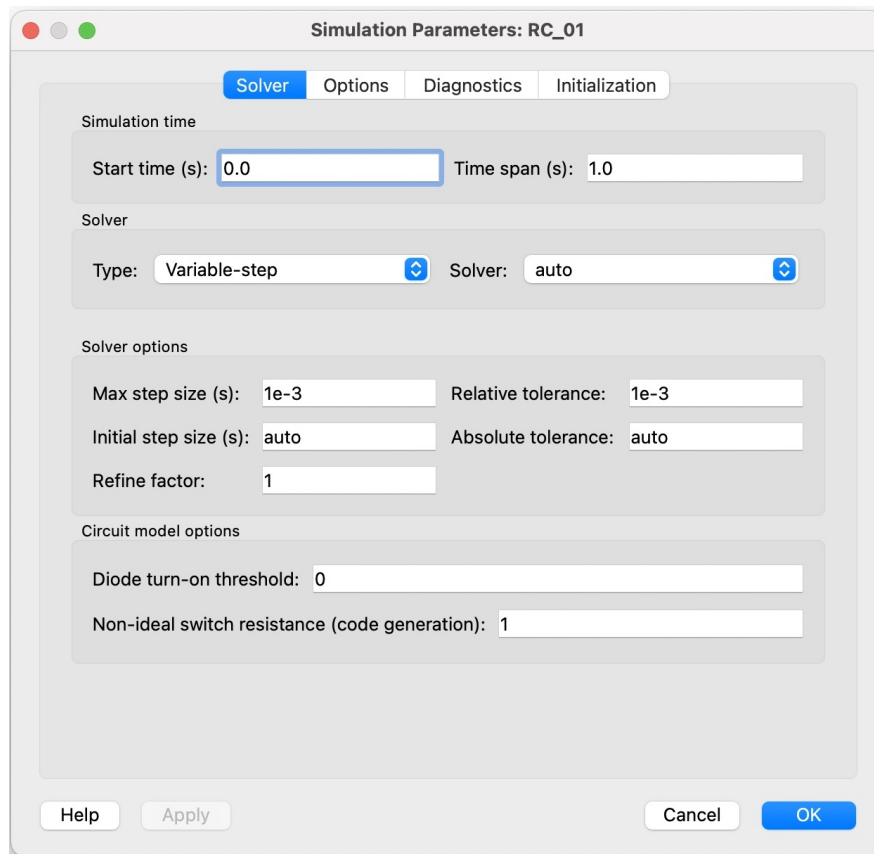


Fig. A.4: Dialog to set the simulation properties.



A.4

Lab. A.4: choose proper simulation parameters to reach steady state after transient (RC_02)

With the given parameters, the time constant of the circuit is:

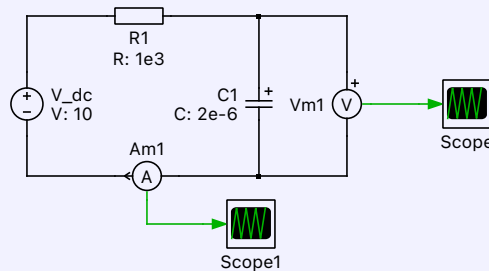
$$\tau = R \cdot C = 1000 \cdot 2 \cdot 10^{-6} = 2 \text{ ms}$$

Therefore, the steady state should be reached in a time span of about:

$$10 \cdot \tau = 20 \text{ ms} \quad (\text{A.1})$$

- Set a time span for the simulation according to (A.1).
- Run a simulation of the model selecting Start from the Simulation menu or by pressing **Ctrl-T**.
- Inspect the simulation results double-clicking the scopes in the schematic window.

The complete circuit should appear as in the following model:



Post computations

Once the model has been simulated, the scopes inserted into the schematic contain the results which can be used to perform additional computations and considerations. The scopes in PLECS are quite powerful allowing several post computations and inspection of the results. A full description is not addressed here but several resources and video which show the different features are available on the PLECS website.



A.5

Lab. A.5: post computation on the RC network (RC_03)

- Check the voltage response of the capacitor and compare it with Fig. A.2. Verify the steady state value on the basis of the adopted model parameters, i.e. $v_c(\infty) = E$.
- Check the current response of the circuit and check its initial value with the theoretical one, i.e. $i(0) = \frac{E}{R}$
- Include in the model a multiplication block to compute the instantaneous power

$p(t) = v_c \cdot i(t)$ and visualize it on a scope.

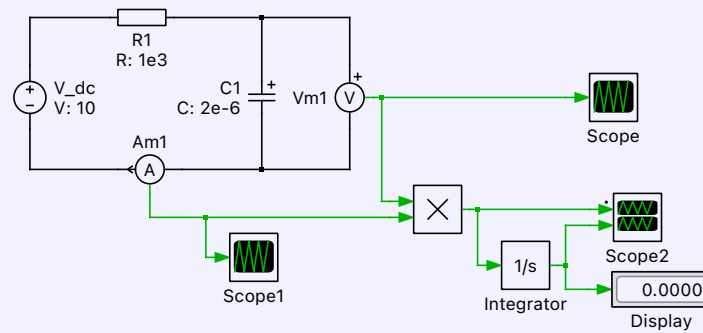
- Add an integrator block to compute the electrical work W_e entering the capacitor and visualize its value in the scope along with the instantaneous power. Add also a display to visualize the final value of the integrator state.
- Check that:

$$W_e = \int_0^{t^*} p(t) dt = \frac{1}{2} C v_c^2(t^*)$$

in particular at the end of the simulation when steady state is reached:

$$W_e = \frac{1}{2} C E^2$$

The model should appear as in the following figure:



Studying an ac network

The next example regards the simulation of an ac circuit with an RL load. In particular the circuit of Fig. A.5 is considered.

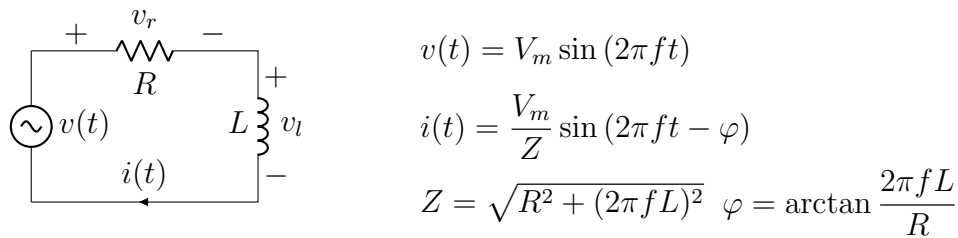


Fig. A.5: RL ac circuit.



A.6

Lab. A.6: simulation of an ac circuit (RL)

- Build and simulate in PLECS the circuit of Fig. A.5 considering the following parameters: $V_m=1000$ V; $2\pi f=100$ rad/s; $R=300$ Ω ; $L=4$ H. Choose a simulation time so that 5 ac periods are simulated.
- Check that peak value of the current at steady-state with the theoretical one $I_m = V_m/Z$.
- Include in the model a RMS block to evaluate the root mean square value of the current in the circuit (I_{rms}) and compare this value with the theoretical one and the computation from a scope. Note that the computation of the RMS value in PLECS requires in both case to specify the frequency/period on which compute the RMS of the quantity.
- Adopting two multiplication blocks, compute the instantaneous power on R and L , visualize them on a scope and compute their average value. Compare the value for the resistor with the value RI_{rms}^2 .
- Using the scope, measure the current lag with respect the generator voltage (using cursors) and then compute the active power delivered by the generator $v(t)$. Compare this value with the average power computed on the resistor.

The model should appear as in the following figure:

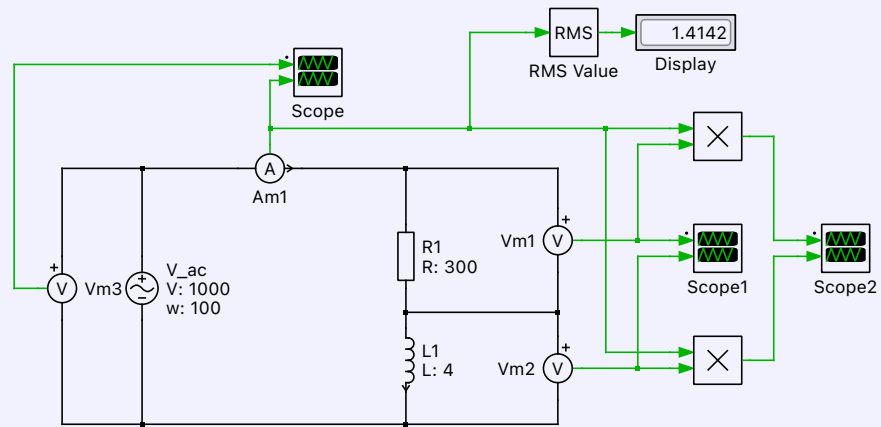
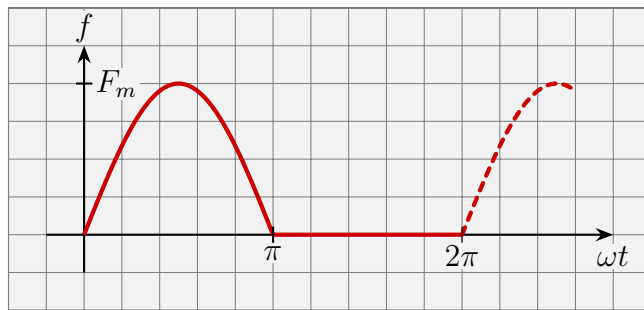


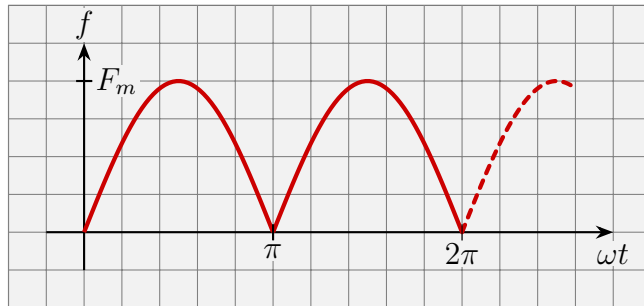
Table of useful Fourier series expansion

Half sinewave



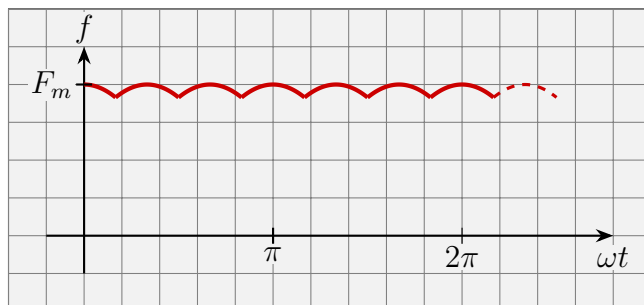
$$f(t) = \frac{F_m}{\pi} + \frac{F_m}{2} \sin(\omega t) - \sum_{n=2,4,6,\dots}^{\infty} \frac{2F_m}{(n^2 - 1)\pi} \cos(n\omega t)$$

Rectified sinewave



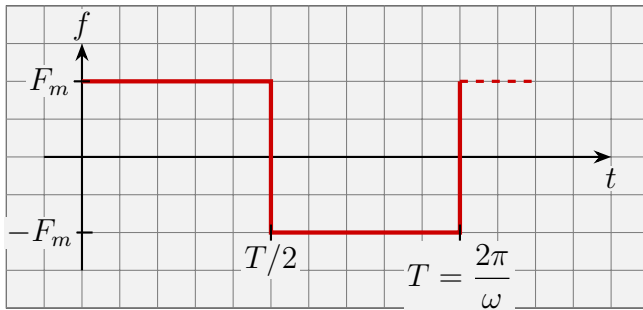
$$f(t) = \frac{2 F_m}{\pi} \left\{ 1 + \sum_{n=2,4,6,\dots}^{\infty} \frac{2}{n^2 - 1} \cos(n\omega t + \pi) \right\}$$

Terna trifase raddrizzata



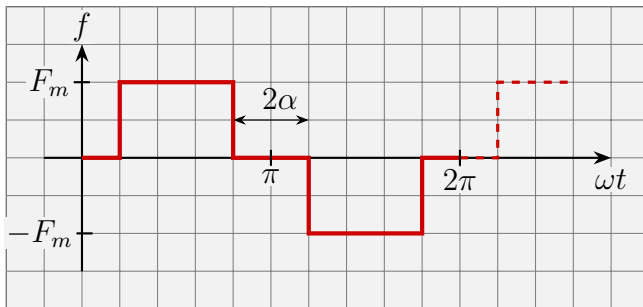
$$f(t) = \frac{3}{\pi} F_m + \sum_{n=6,12,18,\dots}^{\infty} \frac{6 F_m}{\pi(n^2 - 1)} \cos(n\omega t + \pi)$$

Onda quadra



$$f(t) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{\pi} \frac{F_m}{n} \sin(n\omega t)$$

Onda quadra modificata



$$f(t) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{\pi} \frac{F_m}{n} \cos(n\alpha) \sin(n\omega t)$$